Skip Connections Unlock d^2_{model} Parameters per Layer

Marko Karbevski*†
In Simplicity Technologies
markokarbevski@insimplicity.tech
markokarbevski@gmail.com

Antonij Mijoski*
IRMA, Université de Strasbourg
antonij.mijoski@unistra.fr
antonijmijo@gmail.com

Abstract

Skip connections change the function class of an MLP block. By how much? Here we show that they unlock d^2_{model} parameters per block, or about $d^2_{\text{model}}/2$ under pre-RMSNorm. In a network without skips, the function does not change when the output weights of a block are multiplied by any invertible matrix and the input weights of the next block by its inverse. This freedom makes d^2_{model} parameters redundant at every hidden state. Under pre-RMSNorm the matrix must be orthogonal, and about half as many are redundant: 8.5 percent of a skipless ViT-B, whose LayerNorm is a form of RMSNorm. Without normalization, the same freedom turns a trainable linear skip into the identity, so a network with trainable skips reduces exactly to a network with ordinary skip connections. We then prove that the parameters that are redundant without skips are meaningful with them: skips raise the functional dimension of the network, for MLPs with standard activations and for attention blocks.

1 Introduction

Skip connections are a building block of almost every deep network trained today (He et al., 2016). The usual reason is training: skips make deep networks easier to optimize (Balduzzi et al., 2017; He et al., 2016; Orhan and Pitkow, 2018). Mijoski and Karbevski (2026) show that skips also change which functions a network can represent: for most kinds of block, almost every network with skips computes a function that no network of such blocks without skips can compute, whatever its depth and widths. Their result establishes that there is a difference in representation, but does not quantify it. In the same direction, we show that skip connections add at least d^2_{model} degrees of freedom per block, and about half as many under pre-RMSNorm, where d_{model} is the model dimension.

Skipless networks are now trained as well. He et al. (2023) train deep transformers without skips by modifying the attention, and He and Hofmann (2024) remove the skips of transformer blocks without losing training speed. Ji et al. (2025) find that, with standard training, the skips around the MLP blocks of a vision transformer can be removed with little loss of accuracy. Ji et al. (2026) then remove the skips around the attention blocks as well, with a suitable initialization, and, notably, their ViT-B with no skips at all outperforms the residual one on ImageNet-1k, at the same parameter budget. Their explanation is that a skip keeps bringing the information of early layers back into later ones, which disrupts the hierarchy of features that depth should build.

Where the degrees of freedom come from. Without skips, the basis in which a hidden state is written is free: if the weights that write the output of one block are multiplied by an invertible matrix, and the weights through which the next block reads it by the inverse, the function does not change. Such a change of basis can set a $d_{\text{model}} \times d_{\text{model}}$ submatrix of the weights to the identity, so every hidden state of a skipless network carries d^2_{model} redundant parameters. Under pre-RMSNorm only orthogonal changes of basis preserve the function, and a skipless network carries

*Author contributions. Marko Karbevski (MK) conceived and led the study, proved the results at depth (Theorems 4.7 to 4.9), and worked on the symmetry breaking across layers and its counting, the parameter counts at model scale, the experiments and the applications. Antonij Mijoski (AM) worked on the writing. All of MK's work on the results, the proofs and the experiments was done with AI assistants.

†Corresponding author.

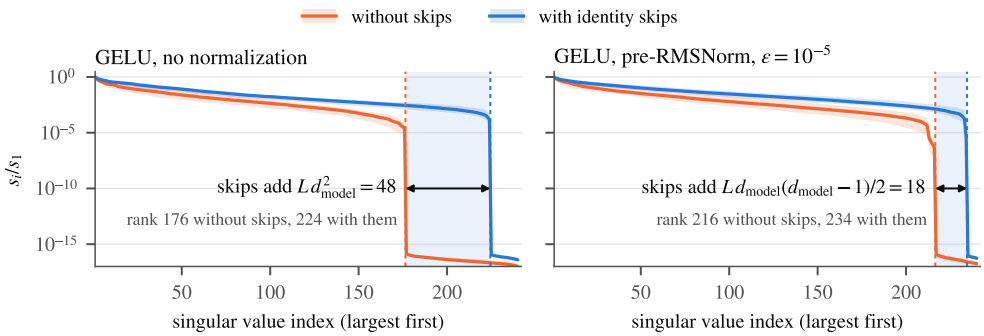


Figure 1: Singular values of the Jacobian of the parameter-to-output map on 4000 Gaussian inputs, with and without skips at the same weights, for three GELU blocks of 8 units with $d_{\text{model}} = 4$; median and range over 20 draws of weights and inputs. Skips raise the rank, the functional dimension.

$d_{\text{model}}(d_{\text{model}} - 1)/2$ redundant parameters per hidden state. A skip adds each hidden state to the output of the next block, and the sum is the next hidden state, so the function is preserved only when two consecutive hidden states have the same change of basis. With skips, one change of basis is therefore left for the whole network, whereas a skipless network of L blocks has one at each of its $L + 1$ hidden states, from the one after the embedding to the one before the readout. The difference is L changes of basis: d_{model}^2 parameters per block, or $d_{\text{model}}(d_{\text{model}} - 1)/2$ under pre-RMSNorm. In the skipless ViT-B of Ji et al. (2026), whose LayerNorm is RMSNorm applied to centered tokens, the redundant parameters at its 25 hidden states are 8.5 percent of the model (Section 3.4).

Functional dimension. By the degrees of freedom of a network we mean its *functional dimension*, the number of independent directions in which the parameters move the function (Grigsby et al., 2025). We use the two terms interchangeably. At generic parameters, the functional dimension is the local dimension of the set of functions that the network can represent. Redundant parameters do not move the function, so the functional dimension is at most the number of parameters minus the number of redundant ones, those inside each block included (Section 4.2). For networks without biases and with continuous inputs, which excludes one-hot tokens, Section 4 proves that a residual network of attention, $\tanh$, GELU, SwiGLU or GeGLU blocks attains this bound, at any depth and with or without pre-RMSNorm. So parameters that are redundant without skips move the function with skips, which is the sense in which skips unlock them (Figure 1). The increase in functional dimension is exactly d_{model}^2 per block, or $d_{\text{model}}(d_{\text{model}} - 1)/2$ under pre-RMSNorm, when the skipless network attains its bound too. We prove that it does in three cases, all without normalization: networks of one block, of $\tanh$ blocks, and of GELU blocks with at least $2d_{\text{model}}$ hidden units (Section 4.3).

Functional dimension and equality of functions. The result of Mijoski and Karbevski (2026) does not cover GELU blocks without normalization: there, they show that almost every residual network computes the same function as a skipless network with d_{model} more units per block. Functional dimension shows that extra units are needed. At the same widths, skipless networks have the lower functional dimension, and a set of functions cannot lie inside a set of lower dimension. From this, Theorem 4.9 deduces that almost no residual network of attention, $\tanh$, GELU, SwiGLU or GeGLU blocks computes the same function as a skipless network of the same widths. For GELU networks of L blocks without normalization and almost every residual network, it also deduces that a skipless network that computes the same function has at least $Ld_{\text{model}}/2$ more units in total. A greater functional dimension does not guarantee the inclusion in the other direction, that residual networks compute every function of the skipless networks of the same widths, or even one of them. Without normalization, every skipless network is nevertheless a limit of residual networks of the same widths (Mijoski and Karbevski, 2026).

Contributions and related work. Table 1 lists the results, and the proofs are in the appendix.

1. *A count of changes of basis, with and without skips* (Section 3). Theorems 3.3 and 3.4 prove that the changes of basis are one per hidden state without skips and one in all with them. The change of basis itself is known. Zhao et al. (2023) define a change of basis at every hidden layer of an

MLP, which preserves the function where the activation is linear and, for orthogonal changes, where it is a radial rescaling (their Proposition C.1), and they remark that it adapts to networks with skips (their Appendix C.5). Graef (2024) arrived at it independently and used it to remove weights in a restricted case, skipless transformers without normalization. Changes of basis can also set the values and projections of skipless blocks to the identity, as He and Hofmann (2024) do. In residual ReLU networks, skips are known to tie the permutations and rescalings of the hidden states (Godfrey et al., 2022, Remark 3.6). In the language of gauge theory, skips turn a symmetry that is local in depth into a global one (Hashimoto et al., 2024). Where a block outweighs its skip, the local changes of basis remain approximate symmetries (Gong et al., 2026, Section 6.4).

2. *The one change of basis that the skips leave* (Sections 3.3 and 3.4). We show that the changes of basis that preserve the function of a residual network are the same at all hidden states, and orthogonal under pre-RMSNorm (Theorems 3.3 and 3.4). This one change of basis also changes the embedding, so it is not a product of symmetries of the blocks, which Wang and Wang (2025) characterize (Appendix C). Earlier work uses it. The residual stream has no privileged basis (Elhage et al., 2021). SliceGPT proves that an RMSNorm transformer computes the same function after one rotation of every matrix that touches the stream (Ashkboos et al., 2024a, Theorem 1), QuaRot and SpinQuant use this invariance for quantization (Ashkboos et al., 2024b; Liu et al., 2025), and model merging aligns streams with one global permutation (Verma and Elbayad, 2024, Section 3.3) or orthogonal matrix (Theus et al., 2025). van Nierop (2024) counts the symmetries of LayerNorm transformers, with one rotation of the stream for the whole model.
3. *Learned skips* (Section 3.2). A learned skip multiplies the hidden state by a trainable $d_{\text{model}} \times d_{\text{model}}$ matrix A before adding it, and the usual skip, with $A = I$, is an *identity skip*. A block with a learned skip is a block with d_{model} additional identity units, hidden units whose activation is the identity. Without normalization, changes of basis turn invertible learned skips into identity skips without changing the function (Theorem 3.2). So the Ld_{model}^2 entries of the learned skips can be removed, and what is left is the network with skip connections. Under pre-RMSNorm, the learned skips can be made triangular. Other work trains or emulates skips: LAuReL trains skips that are the identity plus a low-rank matrix (Menghani et al., 2025), RMNet emulates the skips of convolutional networks with ReLU units (Meng et al., 2021), and SliceGPT inserts rotations into identity skips and deletes rows and columns, approximating the model (Ashkboos et al., 2024a, Sections 3.3 and 3.4).
4. *Functional dimension* (Section 4). Beyond the cases proved above (Theorems 4.6 to 4.8), we measure the increase in functional dimension in MLPs with other activations and in small transformers (Section 4.5). Grigsby et al. (2025) define the functional dimension of ReLU networks and bound it by their rescaling symmetries; it is lower on sets of parameters of positive measure (Grigsby et al., 2023), and it is observed to fall during training (Bona-Pellissier et al., 2024). It bounds a local pseudodimension from below (Grigsby and Lindsey, 2024), and it accounts for the memorization capacity of attention layers better than the parameter count does (Henry, 2026). For polynomial networks it is the dimension of the functional variety, a measure of expressive power that Kileel et al. (2019) compute, as we do, as the rank of a Jacobian over a finite field. For attention with one head and without softmax, Henry et al. (2025) prove that a generic skipless network has no redundancy beyond scalings and changes of basis inside and between its layers, and they conjecture the same with softmax; with several heads, one attention layer has no continuous symmetries beyond those inside its heads (Henry, 2026). MacDonald et al. (2023) show that identity skips keep the Jacobian of the parameter-to-output map of full row rank on linearly independent inputs. We study the kernel of the same Jacobian. Marcotte et al. (2025) show that a block has the same conservation laws of gradient flow with and without its skip. This is consistent with our result, which needs an embedding and a readout around the block: without them, the skip does not change the derivative of the block with respect to its parameters.
5. *Identifiability* (Section 4.2). For attention, $\tanh$, GELU, SwiGLU and GeGLU blocks, we show that the symmetries are the only redundancy of a network exactly when, near generic parameters, the function determines the parameters up to these symmetries. Residual networks have this property (Theorem 4.7). Dahlgren et al. (2026)[1] prove that residual networks with polynomial

[1]Concurrently with and independently of our first version (7 October 2026, https://github.com/

Proven	Value	Blocks	Result
Parameters removed without changing the function			
without skips	$(L+1)d_{\text{model}}^2$	all	Prop. 3.1
with pre-RMSNorm	$(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$	all	Thm. 3.4(i)
learned skips	Ld_{model}^2, every skip entry	all	Thm. 3.2
with pre-RMSNorm	$Ld_{\text{model}}(d_{\text{model}}-1)/2$	all	Thm. 3.4(iv)
identity skips	d_{model}^2	all	Thm. 3.3(iii)
with pre-RMSNorm	$d_{\text{model}}(d_{\text{model}}-1)/2$	all	Thm. 3.4(iii)
Dimensions of symmetry removed by the skips			
any L	Ld_{model}^2	all	Thm. 3.3(ii)
with pre-RMSNorm	$Ld_{\text{model}}(d_{\text{model}}-1)/2$	all	Thm. 3.4(ii)
Functional dimension added by the skips			
$L=1$	exactly d_{model}^2	all	Thm. 4.6(i)
with pre-RMSNorm	at least $d_{\text{model}}(d_{\text{model}}-1)/2$	all	Thm. 4.6(ii)
$L\geq 2$	at least Ld_{model}^2	analytic	Thm. 4.7
with pre-RMSNorm	at least $Ld_{\text{model}}(d_{\text{model}}-1)/2$	analytic	Thm. 4.7
$L\geq 2$	exactly Ld_{model}^2	$\tanh$	Thm. 4.8(i)
$L\geq 2$	exactly Ld_{model}^2	GELU, $\geq 2d_{\text{model}}$ units	Thm. 4.8(ii)
Residual networks that equal a skipless network			
same widths, any L	measure zero	analytic	Thm. 4.9(i)
with pre-RMSNorm	measure zero	analytic	Thm. 4.9(i)
$< Ld_{\text{model}}/2$ more units	measure zero	GELU	Thm. 4.9(ii)

Table 1: The results at generic parameters, for networks of L blocks with model dimension d_{model}; scope in Section 2. All: every block of Section 2. Analytic: attention, $\tanh$, GELU, SwiGLU and GeGLU blocks.

activations and without write matrices are generically identifiable up to finitely many sign flips. The changes of basis that the skips remove do not arise there.

2 Setup

Blocks. A hidden state is a matrix $h \in \mathbb{R}^{T\times d_{\text{model}}}$ whose rows are the T tokens of a sequence. A block reads it with a trainable matrix W_{read}, the *read*, applies a fixed map f, and writes the result with a trainable matrix W_{write}, the *write*:

$$B(h) = f(hW_{\text{read}})\, W_{\text{write}}. \tag{1}$$

In an MLP block with m hidden units, its width, $W_{\text{read}} \in \mathbb{R}^{d_{\text{model}}\times m}$, $W_{\text{write}} \in \mathbb{R}^{m\times d_{\text{model}}}$, and f applies an activation to every unit of every token: $\tanh$, GELU, ReLU, or its square, ReLU^2. In a gated block the read has $2m$ columns, which give each token a gate g_i and a value v_i for each unit, and f maps them to $\sigma(g_i)\, v_i$, with σ the ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). In an attention block with H heads of width d_h, the read holds the query, key and value matrices W_Q, W_K and W_V of all heads, f is the attention map, and the write is the output projection W_O; the width of the block is Hd_h.

Networks. A network has a linear embedding $E \in \mathbb{R}^{d_{\text{in}}\times d_{\text{model}}}$, then L blocks $B_1, \ldots, B_L$, each with a skip A_ℓ, then a linear readout, or unembedding, $D \in \mathbb{R}^{d_{\text{model}}\times d_{\text{out}}}$, and an output function C. It maps an input $x \in \mathbb{R}^{T\times d_{\text{in}}}$ to

$$\mathcal{M}(x) = C \circ D \circ (A_L + B_L) \circ \cdots \circ (A_1 + B_1) \circ E\,(x), \tag{2}$$

where E, D and A_ℓ stand for the maps $x \mapsto xE$, $h \mapsto hD$ and $h \mapsto hA_\ell$: $A_\ell = 0$ in a *skipless* network, $A_\ell = I$ in a *residual* network, and A_ℓ is a trainable $d_{\text{model}} \times d_{\text{model}}$ matrix in a network

`MarkoKarbevski/skips_fix_gauges/commit/8128617`).

with *learned skips*. So $h_0 = xE$, $h_\ell = h_{\ell-1}A_\ell + B_\ell(h_{\ell-1})$ and $y = h_L D$. The output function C is set by the task, for example the identity for regression and the softmax for multiclass classification (Bishop, 2006, Sections 5.1 and 5.2). It has no weights, and we ignore it: functions, equality and functional dimension below are those of the map $x \mapsto y$. The skipless and the residual network have the same P parameters. A transformer layer has two blocks, its attention and MLP sub-layers: a transformer of 12 layers has $L = 24$, and *per block* means per sub-layer.

Normalization. RMSNorm, written N, divides each row v of h by $\sqrt{\|v\|^2/d_{\text{model}} + \varepsilon}$, with $\varepsilon > 0$. In a *pre-RMSNorm* network the blocks and the readout read $N(h)$ in place of h. Skips read h itself, so a residual network computes $h_\ell = h_{\ell-1} + B_\ell\big(N(h_{\ell-1})\big)$ and $y = N(h_L)D$. A learned gain after N is folded into the next read. Below, *under RMSNorm* means in a pre-RMSNorm network.

Scope. Networks have no biases, and their inputs are continuous: they range over all of $\mathbb{R}^{T\times d_{\text{in}}}$, which excludes one-hot tokens. Every MLP block has at least d_{model} hidden units, and all MLP blocks are of one kind. All attention blocks have the same number of heads, of width at most d_{model} and of total width $Hd_h \geq d_{\text{model}}$, and act on sequences of at least two tokens. The input and the output satisfy $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$, which excludes a classifier with fewer classes than d_{model}. The reductions of Section 3 hold for any inputs, one-hot tokens included, and with biases. For one-hot tokens the counts of Section 4 are measured (Section 4.5), and with biases the skips are measured to add the same Ld^2_{model} (Table 6). The appendix states each result in greater generality. A statement holds at *generic* parameters if it holds outside a set of Lebesgue measure zero, so with probability one for weights drawn from a density. Attention, $\tanh$, GELU, SwiGLU and GeGLU blocks are real-analytic, *analytic* for short, and for them any open set of inputs gives the same results.

3 Changes of basis that preserve the function

3.1 Skipless networks: one change of basis per hidden state

Without normalization, nothing nonlinear happens between the write of one block and the read of the next, so the basis of the hidden state between them is free. For an invertible $d_{\text{model}} \times d_{\text{model}}$ matrix G, the *change of basis* G at h_ℓ (Zhao et al., 2023) maps

$$W \mapsto WG \text{ for the writes into } h_\ell, \qquad W \mapsto G^{-1}W \text{ for the reads of } h_\ell. \tag{3}$$

The writes into h_ℓ are W^ℓ_{write}, or E if $\ell = 0$, and the reads of h_ℓ are $W^{\ell+1}_{\text{read}}$, or D if $\ell = L$. The hidden state becomes $h_\ell G$, and every read of it computes the same product as before, so the function does not change (Figure 2b). A suitable G removes parameters.

Proposition 3.1 (Exact reduction without skips)**.** *Without normalization, let d_{model} columns of the read of h_ℓ in a skipless network form an invertible matrix W_1. The change of basis $G = W_1$ at h_ℓ preserves the function and turns W_1 into the identity. At generic parameters every hidden state has such columns, and these changes of basis at all $L+1$ hidden states remove $(L+1)d^2_{\text{model}}$ parameters.*

A special case of Proposition A.9, proved in Appendix A.3.

Rows of a write work too: if d_{model} rows of the write into h_ℓ form an invertible matrix, the change of basis by its inverse turns them into the identity. In a skipless transformer with $Hd_h = d_{\text{model}}$, $G = W_Q$ before an attention block sets its queries to the identity, and $G = W_O^{-1}$ after it sets its output projection to the identity: $2d^2_{\text{model}}$ parameters per layer. This is how Graef (2024) removes 15 percent of the weights of a skipless Mistral-7B.

3.2 Learned skips: reduction to identity skips

A learned skip, which adds hA in place of h, is a block of d_{model} *identity units*, hidden units whose activation is the identity (Lin and Jegelka, 2018, Section 4), read by A and written by I. The block with this skip is then a wider block of the form (1) (Figure 2a):

$$hA + f(hW_{\text{read}})\,W_{\text{write}} \;=\; \tilde{f}\big(h\,[\,A \mid W_{\text{read}}\,]\big)\begin{bmatrix} I \\ W_{\text{write}} \end{bmatrix}, \qquad \tilde{f}\big([\,u \mid z\,]\big) = \big[\,u \mid f(z)\,\big]. \tag{4}$$

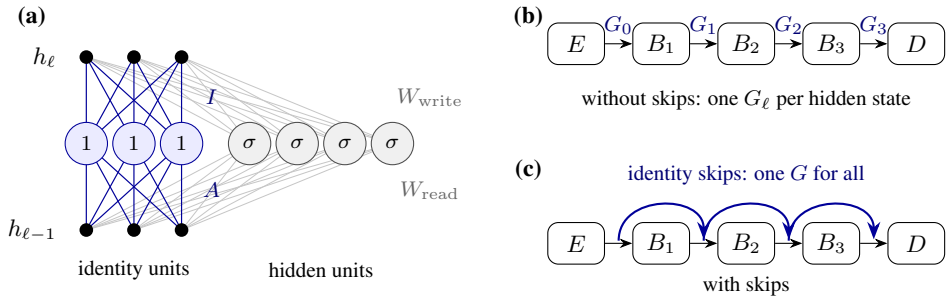


Figure 2: (a) A learned skip A acts as d_{model} more hidden units, with the identity as activation and A as read. (b) Without skips, every hidden state has its own change of basis G_ℓ. (c) Identity skips carry each hidden state into the next, so one change of basis is left.

So a skip is a $d_{\text{model}} \times d_{\text{model}}$ submatrix of a read, and a change of basis can turn it into the identity, as in Proposition 3.1. The skip A_ℓ reads $h_{\ell-1}$ and writes h_ℓ, so changes of basis at the two hidden states map it to $G_{\ell-1}^{-1} A_\ell G_\ell$, and $G_{\ell-1} = A_\ell G_\ell$ turns it into the identity. This multiplies the skip before it, $A_{\ell-1}$, by $A_\ell G_\ell$. Sweeping from the readout back, with $G_L = I$, gives $G_\ell = A_{\ell+1} \cdots A_L$, called Λ_ℓ below: every skip becomes the identity, and their product moves into the embedding.

Theorem 3.2 (Learned skips reduce to identity skips)**.** *Without normalization, let every learned skip A_ℓ be invertible, and let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$. The network with learned skips computes the same function as the residual network with the embedding $E\Lambda_0$, the reads $\Lambda_{\ell-1}^{-1} W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Lambda_\ell$ and the readout D.*

A special case of Theorem A.5, proved in Appendix A.2.

So all $L d_{\text{model}}^2$ parameters of invertible learned skips can be removed, and the functional dimension is that of the residual network (Theorem A.5(ii)).

3.3 Identity skips: one change of basis for the whole network

A residual network keeps one change of basis, the same at all its hidden states. With a change of basis $G_{\ell-1}$ at $h_{\ell-1}$ and G_ℓ at h_ℓ, a residual block computes

$$h_{\ell-1} G_{\ell-1} + B_\ell(h_{\ell-1})\, G_\ell \;=\; h_\ell G_\ell + h_{\ell-1}(G_{\ell-1} - G_\ell). \tag{5}$$

The skip carries $h_{\ell-1}$ in the basis $G_{\ell-1}$, and the block writes in the basis G_ℓ. Unless the two agree, the defect $h_{\ell-1}(G_{\ell-1} - G_\ell)$ reaches every later block.

Theorem 3.3 (One change of basis for the whole network)**.** *Without normalization, let $G_0, \ldots, G_L$ be invertible.*
(i) *The map $E \mapsto EG_0$, $W_{\text{read}}^\ell \mapsto G_{\ell-1}^{-1} W_{\text{read}}^\ell$, $W_{\text{write}}^\ell \mapsto W_{\text{write}}^\ell G_\ell$, $D \mapsto G_L^{-1} D$ preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$.*
(ii) *The changes of basis that preserve the function at all parameters form a group of dimension $(L+1) d_{\text{model}}^2$ in a skipless network and d_{model}^2 in a residual one.*
(iii) *In a residual network, let d_{model} columns of any one read form an invertible matrix W_1. The same change of basis W_1 at all hidden states preserves the function and turns W_1 into the identity.*

Proof in Appendix A.2.

So the L skips remove $L d_{\text{model}}^2$ dimensions from this group, and Section 4 shows when they become functional dimension.

3.4 RMSNorm and LayerNorm: orthogonal changes of basis

With pre-RMSNorm, the network with learned skips computes

$$h_\ell = h_{\ell-1} A_\ell + B_\ell\big(N(h_{\ell-1})\big), \qquad y = N(h_L)\, D. \tag{6}$$

A change of basis G at h_ℓ now reaches the reads as $N(h_\ell G)$. For an orthogonal Q, $N(hQ) = N(h)Q$, since Q preserves the norm of every row, and the read undoes Q by $W \mapsto Q^\top W$. Since $\varepsilon > 0$, a read can undo no other invertible matrix (Lemma A.2). So every hidden state of a skipless network, or of one with learned skips, keeps an orthogonal change of basis, $d_{\text{model}}(d_{\text{model}} - 1)/2$ dimensions. It cannot turn a submatrix into the identity, but a QR factorization makes it triangular.

Theorem 3.4 (Orthogonal changes of basis under RMSNorm)**.** *Consider pre-RMSNorm networks.*
(i) *In a skipless network, let d_{model} columns of the read of h_ℓ form a matrix $W_1 = QR$, with Q orthogonal and R upper triangular. The change of basis Q at h_ℓ preserves the function and turns W_1 into R; at all $L+1$ hidden states, they remove $(L+1)d_{\text{model}}(d_{\text{model}} - 1)/2$ parameters.*
(ii) *The changes of basis that preserve the function at all parameters are orthogonal, and in a residual network they agree at all hidden states. They form a group of dimension $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ in a skipless network and $d_{\text{model}}(d_{\text{model}} - 1)/2$ in a residual one.*
(iii) *In a residual network, let d_{model} columns of any one read form a matrix $W_1 = QR$. The same change of basis Q at all hidden states preserves the function and turns W_1 into R.*
(iv) Triangular skips. *If every learned skip A_ℓ is invertible, there are orthogonal $Q_1, \dots, Q_L$, with $Q_0 = I$, such that every $Q_{\ell-1}^\top A_\ell Q_\ell$ is upper triangular, and the network with these skips, the reads $Q_{\ell-1}^\top W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function.*

Proof in Appendix A.4.

Learned skips under RMSNorm. Under RMSNorm, part of a learned skip cannot be removed. Write a skip as an upper triangular matrix times an orthogonal one. The orthogonal factor is removable, by Theorem 3.4(iv). The $d_{\text{model}}(d_{\text{model}} + 1)/2$ entries of the triangular factor stay, and an identity skip fixes them as well (Theorem A.10(iii)).

LayerNorm. LayerNorm is RMSNorm after a projection that removes the mean of each row, so every RMSNorm count above holds with $d_{\text{model}} - 1$ in place of d_{model} (Proposition A.12).

Trained models. The reduction applies to trained weights. In the skipless ViT-B (Dosovitskiy et al., 2021) of Ji et al. (2026), with standard pre-LayerNorm blocks, the changes of basis at its 25 hidden states remove 7.3M parameters, 8.5 percent of the model, without changing its function. A residual model keeps one change of basis, and Appendix D gives the counts for the residual ViT-B and for Llama-3.1 (Grattafiori et al., 2024).

4 Functional dimension with and without skips

The parameters removed in Section 3 did not change the function, so the functional dimension is at most the number that remain. This section shows when the two are equal.

4.1 Definition and a linear example

Definition 4.1 (Functional dimension (Grigsby et al., 2025))**.** Let F_θ be the function computed at parameters $\theta \in \mathbb{R}^P$. The *functional dimension* at θ is the largest rank of the Jacobian of $\theta \mapsto (F_\theta(x_1), \dots, F_\theta(x_n))$ over finite batches of inputs $x_1, \dots, x_n$. The *redundancy* at θ is P minus the functional dimension.

A continuous symmetry adds to the redundancy: if a continuous family of maps of the parameters preserves the function, the directions in which it moves θ lie in the kernel of every such Jacobian.

Example 4.2 (A product of three matrices)**.** The linear network $x \mapsto xM_1M_2M_3$, with three $d_{\text{model}} \times d_{\text{model}}$ matrices, has $3d_{\text{model}}^2$ parameters and computes every linear map $x \mapsto xM$. So its functional dimension is d_{model}^2 at generic parameters, and its redundancy is $2d_{\text{model}}^2$: one change of basis at each of its two hidden states. Merging M_2 and M_3 into $M_2' = M_2M_3$ leaves $2d_{\text{model}}^2$ parameters, the same functions and the same functional dimension. Using a symmetry in this way lowers the parameter count and leaves the functional dimension unchanged.

4.2 Upper bounds from the symmetries

Two kinds of symmetry bound the functional dimension. The changes of basis of one hidden state form a group $\mathcal{G}$: $\mathrm{GL}(d_{\mathrm{model}})$ without normalization, of dimension d_{model}^2, and $\mathrm{O}(d_{\mathrm{model}})$ under RMSNorm, of dimension $d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$. The *symmetries inside the blocks* change the read and the write of one block and leave the map of that block unchanged. Permutations of hidden units or heads and sign flips of $\tanh$ units are finite and remove no parameters (Fefferman, 1994; Sussmann, 1992). The others are continuous. Each hidden unit of a ReLU or ReLU^2 block can be rescaled, and so can the value of each gated unit and, in ReGLU, its gate (Godfrey et al., 2022; Rolnick and Kording, 2020). Each attention head has a change of basis between its queries and keys and one between its values and outputs, of dimension d_h^2 each (Elhage et al., 2021; Theus et al., 2025; Wang and Wang, 2025). A block of $4d_{\mathrm{model}}$ ReLU units has $4d_{\mathrm{model}}$ rescalings, against d_{model}^2 dimensions for a change of basis. Let s be the total dimension of the symmetries inside the blocks, summed over the blocks. For $\tanh$ and GELU blocks, $s = 0$. These symmetries leave the map of every block unchanged, so they preserve the function with skips and without them, and s is the same number for both networks.

Theorem 4.3 (Upper bounds)**.** *Let $\mathcal{G}$ be the group of changes of basis of one hidden state, and s the dimension of the symmetries inside the blocks, the same with and without skips. With or without pre-RMSNorm, at generic parameters, the functional dimension is at most $P - (L+1)\dim\mathcal{G} - s$ without skips and at most $P - \dim\mathcal{G} - s$ with identity skips.*

A special case of Theorem A.13, proved in Appendix A.6.

Definition 4.4 (Equality condition)**.** A network satisfies the *equality condition* if its bound in Theorem 4.3 is attained at generic parameters, that is, if its symmetries are its only redundancy.

When the residual network satisfies it, each skip unlocks $\dim\mathcal{G}$ parameters, d_{model}^2 without normalization and $d_{\mathrm{model}}(d_{\mathrm{model}}-1)/2$ under RMSNorm: they are redundant in the skipless network and count in the residual one.

Definition 4.5 (Identifiability up to symmetries)**.** A network is *identifiable up to its symmetries* at θ if every θ' with $F_{\theta'} = F_\theta$ is the image of θ under a symmetry: a change of basis that preserves the function, a symmetry inside a block, or a composition of these. It is *locally identifiable* at θ if this holds for every θ' in some neighborhood of θ.

Functional dimension and identifiability. The two notions meet in the equality condition. For analytic blocks, a network satisfies it if and only if it is locally identifiable up to its symmetries at generic parameters (Proposition A.16): the symmetries are the only redundancy exactly when, near θ, they reach every network with the same function. So identifiability implies the equality condition, which is how Theorem 4.8 treats skipless $\tanh$ networks. The equality condition does not give identifiability in return, because it says nothing about parameters far from θ. The residual networks of Theorem 4.7 are thus locally identifiable, and whether they are identifiable is open.

The gain in a transformer. Under the equality condition, a residual network with P parameters has the functional dimension of a skipless network with $L\dim\mathcal{G}$ more parameters. Under normalization, this is about a twelfth more for transformer layers with an MLP of width $4d_{\mathrm{model}}$ (Appendix D). What functional dimension predicts is only partly known (Section 1), and its link to accuracy is open. The skipless ViT-B of Ji et al. (2026) outperforms the residual one. If it satisfies the equality condition, it does so with less functional dimension, by the changes of basis at 24 of its 25 hidden states.

4.3 When the bounds are attained

The equality condition is a hypothesis, and it can fail (Section 4.5). We prove it in three cases.

Theorem 4.6 (Equality for one block)**.** *Let $L = 1$, with a $\tanh$, GELU, ReLU, ReLU^2, ReGLU, SwiGLU, GeGLU or attention block.*

(i) *Without normalization, the skipless and the residual network satisfy the equality condition, and the skip raises the functional dimension by exactly* d_{model}^2.
(ii) *Under RMSNorm, the residual network satisfies the equality condition, and the skip raises the functional dimension by at least* $d_{\text{model}}(d_{\text{model}}-1)/2$. *A learned skip raises it by exactly* $d_{\text{model}}(d_{\text{model}}+1)/2$ *more than an identity skip.*

A special case of Theorem B.1, proved in Appendix B.

Theorem 4.7 (Equality for residual networks at any depth)**.** *Let the blocks be analytic, with or without pre-RMSNorm. The residual network satisfies the equality condition: at generic parameters its functional dimension is* $P - \dim \mathcal{G} - s$. *So the skips raise the functional dimension by at least* $L \dim \mathcal{G}$, *and by exactly* $L \dim \mathcal{G}$ *when the skipless network satisfies the equality condition too.*

Proof in Appendix A.7.

Theorem 4.8 (Exact gain for $\tanh$ and GELU)**.** *Without normalization, in the two cases below the skipless network satisfies the equality condition too, so the skips raise the functional dimension by exactly* Ld_{model}^2.
(i) *The blocks are* $\tanh$ *blocks.*
(ii) *The blocks are GELU blocks with at least* $2d_{\text{model}}$ *hidden units.*

Proof in Appendix A.7.

Proof sketches. With one block, the parameters enter the function only through the products ED, $W_{\text{write}}D$ and EW_{read}. A factorization of a matrix of rank d_{model} into two factors is unique up to a change of basis between them, so the skipless network keeps one change of basis at h_0 and one at h_1, and the term xED of the skip ties them into one. At any depth, multiply the writes by a small τ: to first order in τ every block reads the embedding xE, and the residual network is the linear map xED plus one wide block that holds all its units and heads. This brings the one-block argument to every depth, and for analytic blocks a rank that is reached at one choice of weights is reached at almost all. For skipless $\tanh$ networks, the function determines the product of each write with the next read, up to permutations and sign flips (Fefferman, 1994), and these products determine the weights up to the changes of basis. For skipless GELU networks, the proof starts from a residual network and absorbs each skip with d_{model} more units per block, so it reaches only blocks of at least $2d_{\text{model}}$ units.

4.4 Residual and skipless networks compute different functions

A set of functions cannot lie inside a set of lower dimension, so for analytic blocks the gain in functional dimension excludes almost every residual network from the skipless networks of the same widths. It guarantees no inclusion in the other direction, although without normalization every skipless network is a limit of residual networks of the same widths (Mijoski and Karbevski, 2026).

Theorem 4.9 (Different function classes)**.** **(i)** *For analytic blocks, with or without pre-RMSNorm, a residual network at generic parameters equals no skipless network of the same widths.*
(ii) Too few extra units. *For networks of GELU blocks without normalization, a residual network of* L *blocks at generic parameters equals no skipless network of* L *blocks with, in total, fewer than* $Ld_{\text{model}}/2$ *more hidden units.*

A special case of Theorem A.19, proved in Appendix A.7.

In (ii), each extra GELU unit adds $2d_{\text{model}}$ parameters against a gain of Ld_{model}^2. Since d_{model} more units per block suffice (Mijoski and Karbevski, 2026), a skipless GELU network needs between $Ld_{\text{model}}/2$ and Ld_{model} more units in total.

4.5 Numerical ranks at depth, and a conjecture

At depth we compute the rank of the Jacobian on a large batch of inputs (Appendix E), which is a lower bound on the functional dimension. Where this rank meets the upper bound of Theorem 4.3, the functional dimension is known exactly. The two meet in every row of Table 2, and also for SwiGLU

Measured	d_{model}	L	Normalization	P	Rank		Added
					no skips	skips	
$\tanh$, GELU	4	3	none	240	176	224	48
$\tanh$	4	3	RMSNorm	240	216	234	18
$\tanh$	16	6	RMSNorm	3712	2872	3592	720
ReLU^2, exact rank	4	8	none	560	352	480	128
ReGLU, exact rank	4	8	none	816	544	672	128
nanoGPT: attention and GELU	8	8	LayerNorm	3336	2532	2700	168

Table 2: Rank of the Jacobian at random weights, without biases (Appendix E); exact rank: computed modulo a prime. In every row the skips add $L \dim \mathcal{G}$, where $\dim \mathcal{G}$ is d_{model}^2 without normalization, $d_{\text{model}}(d_{\text{model}}-1)/2$ under RMSNorm and $(d_{\text{model}}-1)(d_{\text{model}}-2)/2$ under LayerNorm.

and GeGLU networks and for nanoGPT (Karpathy, 2023) at other depths and without normalization (Tables 6 and 7). After training on four datasets, 398 of 436 small MLPs keep the rank that they have at random weights (Figure 4), and 36 of the other 38 are the skipless ReGLU networks under RMSNorm treated below.

In two cases the measured rank stays below the upper bound. In skipless ReGLU networks under RMSNorm with $\varepsilon = 10^{-5}$, some singular values fall below double precision. In one of eight skipless ReLU networks, five units active on the same inputs act as one linear map, as can happen on sets of positive measure (Grigsby et al., 2023).

Conjecture 1 (Equality without skips)**.** *With or without pre-RMSNorm, skipless networks of* $\tanh$*, GELU, SwiGLU or GeGLU blocks satisfy the equality condition. By Theorem 4.7, the skips then raise the functional dimension by exactly* $L \dim \mathcal{G}$*.*

For ReLU^2 and ReGLU blocks we expect the equality condition to hold only for residual networks: without skips, a unit can be inactive at every input on an open set of weights (Proposition A.25). For ReLU blocks the condition fails on an open set of weights with skips as well (Appendix A.8).

One-hot tokens. With one-hot tokens under RMSNorm, the rows of the embedding table can be moved one by one to undo any change of basis near the identity, so the change of basis at the embedding of a skipless network need not be orthogonal, which adds $d_{\text{model}}(d_{\text{model}}+1)/2$ dimensions of symmetry (Table 6). The nanoGPT models take one-hot tokens too, but add a position embedding to each token embedding, and show no such addition (Table 7).

5 Conclusion

Skip connections change how a network uses its parameters. Without skips, the basis of every hidden state is free, and d_{model}^2 parameters per hidden state are redundant. With skips, all hidden states share one basis. Where the symmetries are the only redundancy, the parameters that are redundant without skips become functional dimension. We prove this gain in the cases of Table 1, exactly or as a lower bound, and measure it in further networks. Mijoski and Karbevski (2026) establish that networks with and without skips differ in what they can represent, and this gain gives the difference a size. The changes of basis also apply to trained models: the redundant parameters of a skipless network and every invertible learned skip without normalization can be removed without changing the function.

Open questions. The equality condition at depth is open for the skipless networks of Conjecture 1 beyond those of Theorem 4.8, and for residual ReLU^2 and ReGLU networks. Whether the reduced forms, such as triangular learned skips, train better is untested. Input-dependent skips, as in mHC (Xie et al., 2025), are nonlinear in the hidden state, and we do not treat them.

Use of AI

The conception of the study is the authors' own. One author obtained their results, proofs and experiments working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs and to write the code of the experiments, which the authors validated.

Code availability

The notebook that reproduces the numbers, tables and figures of this paper (Appendix E) will be released in a public Git repository, coming soon.

References

Saleh Ashkboos, Maximilian L. Croci, Marcelo Gennari do Nascimento, Torsten Hoefler, and James Hensman. SliceGPT: Compress large language models by deleting rows and columns. In *ICLR*, 2024a.

Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Pashmina Cameron, Martin Jaggi, Dan Alistarh, Torsten Hoefler, and James Hensman. QuaRot: Outlier-free 4-bit inference in rotated LLMs. In *NeurIPS*, 2024b.

David Balduzzi, Marcus Frean, Lennox Leary, J. P. Lewis, Kurt Wan-Duo Ma, and Brian McWilliams. The shattered gradients problem: If resnets are the answer, then what is the question? In *ICML*, pages 342–350, 2017.

Jens Behrmann, Will Grathwohl, Ricky T. Q. Chen, David Duvenaud, and Jörn-Henrik Jacobsen. Invertible residual networks. In *ICML*, volume 97 of *PMLR*, 2019.

Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

Joachim Bona-Pellissier, François Malgouyres, and François Bachoc. Geometry-induced implicit regularization in deep ReLU neural networks. *arXiv:2402.08269*, 2024.

Isabel Dahlgren, Giovanni Luca Marchetti, and Kathlén Kohn. Polynomial residual networks are generically identifiable. *arXiv:2610.33738*, 2026.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=YicbFdNTTy.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. https://transformer-circuits.pub/2021/framework/index.html.

Charles Fefferman. Reconstructing a neural net from its output. *Revista Matemática Iberoamericana*, 10(3): 507–555, 1994.

Charles Godfrey, Davis Brown, Tegan Emerson, and Henry Kvinge. On the symmetries of deep learning models and their internal representations. In *NeurIPS*, 2022.

Wenbo Gong, Javier Zazo, Qijun Luo, Puqian Wang, James Hensman, and Chao Ma. ARO: A new lens on matrix optimization for large models. *arXiv:2602.09006*, 2026.

Nils Graef. Transformer tricks: Removing weights for skipless transformers. *arXiv:2404.12362v1*, 2024. Version 2 (2025) is titled "KV-weights are all you need for skipless transformers".

Aaron Grattafiori, Abhimanyu Dubey, et al. The Llama 3 herd of models. *arXiv:2407.21783*, 2024.

J. Elisenda Grigsby and Kathryn Lindsey. On functional dimension and persistent pseudodimension. *arXiv:2410.17191*, 2024.

J. Elisenda Grigsby, Kathryn Lindsey, and David Rolnick. Hidden symmetries of ReLU networks. In *ICML*, 2023.

J. Elisenda Grigsby, Kathryn Lindsey, Robert Meyerhoff, and Chenxi Wu. Functional dimension of feedforward ReLU neural networks. *Advances in Mathematics*, 482:110636, 2025. arXiv:2209.04036.

Koji Hashimoto, Yuji Hirono, and Akiyoshi Sannai. Unification of symmetries inside neural networks: Transformer, feedforward and neural ODE. *Machine Learning: Science and Technology*, 5:025079, 2024.

Bobby He and Thomas Hofmann. Simplifying transformer blocks. In *ICLR*, 2024.

Bobby He, James Martens, Guodong Zhang, Aleksandar Botev, Andrew Brock, Samuel L. Smith, and Yee Whye Teh. Deep transformers without shortcuts: Modifying self-attention for faithful signal propagation. In *ICLR*, 2023.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016.

Nathan W. Henry. Generic fibers and functional dimension of multi-head attention. In *ICML Workshop on Weight-Space Symmetries*, 2026.

Nathan W. Henry, Giovanni Luca Marchetti, and Kathlén Kohn. Geometry of lightning self-attention: Identifiability and dimension. In *ICLR*, 2025.

Yiping Ji, Hemanth Saratchandran, Peyman Moghadam, and Simon Lucey. Always skip attention. In *ICCV*, 2025.

Yiping Ji, James Martens, Jianqiao Zheng, Ziqin Zhou, Peyman Moghadam, Xinyu Zhang, Hemanth Saratchandran, and Simon Lucey. Cutting the skip: Training residual-free transformers. In *ICLR*, 2026.

Andrej Karpathy. nanoGPT, files `model.py` and `train.py`. https://github.com/karpathy/nanoGPT, 2023. Accessed 28 September 2026.

Joe Kileel, Matthew Trager, and Joan Bruna. On the expressive power of deep polynomial neural networks. In *NeurIPS*, 2019.

Hongzhou Lin and Stefanie Jegelka. ResNet with one-neuron hidden layers is a universal approximator. In *NeurIPS*, 2018.

Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. SpinQuant: LLM quantization with learned rotations. In *ICLR*, 2025.

Lachlan E. MacDonald, Jack Valmadre, Hemanth Saratchandran, and Simon Lucey. On skip connections and normalisation layers in deep optimisation. In *NeurIPS*, 2023.

Sibylle Marcotte, Rémi Gribonval, and Gabriel Peyré. Transformative or conservative? conservation laws for ResNets and transformers. In *ICML*, 2025.

Fanxu Meng, Hao Cheng, Jiaxin Zhuang, Ke Li, and Xing Sun. RMNet: Equivalently removing residual connection from networks. *arXiv:2111.00687*, 2021.

Gaurav Menghani, Ravi Kumar, and Sanjiv Kumar. LAuReL: Learned augmented residual layer. In *ICML*, 2025.

Antonij Mijoski and Marko Karbevski. Can an MLP absorb its skip connection exactly? Companion paper, 2026.

A. Emin Orhan and Xaq Pitkow. Skip connections eliminate singularities. In *ICLR*, 2018.

David Rolnick and Konrad P. Kording. Reverse-engineering deep ReLU networks. In *ICML*, 2020.

Noam Shazeer. GLU variants improve transformer. *arXiv:2002.05202*, 2020.

Héctor J. Sussmann. Uniqueness of the weights for minimal feedforward nets with a given input-output map. *Neural Networks*, 5(4):589–593, 1992.

Alexander Theus, Alessandro Cabodi, Sotiris Anagnostidis, Antonio Orvieto, Sidak Pal Singh, and Valentina Boeva. Generalized linear mode connectivity for transformers. In *NeurIPS*, 2025.

Viet-Hoang Tran, Vinh Khanh Bui, Van-Hoan Trinh, Tan Lai Ngoc, and Tan M. Nguyen. Functional equivalence in attention: A comprehensive study with applications to linear mode connectivity. In *ICML*, 2026.

Leo van Nierop. Transformer models are gauge invariant: A mathematical connection between AI and particle physics. *arXiv:2412.14543*, 2024.

Neha Verma and Maha Elbayad. Merging text transformer models from different initializations. *Transactions on Machine Learning Research*, 2024. arXiv:2403.00986.

Hong Wang and Kelly Wang. Complete characterization of gauge symmetries in transformer architectures. In *Symmetry and Geometry in Neural Representations (NeurReps), Proceedings Track*, volume 282 of *PMLR*, pages 622–663, 2025.

Zhenda Xie, Yixuan Wei, Huanqi Cao, Chenggang Zhao, Chengqi Deng, Jiashi Li, Damai Dai, Huazuo Gao, Jiang Chang, Kuai Yu, Liang Zhao, Shangyan Zhou, Zhean Xu, Zhengyan Zhang, Wangding Zeng, Shengding Hu, Yuqing Wang, Jingyang Yuan, Lean Wang, and Wenfeng Liang. mHC: Manifold-constrained hyper-connections. *arXiv:2512.24880*, 2025.

Bo Zhao, Iordan Ganev, Robin Walters, Rose Yu, and Nima Dehmamy. Symmetries, flat minima, and the conserved quantities of gradient flow. In *ICLR*, 2023.

Jiachen Zhu, Xinlei Chen, Kaiming He, Yann LeCun, and Zhuang Liu. Transformers without normalization. In *CVPR*, 2025.

A Proofs

Throughout, the tokens of a hidden state are its rows; where every block acts on each token separately, as MLP blocks do, it suffices to take one token, $T = 1$, so that $h \in \mathbb{R}^{1\times d_{\text{model}}}$ is a row vector. Biases, when present, enter as $B_\ell(h) = f_\ell(hW^\ell_{\text{read}} + b^\ell_{\text{read}})W^\ell_{\text{write}} + b^\ell_{\text{write}}$, with row vectors b added to every token. Each subsection recalls the results of the main text named in its title, explains what the proof does, and then proves them; auxiliary results are stated where they are first used. Here a *mixed block* is a block with d_{model} identity units with trainable read U_ℓ and write V_ℓ; in (4), $U_\ell = A_\ell$ and $V_\ell = I$. Further, $\mathcal{G}$ is $\mathrm{GL}(d_{\text{model}})$ without normalization, $\mathrm{O}(d_{\text{model}})$ under RMSNorm with $\varepsilon > 0$, and $\mathrm{CO}(d_{\text{model}})$, the positive multiples of orthogonal matrices, under RMSNorm with $\varepsilon = 0$. The one-block theorem, Theorem B.1, is proved in Appendix B.

A.1 Changes of basis at junctions

The main text uses changes of basis at hidden states (Sections 3.1 and 3.2); the proofs also use them at the identity units of a mixed block. A *junction* is a place where one linear map feeds the next: a hidden state, or the identity units of a mixed block. The following proposition states both kinds of change of basis, for skipless networks and networks of mixed blocks.

Proposition A.1 (Junctions)**. (i) Only UV matters.** *In a mixed block, $(U, V) \mapsto (UK, K^{-1}V)$ preserves the function for every $K \in \mathrm{GL}(d_{\text{model}})$.*
(ii) Hidden states are junctions. *In a skipless network, and in a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write a hidden state and $W \mapsto G^{-1}W$ on those that read it preserves the function for every $G \in \mathcal{G}$. For $\varepsilon = 0$ and $G = cQ$, a read through RMSNorm maps to $Q^\top W$ instead.*
(iii) One group per junction. *The maps of (i) and (ii) commute when they act at different junctions. A skipless network therefore carries one copy of $\mathcal{G}$ per hidden state, and a network of mixed blocks one more copy of $\mathrm{GL}(d_{\text{model}})$ per block.*

Overview. The only real work is under normalization, where a read sees $N(hG)$ rather than hG: Lemma A.2 determines which changes of basis can be moved through N.

Lemma A.2 (RMSNorm equivariance)**.** *Let $G \in \mathrm{GL}(d_{\mathrm{model}})$.*
(i) *There is a matrix M with $N(hG) = N(h)M$ for all row vectors $h \neq 0$ if and only if $G \in \mathrm{CO}(d_{\mathrm{model}})$ when $\varepsilon = 0$, and $G \in \mathrm{O}(d_{\mathrm{model}})$ when $\varepsilon > 0$. Then $M = Q$, where $G = cQ$ with Q orthogonal.*
(ii) *If $\varepsilon > 0$ and $G = cQ$, then $\|N(hG) - N(h)Q\|/\|N(h)\| = \frac{\varepsilon}{2r^2}|1 - c^{-2}| + O(\varepsilon^2)$, where $r^2 = \|h\|^2/d_{\mathrm{model}}$.*

Proof. The case $\varepsilon = 0$. Here $N(h) = \sqrt{d}_{\mathrm{model}}\, h/\|h\|$. If $G = cQ$, the factor c cancels in $hG/\|hG\|$, so $N(hG) = N(h)Q$. Conversely, if $hG/\|hG\| = hM/\|h\|$ for all $h \neq 0$, taking norms gives $\|hM\| = \|h\|$, so M is orthogonal, and $hGM^\top = (\|hG\|/\|h\|)\, h$: every nonzero row vector is a left eigenvector of $GM^\top$, so $GM^\top = cI$ with $c = \|hG\|/\|h\| > 0$, and $G = cM$.

The case $\varepsilon > 0$. For $G = cQ$, $N(h\,cQ) = c\,hQ/\sqrt{c^2\|h\|^2/d_{\mathrm{model}} + \varepsilon}$ and $N(h)Q = hQ/\sqrt{\|h\|^2/d_{\mathrm{model}} + \varepsilon}$; these agree for all h if and only if $c = 1$. If $N(hG) = N(h)M$ for all h with $\varepsilon > 0$, letting $\|h\| \to \infty$ along rays reduces to the case $\varepsilon = 0$, so $G = cM$ with M orthogonal, and the computation just made forces $c = 1$. The expansion of the deviation is the Taylor expansion of $c\sqrt{r^2 + \varepsilon}/\sqrt{c^2 r^2 + \varepsilon} - 1$ in ε. □

For positively homogeneous blocks and $\varepsilon = 0$, more changes of basis preserve the function: every invertible one before the last hidden state. The next lemma shows this and measures how $\varepsilon > 0$ breaks it; Theorem A.13, Theorem B.1(iv) and the checks of Appendix E use it.

Lemma A.3 (Homogeneous blocks under RMSNorm)**.** *Let a skipless network with pre-RMSNorm, $h_\ell = B_\ell(N(h_{\ell-1}))$ for $\ell = 1, \dots, L$ and $y = N(h_L)D$, have blocks that are positively homogeneous of degree k, and let $r_i^2 = \|h_i\|^2/d_{\mathrm{model}}$. Apply the change of basis $G \in \mathrm{GL}(d_{\mathrm{model}})$ at h_ℓ with $\ell < L$, or the scaling $G = cI$ at h_L with its read D unchanged. At every input whose hidden states are nonzero, the output becomes Λy with a scalar $\Lambda > 0$, and*
(i) *$\Lambda = 1$ if $\varepsilon = 0$ or G is orthogonal;*
(ii) *for $G = e^{tS}$ with S symmetric and $\ell < L$,*

$$\frac{d}{dt}\log\Lambda\Big|_{t=0} = -\,k^{L-\ell}\,\frac{h_\ell S h_\ell^\top}{d_{\mathrm{model}}\,(r_\ell^2 + \varepsilon)}\prod_{i=\ell+1}^{L}\frac{\varepsilon}{r_i^2 + \varepsilon},$$

and for the scaling $c = e^t$ at h_L it is $\varepsilon/(r_L^2 + \varepsilon)$.

Proof. Since $N(h)$ is a positive multiple of h,

$$N(h_\ell G)\, G^{-1} = \gamma N(h_\ell), \qquad \gamma = \sqrt{\frac{r_\ell^2 + \varepsilon}{\|h_\ell G\|^2/d_{\mathrm{model}} + \varepsilon}}.$$

Block $\ell + 1$ thus reads $\gamma N(h_\ell)$ and, being homogeneous, writes $\mu_{\ell+1} h_{\ell+1}$ with $\mu_{\ell+1} = \gamma^k$; for the scaling at h_L, set $\mu_L = c$. If a hidden state becomes $\mu_i h_i$ with $\mu_i > 0$, then $N(\mu_i h_i) = \beta_i N(h_i)$ with

$$\beta_i = \mu_i\sqrt{\frac{r_i^2 + \varepsilon}{\mu_i^2 r_i^2 + \varepsilon}},$$

so the next block writes $\mu_{i+1} h_{i+1}$ with $\mu_{i+1} = \beta_i^k$, and the readout gives $\Lambda = \beta_L$.

(i) If $\varepsilon = 0$, every β_i is 1. If G is orthogonal, $\gamma = 1$, so every μ_i and β_i is 1.

(ii) At $\mu_i = 1$, $\partial \log \beta_i / \partial \log \mu_i = \varepsilon/(r_i^2 + \varepsilon)$, and $\frac{d}{dt}\log\gamma\big|_{t=0} = -h_\ell S h_\ell^\top/(d_{\mathrm{model}}(r_\ell^2 + \varepsilon))$, since $\|h_\ell e^{tS}\|^2 = \|h_\ell\|^2 + 2t\, h_\ell S h_\ell^\top + O(t^2)$. The chain rule along $\log\mu_{\ell+1} = k\log\gamma$, $\log\mu_{i+1} = k\log\beta_i$ and $\log\Lambda = \log\beta_L$ gives the product. □

Proof of Proposition A.1. (i) $(hUK)K^{-1}V = hUV$, and no other part of the network involves U or V. (ii) Fix a hidden state h_ℓ and apply the change of basis G there (Section 3.1). The writes into h_ℓ produce $h_\ell G$ in place of h_ℓ. A read that sees h_ℓ itself, such as the read of the identity units, computes $(h_\ell G)G^{-1}W = h_\ell W$. With pre-RMSNorm reads, a read sees $N(h_\ell G)$, which by Lemma A.2 equals $N(h_\ell)Q$ exactly when $G = cQ$ with Q orthogonal and $c = 1$ if $\varepsilon > 0$, and is then absorbed by $W \mapsto Q^\top W$. So nothing downstream changes, for every G in $\mathrm{GL}(d_{\mathrm{model}})$ without normalization, in $\mathrm{CO}(d_{\mathrm{model}})$ for $\varepsilon = 0$ and in $\mathrm{O}(d_{\mathrm{model}})$ for $\varepsilon > 0$. With biases and without normalization, the affine change $h_\ell \mapsto h_\ell G + g$ is absorbed by $b_{\mathrm{write}} \mapsto b_{\mathrm{write}}G + g$ on the writes and by $(W, b) \mapsto (G^{-1}W,\ b - gG^{-1}W)$ on the reads, since $h_\ell W + b = (h_\ell G + g)G^{-1}W + b - gG^{-1}W$; this gives $\mathrm{Aff}(d_{\mathrm{model}})$. (iii) The change of basis at the identity units of block ℓ multiplies U_ℓ on the right and V_ℓ on the left. The change at $h_{\ell-1}$ multiplies the reads of block ℓ, U_ℓ included, on the left, and the change at h_ℓ multiplies the writes of block ℓ, V_ℓ included, on the right. Multiplications on the left and on the right commute, and every matrix is multiplied on each side by the change at one junction only, so all these changes commute, and a matrix of the skipless network is touched by one hidden state only. The symmetries inside blocks (Section 4.2) multiply W_{read} on the right and W_{write} on the left, and commute with all of the above. □

Under RMSNorm with $\varepsilon > 0$, the orthogonal changes of basis are the only ones, also for skipless networks.

Lemma A.4 (Only orthogonal changes of basis pass)**.** *Consider a skipless pre-RMSNorm network with $\varepsilon > 0$ whose blocks are those of Section 2, and let $G_0, \ldots, G_L$ be invertible. Multiply the writes into every hidden state h_ℓ by G_ℓ on the right and its reads by G_ℓ^{-1} on the left. If this preserves the function at all parameters, every G_ℓ is orthogonal.*

Proof. For a row u and an invertible G, let $N^G(u) = N(uG)\,G^{-1} = u/\sqrt{\|uG\|^2/d_{\mathrm{model}} + \varepsilon}$. The changed network is the original one with N^{G_ℓ} in place of N at the hidden state h_ℓ, for every ℓ. Take inputs whose tokens are all equal. Then the tokens of every hidden state are equal, an MLP block acts on each of them, and an attention block maps equal rows a to aS, with S the sum of its value–output products. Let u_ℓ and $\tilde{u}_\ell$ be a row of h_ℓ in the original network and, before the factor G_ℓ, in the changed one. We use three facts.

(a) Scaling. If $N^G(c\tilde{u}) = N(cu)$ for all $c > 0$, then $\tilde{u} = u$ and $\|uG\| = \|u\|$. Indeed, both sides are positive multiples of $\tilde{u}$ and of u, so $\tilde{u} = \lambda u$ with $\lambda > 0$, or both vanish. Then $\lambda^2(c^2\|u\|^2/d_{\mathrm{model}} + \varepsilon) = c^2\lambda^2\|uG\|^2/d_{\mathrm{model}} + \varepsilon$ for all $c > 0$. The constant terms give $\lambda = 1$, as $\varepsilon > 0$, and the terms in c^2 give $\|uG\| = \|u\|$.

(b) Blocks separate their inputs. If rows a and a' satisfy $B(a) = B(a')$ for all parameters of a block B, then $a = a'$. For an attention block on equal tokens, $aS = a'S$ for every matrix S. For a vanilla block, $\sigma(aw) = \sigma(a'w)$ for every column w. If $a \neq a'$, choose w with $\alpha = aw \neq a'w = \beta$, and replace w by tw: $\sigma(t\alpha) = \sigma(t\beta)$ for all t. If $\beta = -\alpha$, σ is even. Otherwise $\sigma(z) = \sigma(qz)$ for all z with some $|q| < 1$, so $\sigma(z) = \sigma(q^k z) \to \sigma(0)$, and σ is constant. None of $\tanh$, GELU, ReLU and ReLU^2 is even or constant. For a gated block, $\sigma(ag)\,(av) = \sigma(a'g)\,(a'v)$ for all columns g and v, so $\sigma(ag)\,a = \sigma(a'g)\,a'$ for all g. If $a = 0$ then $a' = 0$. Otherwise $a' = \kappa a$ for a number $\kappa \neq 0$, and $\sigma(z) = \kappa\,\sigma(\kappa z)$ for all z. For the ReLU gate this gives $\kappa = 1$. The SiLU and GELU gates have $\sigma'(0) = 1/2$, so $\kappa^2 = 1$, and they are not odd, so $\kappa = 1$.

(c) Hidden states reach every row. As the input and the parameters vary, u_ℓ takes every row of $\mathbb{R}^{1\times d_{\mathrm{model}}}$. For $u_0 = xE$ this is clear. If $u_{\ell-1} \neq 0$, some read of block ℓ gives hidden units that do not all vanish, and its write then sends them to any row.

Induction from the last hidden state. The outputs agree for all D, so $N^{G_L}(\tilde{u}_L) = N(u_L)$. Multiplying the write of block L by $c > 0$ multiplies $\tilde{u}_L$ and u_L by c, so (a) gives $\tilde{u}_L = u_L$ and $\|u_L G_L\| = \|u_L\|$. By (c), u_L is any row, so G_L is orthogonal. Now suppose that $\tilde{u}_\ell = u_\ell$ at all parameters, for some $\ell \geq 1$. As $u_\ell = B_\ell\big(N(u_{\ell-1})\big)$ and $\tilde{u}_\ell = B_\ell\big(N^{G_{\ell-1}}(\tilde{u}_{\ell-1})\big)$, where the inputs of B_ℓ do not depend on its parameters, (b) gives $N^{G_{\ell-1}}(\tilde{u}_{\ell-1}) = N(u_{\ell-1})$. Multiplying the write of block $\ell - 1$, or the embedding if $\ell = 1$, by $c > 0$, (a) gives $\tilde{u}_{\ell-1} = u_{\ell-1}$ and $\|u_{\ell-1}G_{\ell-1}\| = \|u_{\ell-1}\|$. By (c), $u_{\ell-1}$ is any row, so $G_{\ell-1}$ is orthogonal. □

A.2 The gauge fix (Theorems 3.2 and 3.3)

A symmetry of the parameters that preserves the function is also called a *gauge*, and using it to fix parameters, as the sweep of Section 3.2 does, a *gauge fix*. Theorems 3.2 and 3.3 collect what the main text uses of the following theorem. We write the parameters of a network of mixed blocks as $\theta = \big(E,\ (U_\ell, V_\ell, W^\ell_{\text{read}}, W^\ell_{\text{write}})_{\ell=1}^L,\ D\big)$, and those of a network with learned skips as $\big(E,\ (A_\ell, W^\ell_{\text{read}}, W^\ell_{\text{write}})_{\ell=1}^L,\ D\big)$. For invertible $G_0, \dots, G_L$ and $K_1, \dots, K_L$, let $T_{G,K}$ be the map

$$\begin{aligned} E \mapsto EG_0, \qquad W^\ell_{\text{read}} \mapsto G_{\ell-1}^{-1} W^\ell_{\text{read}}, \qquad W^\ell_{\text{write}} \mapsto W^\ell_{\text{write}} G_\ell, \qquad D \mapsto G_L^{-1} D, \\ U_\ell \mapsto G_{\ell-1}^{-1} U_\ell K_\ell, \qquad V_\ell \mapsto K_\ell^{-1} V_\ell G_\ell, \end{aligned} \tag{7}$$

which applies the change of basis G_ℓ at h_ℓ and K_ℓ at the identity units of block ℓ. On a network with learned skips, T_G acts by $A_\ell \mapsto G_{\ell-1}^{-1} A_\ell G_\ell$ in place of the second line. When every skip $A_\ell = U_\ell V_\ell$ is invertible, let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$, and let

$$\Phi(\theta) = \big(E\Lambda_0,\ (\Lambda_{\ell-1}^{-1} W^\ell_{\text{read}},\ W^\ell_{\text{write}} \Lambda_\ell)_{\ell=1}^L,\ D\big), \tag{8}$$

a parameter of the residual network, with the same formula for learned skips. With biases, Φ also sends the embedding bias b_E to $b_E\Lambda_0$ and each write bias b^ℓ_{write} to $b^\ell_{\text{write}}\Lambda_\ell$, and keeps each read bias. We write F_θ for the function of the network of mixed blocks at θ, and $F^{\text{res}}_{\Phi(\theta)}$ for that of the residual network at $\Phi(\theta)$.

Theorem A.5 (The gauge fix)**.** *Consider networks without normalization in (i) and (ii), and parameters θ whose skips A_ℓ are invertible.*
(i) Same function. *The maps $T_{G,K}$ preserve the function. With $G_\ell = \Lambda_\ell$ and $K_\ell = U_\ell^{-1}\Lambda_{\ell-1}$, $T_{G,K}$ sets every U_ℓ and V_ℓ to I and the other parameters to $\Phi(\theta)$, so $F_\theta = F^{\text{res}}_{\Phi(\theta)}$. For learned skips, T_G with $G_\ell = \Lambda_\ell$ sets every A_ℓ to I in the same way.*
(ii) What the gauge fix identifies. *Every parameter of the residual network is $\Phi(\theta)$ for some θ, and $\Phi(\tilde\theta) = \Phi(\theta)$ exactly when $\tilde\theta = T_{G,K}(\theta)$ for some G and K with $G_L = I$. The derivative of Φ is onto at every θ. So the networks of mixed blocks with invertible skips and the residual networks compute the same set of functions, with the same functional dimension at θ and at $\Phi(\theta)$. The same holds for learned skips, without the K_ℓ, and for networks with biases.*
(iii) One change of basis is left. *Let $d_{\text{model}} \geq 2$, and let every block be nonzero for some value of its parameters. Without normalization or with pre-RMSNorm and $\varepsilon > 0$, let $G_0, \dots, G_L$ be invertible. The map T_G of the embedding, the reads, the writes and the readout preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L \in \mathcal{G}$. Under RMSNorm with $\varepsilon = 0$, the same holds for $G_\ell = c_\ell Q_\ell \in \mathrm{CO}(d_{\text{model}})$, with $c_\ell > 0$ and Q_ℓ orthogonal, when T_G multiplies the reads and the readout by $Q_{\ell-1}^\top$ and $Q_L^\top$ in place of $G_{\ell-1}^{-1}$ and G_L^{-1}.*

Overview. Part (i) is an induction along the network: in the coordinates $h'_\ell = h_\ell\Lambda_\ell$ the identity units become identity skips, and we check that the residual recursion with transformed weights produces exactly these coordinates. Part (ii) lifts every direction on the residual side to one with the identity units held fixed, which makes the derivative of Φ onto, and solves for the maps $T_{G,K}$ that relate two parameters with the same image. Part (iii) switches off all branches but one, so that the output compares the bases on the two sides of that block, and a branch that is not identically zero forces them to agree.

Proof of Theorem A.5. Part (i), with biases and a skip bias. Merge the identity units, $A_\ell = U_\ell V_\ell$, and write the network as $h_\ell = h_{\ell-1}A_\ell + k_\ell + f_\ell(h_{\ell-1}W^\ell_{\text{read}} + b^\ell_{\text{read}})W^\ell_{\text{write}} + b^\ell_{\text{write}}$ with $h_0 = xE + b_E$, where $k_\ell \in \mathbb{R}^{1\times d_{\text{model}}}$ is a skip bias ($k_\ell = 0$ is allowed). Every A_ℓ is invertible, hence every Λ_ℓ, and $\Lambda_{\ell-1} = A_\ell\Lambda_\ell$. Define new variables and new weights

$$\begin{gathered} h'_\ell = h_\ell\Lambda_\ell, \qquad E' = E\Lambda_0, \quad b'_E = b_E\Lambda_0, \qquad W'^\ell_{\text{read}} = \Lambda_{\ell-1}^{-1} W^\ell_{\text{read}}, \quad b'^\ell_{\text{read}} = b^\ell_{\text{read}}, \\ W'^\ell_{\text{write}} = W^\ell_{\text{write}}\Lambda_\ell, \quad b'^\ell_{\text{write}} = (b^\ell_{\text{write}} + k_\ell)\Lambda_\ell, \qquad D' = D. \end{gathered}$$

We show by induction that the residual network with primed weights computes h'_ℓ. For $\ell = 0$, $h'_0 = (xE + b_E)\Lambda_0 = xE' + b'_E$. Assume $h'_{\ell-1} = h_{\ell-1}\Lambda_{\ell-1}$. The residual recursion with primed weights gives

$$\begin{aligned} &h'_{\ell-1} + f_\ell(h'_{\ell-1}W'^{\ell}_{\text{read}} + b'^{\ell}_{\text{read}})W'^{\ell}_{\text{write}} + b'^{\ell}_{\text{write}} \\ &\qquad = h_{\ell-1}\Lambda_{\ell-1} + f_\ell(h_{\ell-1}W^{\ell}_{\text{read}} + b^{\ell}_{\text{read}})W^{\ell}_{\text{write}}\Lambda_\ell + (b^{\ell}_{\text{write}} + k_\ell)\Lambda_\ell, \end{aligned}$$

and since $\Lambda_{\ell-1} = A_\ell\Lambda_\ell$ the right-hand side equals $h_\ell\Lambda_\ell$. Finally $h'_L D' = h_L D$ because $\Lambda_L = I$. This is the map $T_{G,K}$ of (7) with $G_\ell = \Lambda_\ell$ and $K_\ell = U_\ell^{-1}\Lambda_{\ell-1}$: it sends U_ℓ to $G_{\ell-1}^{-1}U_\ell K_\ell = I$ and V_ℓ to $K_\ell^{-1}V_\ell G_\ell = \Lambda_{\ell-1}^{-1}A_\ell\Lambda_\ell = I$, and the other matrices as above.

Part (ii). The identity units carry no biases of their own, since these would only add to the write bias. For parameters θ of the mixed network with invertible A_ℓ, the map Φ of (8) is the change of variables of part (i) with $k_\ell = 0$, so $F_\theta = F^{\text{res}}_{\Phi(\theta)}$. Every parameter of the residual network is reached: the θ with the same entries and $U_\ell = V_\ell = I$ has every $\Lambda_\ell = I$, so Φ returns these entries. The Fréchet derivative $\mathrm{D}_\theta\Phi(\theta)$ is onto at every θ. A direction η' of the residual parameters, with the entries η'_E, $\eta'^{\ell}_{\text{read}}$, $\eta'^{\ell}_{\text{write}}$ and η'_D at E', W'^{ℓ}_{read}, W'^{ℓ}_{write} and D', is $\mathrm{D}_\theta\Phi(\theta)\,\eta$ for the direction η of θ that is zero at every U_ℓ and V_ℓ and has the entries $\eta'_E\Lambda_0^{-1}$ at E, $\Lambda_{\ell-1}\eta'^{\ell}_{\text{read}}$ at W^{ℓ}_{read}, $\eta'^{\ell}_{\text{write}}\Lambda_\ell^{-1}$ at W^{ℓ}_{write} and η'_D at D, since along it every Λ_ℓ stays fixed; the biases are treated in the same way. The maps $T_{G,K}$ with $G_L = I$ send A_ℓ to $G_{\ell-1}^{-1}A_\ell G_\ell$ and hence Λ_ℓ to $G_\ell^{-1}\Lambda_\ell$, so they leave Φ unchanged. Conversely, let $\Phi(\tilde\theta) = \Phi(\theta)$ for a second parameter $\tilde\theta$ of the mixed network, whose entries and products $\tilde\Lambda_\ell$ carry a tilde. Set $G_\ell = \Lambda_\ell\tilde\Lambda_\ell^{-1}$, so that $G_L = I$, and $K_\ell = U_\ell^{-1}G_{\ell-1}\tilde U_\ell$. Then $\tilde E = EG_0$, $\widetilde{W}^{\ell}_{\text{read}} = G_{\ell-1}^{-1}W^{\ell}_{\text{read}}$, $\widetilde{W}^{\ell}_{\text{write}} = W^{\ell}_{\text{write}}G_\ell$ and $\tilde D = D$; $\tilde U_\ell = G_{\ell-1}^{-1}U_\ell K_\ell$ by the choice of K_ℓ; and $\tilde A_\ell = \tilde\Lambda_{\ell-1}\tilde\Lambda_\ell^{-1} = G_{\ell-1}^{-1}A_\ell G_\ell$ gives $\tilde V_\ell = \tilde U_\ell^{-1}\tilde A_\ell = K_\ell^{-1}V_\ell G_\ell$. So $\tilde\theta = T_{G,K}(\theta)$. Since $F_\theta = F^{\text{res}}_{\Phi(\theta)}$ and every residual parameter is reached, the two networks compute the same functions. For every batch of inputs, the Jacobian of the mixed network at θ is the Jacobian of the residual network at $\Phi(\theta)$ times the derivative of Φ, by the chain rule; a matrix times a matrix that is onto has the rank of the first, so the two Jacobians have the same rank, and the two networks the same functional dimension. For learned skips the same argument applies without the K_ℓ.

Part (iii). A single global G is a symmetry: the skip carries $h_{\ell-1}G$ to $h_{\ell-1}G$, and the branch becomes $B_\ell(\,\cdot\,G^{-1})\,G$. Conversely, let $G_0, \ldots, G_L$ preserve the function for all parameter values; the change of basis conjugates each branch, $B_\ell \mapsto B_\ell(\,\cdot\,G_{\ell-1}^{-1})\,G_\ell$. First set every write W^k_{write}, and its bias if there is one, to zero. The network computes $x \mapsto xED$ and the transformed network $x \mapsto xEG_0G_L^{-1}D$, and these agree for all D and E only if $G_L = G_0$. Now suppose $G_0 = \cdots = G_{\ell-1} =: G$ and set every write except that of block ℓ to zero. Then $h_L = h_{\ell-1} + B_\ell(h_{\ell-1})$ with $h_{\ell-1} = xE$, block ℓ reads $h_{\ell-1}GG_{\ell-1}^{-1} = h_{\ell-1}$, and the transformed network outputs $\big(h_{\ell-1}G + B_\ell(h_{\ell-1})\,G_\ell\big)G^{-1}D$. Equality with $\big(h_{\ell-1} + B_\ell(h_{\ell-1})\big)D$ for all D gives $B_\ell(h)\,(G_\ell G^{-1} - I) = 0$ for all h and all parameters of block ℓ. As the write of block ℓ ranges over all matrices, the outputs of a block whose computation does not vanish identically span $\mathbb{R}^{d_{\text{model}}}$, so $G_\ell = G$, and by induction all G_ℓ are equal.

Part (iii) under RMSNorm with $\varepsilon > 0$. A global orthogonal G is a symmetry, since $N(hG) = N(h)G$ (Proposition A.1(ii)). Conversely, with all writes zero the network computes $x \mapsto N(xE)D$ and the transformed network $x \mapsto N(xEG_0)G_L^{-1}D$. These agree for all E and D only if $N(hG_0) = N(h)G_L$ for all rows h, so G_0 is orthogonal and $G_L = G_0$, by Lemma A.2. Now suppose $G_0 = \cdots = G_{\ell-1} = G_L =: G$, and set every write except that of block ℓ to zero. With $h = xE$ and $b = B_\ell(N(h))$, the network computes $N(h + b)D$. In the transformed network block ℓ reads $N(hG)G^{-1}W^{\ell}_{\text{read}} = N(h)W^{\ell}_{\text{read}}$ and writes bG_ℓ, so the output is $N(hG + bG_\ell)G^{-1}D = N(h + bG_\ell G^{-1})D$. For $\varepsilon > 0$ the map N is injective, so equality for all D gives $b\,(G_\ell G^{-1} - I) = 0$, and $G_\ell = G$ as above.

Part (iii) under RMSNorm with $\varepsilon = 0$. Now $N(chQ) = N(h)Q$ for $c > 0$ and orthogonal Q. With all writes zero the transformed network computes $N(xE)Q_0Q_L^\top D$, so $Q_L = Q_0$. Suppose $G_0 = \cdots = G_{\ell-1} = cQ$ and $Q_L = Q$, and set every write except that of block ℓ to zero. With h and b as above, the transformed network computes $N(chQ + c_\ell bQ_\ell)Q^\top D = N(ch + c_\ell bR)D$ with $R = Q_\ell Q^\top$. Equality with $N(h + b)D$ for all D makes $ch + c_\ell bR$ a positive multiple of $h + b$. Fix a token, and an input and reads at which the row of $f(N(h)W^{\ell}_{\text{read}})$ of this token is

nonzero. As the write varies, the row b of this token takes every value. So, with $u = h + b$, the row $c_\ell\, uR + (ch - c_\ell\, hR)$ is a positive multiple of u for every $u \neq 0$. Large u give $R = I$. Then $(c - c_\ell)\, h$ is parallel to every u, so $c_\ell = c$. So $G_\ell = cQ$, and by induction all G_ℓ are equal. □

Remark A.6 (The forward sweep). The sweep of part (i) can also run forward, with $G_\ell = \Pi_\ell^{-1}$ at h_ℓ, with $\Pi_\ell = A_1 \cdots A_\ell$ and $\Pi_0 = I$. By the same induction it gives the residual network with embedding E, reads $\Pi_{\ell-1} W^\ell_{\text{read}}$, writes $W^\ell_{\text{write}} \Pi_\ell^{-1}$, write biases $(b^\ell_{\text{write}} + k_\ell)\Pi_\ell^{-1}$ and readout $\Pi_L D$. Any linear change of variables $h'_\ell = h_\ell G'_\ell$ that yields identity skips must satisfy $G'^{-1}_{\ell-1} A_\ell G'_\ell = I$, so $G'_\ell = \Lambda_\ell G'_L$: two such changes, the forward and the backward sweep among them, differ by a global change of basis and, when biases are present, by a translation at each hidden state (Lemma A.8).

Remark A.7 (The changes of basis of Zhao et al.). For vanilla MLP blocks, the maps $T_{G,K}$ are the changes of basis of Proposition C.1 of Zhao et al. (2023), transposed to row vectors. Read the network of mixed blocks as a feedforward network whose hidden layers alternate between the hidden states h_ℓ, with the identity as activation, and the hidden layers of the blocks, with the mixed activation $\tilde{f}\big(\begin{bmatrix} u \mid z \end{bmatrix}\big) = \begin{bmatrix} u \mid f(z) \end{bmatrix}$ of (4). The identity commutes with every matrix in $\mathrm{GL}(d_{\text{model}})$ (their Example C.3), and $\tilde{f}$ with $\mathrm{diag}(K, I)$ for every $K \in \mathrm{GL}(d_{\text{model}})$, so their proposition gives these changes of basis G_ℓ and K_ℓ; for other blocks the same computation does. Propagating a change of basis through a skip is thus their change of basis at the identity units, chosen together with the changes of basis at the two hidden states that the skip joins.

We recall Theorem 3.2 from Section 3.2, where $\Lambda_\ell = A_{\ell+1} \cdots A_L$, and Theorem 3.3 from Section 3.3.

Theorem 3.2 (Learned skips reduce to identity skips)**.** *Without normalization, let every learned skip A_ℓ be invertible, and let $\Lambda_\ell = A_{\ell+1} \cdots A_L$, with $\Lambda_L = I$. The network with learned skips computes the same function as the residual network with the embedding $E\Lambda_0$, the reads $\Lambda^{-1}_{\ell-1} W^\ell_{\text{read}}$, the writes $W^\ell_{\text{write}} \Lambda_\ell$ and the readout D.*

Proof of Theorem 3.2. This is the statement for learned skips in Theorem A.5(i), with $\Phi(\theta)$ written out by (8). □

Theorem 3.3 (One change of basis for the whole network)**.** *Without normalization, let $G_0, \ldots, G_L$ be invertible.*
(i) *The map $E \mapsto EG_0$, $W^\ell_{\text{read}} \mapsto G^{-1}_{\ell-1} W^\ell_{\text{read}}$, $W^\ell_{\text{write}} \mapsto W^\ell_{\text{write}} G_\ell$, $D \mapsto G_L^{-1} D$ preserves the function of every residual network if and only if $G_0 = G_1 = \cdots = G_L$.*
(ii) *The changes of basis that preserve the function at all parameters form a group of dimension $(L+1) d^2_{\text{model}}$ in a skipless network and d^2_{model} in a residual one.*
(iii) *In a residual network, let d_{model} columns of any one read form an invertible matrix W_1. The same change of basis W_1 at all hidden states preserves the function and turns W_1 into the identity.*

Proof of Theorem 3.3. Part (i) is Theorem A.5(iii) with $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$, part (ii) is Theorem A.13(i) with $\dim \mathcal{G} = d^2_{\text{model}}$, and part (iii) is Proposition A.9(ii). □

With biases every hidden state also carries a translation, which the skips do not tie.

Lemma A.8 (Translations are not tied)**.** *In a residual network with biases and without normalization, adding a row g to the hidden states $h_\ell, \ldots, h_L$ preserves the function once the biases are adjusted, for every $\ell = 0, \ldots, L$. The symmetry group generated by changes of basis, translations and the symmetries inside blocks therefore has dimension $d^2_{\text{model}} + (L+1) d_{\text{model}} + s$, against $(L+1)(d^2_{\text{model}} + d_{\text{model}}) + s$ without skips, and the skips remove $L d^2_{\text{model}}$ dimensions, as without biases.*

Proof. Add g to the write bias of block ℓ, or to the embedding bias when $\ell = 0$; subtract $g W^k_{\text{read}}$ from the read bias of every block $k > \ell$; subtract gD from the readout bias. The skips carry g forward unchanged, so every later hidden state is shifted by g, and every later read removes it. So the function is unchanged. These $L + 1$ translations move the biases in independent directions, as many as in a skipless network, where each hidden state has its own translation (Proposition A.1(ii)). A skip ties only the linear part of $\mathrm{Aff}(d_{\text{model}})$, which gives the counts. □

Table 6 confirms these counts for $\tanh$ networks with biases.

A.3 Exact reduction without skips (Proposition 3.1)

The following proposition is the general form of Proposition 3.1. Let W_1 be an invertible $d_{\text{model}} \times d_{\text{model}}$ submatrix formed by d_{model} columns of a matrix that reads h_ℓ, or by d_{model} rows of a matrix that writes it.

Proposition A.9 (Exact reduction without skips)**.** **(i)** *In a skipless network or a network of mixed blocks, the map $W \mapsto WG$ on the matrices that write h_ℓ and $W \mapsto G^{-1}W$ on those that read it turns W_1 into the identity for $G = W_1$ if W_1 is part of a read, and for $G = W_1^{-1}$ if it is part of a write. Under RMSNorm with $\varepsilon > 0$, $G = Q$ from a QR factorization $W_1 = QT$, or $W_1^\top = QT$ for a write, turns W_1 into a triangular matrix. Done at every hidden state, this removes d_{model}^2, respectively $d_{\text{model}}(d_{\text{model}} - 1)/2$, parameters per hidden state without changing the function.*
(ii) *In a residual network, the same map with one G at all hidden states preserves the function and turns one such submatrix, in any read or write of the network, into the identity or a triangular matrix.*

Proof of Proposition A.9. By Proposition A.1, a change of basis $G \in \mathcal{G}$ at h_ℓ preserves the function and replaces every read W of h_ℓ by $G^{-1}W$ and every write W into h_ℓ by WG. So the columns of a read that form W_1 become $G^{-1}W_1$, and the rows of a write that form W_1 become W_1G. Without normalization $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$, and $G = W_1$ for a read, or $G = W_1^{-1}$ for a write, turns W_1 into I. Under RMSNorm with $\varepsilon > 0$, $\mathcal{G} = \mathrm{O}(d_{\text{model}})$ (Lemma A.2). For a read, let $W_1 = QT$ be a QR factorization, with T upper triangular; $G = Q$ gives $Q^\top W_1 = T$. For a write, let $W_1^\top = QT$ be a QR factorization; then $W_1 = T^\top Q^\top$, and $G = Q$ gives $W_1 Q = T^\top$, which is lower triangular. The identity fixes all d_{model}^2 entries of W_1, and the triangular form fixes the $d_{\text{model}}(d_{\text{model}} - 1)/2$ entries on one side of its diagonal. The change of basis at h_ℓ touches only the writes into h_ℓ and the reads of h_ℓ, and a matrix is touched on each side by one hidden state only (Proposition A.1(iii)), so the reductions at all hidden states can be made together, one submatrix per hidden state. In a residual network, one change of basis $G \in \mathcal{G}$ applied at every hidden state preserves the function (Theorem A.5(iii)); it replaces every read W by $G^{-1}W$ and every write W by WG, and leaves the identity skips unchanged, since $G^{-1}IG = I$. The same choice of G then turns one submatrix W_1, of any one read or write, into the identity or into a triangular matrix. □

We recall Proposition 3.1 from Section 3.1.

Proposition 3.1 (Exact reduction without skips)**.** *Without normalization, let d_{model} columns of the read of h_ℓ in a skipless network form an invertible matrix W_1. The change of basis $G = W_1$ at h_ℓ preserves the function and turns W_1 into the identity. At generic parameters every hidden state has such columns, and these changes of basis at all $L+1$ hidden states remove $(L+1)d_{\text{model}}^2$ parameters.*

Proof of Proposition 3.1. This is Proposition A.9(i) without normalization, for a submatrix of a read. □

A.4 RMSNorm (Theorem 3.4)

Theorem 3.4(iv) is a case of the following theorem. Let $\Pi_\ell = A_1 \cdots A_\ell$, with $\Pi_0 = I$.

Theorem A.10 (RMSNorm with learned skips)**.** *Consider pre-RMSNorm networks with $\varepsilon \geq 0$, and the network with learned skips of (6), whose skips A_ℓ are invertible in (i) and (ii).*
(i) Triangular or symmetric skips. *Let $Q_0 = I$, and for $\ell = 1, \ldots, L$ let $Q_{\ell-1}^\top A_\ell = R_\ell Q_\ell^\top$ be an RQ factorization, with R_ℓ upper triangular with positive diagonal and Q_ℓ orthogonal. The network with the skips R_ℓ, the reads $Q_{\ell-1}^\top W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function. So does the network built in the same way from the polar decompositions $Q_{\ell-1}^\top A_\ell = S_\ell Q_\ell^\top$, with S_ℓ symmetric positive definite in place of R_ℓ.*
(ii) Orthogonal skips can be absorbed. *If every Π_ℓ lies in $\mathcal{G}$, write $\Pi_\ell = c_\ell P_\ell$ with $c_\ell > 0$ and P_ℓ orthogonal, where $c_\ell = 1$ if $\varepsilon > 0$. Then the network computes the same function as the residual network with the embedding E, the reads $P_{\ell-1} W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Pi_\ell^{-1}$ and the readout $P_L D$. Conversely, let the reads and the readout have rank d_{model}, and let invertible $G_0, \ldots, G_L$ turn every skip into the identity, $G_{\ell-1}^{-1} A_\ell G_\ell = I$. If the network, written in the variables $h_\ell G_\ell$, is a*

pre-RMSNorm residual network, then every Π_ℓ lies in $\mathcal{G}$.
(iii) What each skip adds. *Under the equality condition for the networks with learned skips, with identity skips and without skips, the functional dimension of the network with learned skips exceeds that of the residual network by $L\,(d_{\text{model}}^2 - \dim \mathcal{G})$, and that of the residual network exceeds that of the skipless network by $L \dim \mathcal{G}$. These are $L\,d_{\text{model}}(d_{\text{model}}+1)/2$ and $L\,d_{\text{model}}(d_{\text{model}}-1)/2$ for $\varepsilon > 0$, and $L\,(d_{\text{model}}(d_{\text{model}}+1)/2 - 1)$ and $L\,(d_{\text{model}}(d_{\text{model}}-1)/2 + 1)$ for $\varepsilon = 0$. For $\varepsilon = 0$ and positively homogeneous blocks, the skipless network does not satisfy the equality condition (Lemma A.3).*

Overview. Part (i) runs the sweep in the proof of Theorem A.5(i) forward, with orthogonal matrices only, which remove the orthogonal factor of each skip and leave its triangular or symmetric factor. Part (ii) reruns the change of variables of that proof and locates the only step that can fail, the read through N; Lemma A.2 says exactly when a matrix can be moved through N. Part (iii) counts the symmetries with Proposition A.1 and Theorem A.13.

Proof of Theorem A.10. Part (i). For every $\varepsilon \geq 0$, multiplying the writes into h_ℓ, the skip A_ℓ included, on the right by an orthogonal Q_ℓ, and the reads of h_ℓ on the left by $Q_\ell^\top$, preserves the function (Proposition A.1(ii)): the raw read $A_{\ell+1}$ directly, and the reads through N because $N(hQ_\ell) = N(h)Q_\ell$ (Lemma A.2). With $Q_0 = I$ and the Q_ℓ of the statement, the skip A_ℓ becomes $Q_{\ell-1}^\top A_\ell Q_\ell = R_\ell$; the RQ factorization of the invertible matrix $Q_{\ell-1}^\top A_\ell$ exists, and is unique. The same computation with the polar decompositions gives $Q_{\ell-1}^\top A_\ell Q_\ell = S_\ell$: every invertible matrix X is $X = SQ^\top$ with $S = (XX^\top)^{1/2}$ symmetric positive definite and $Q^\top = S^{-1}X$ orthogonal, uniquely. The readout absorbs the last factor, $D \mapsto Q_L^\top D$.

Part (ii). Let the network be $h_\ell = h_{\ell-1}A_\ell + f_\ell(N(h_{\ell-1})W_{\text{read}}^\ell)W_{\text{write}}^\ell$ and set $h'_\ell = h_\ell \Pi_\ell^{-1}$, the forward sweep of Remark A.6, which turns every skip into the identity. The read of block ℓ in the new variables is $N(h'_{\ell-1}\Pi_{\ell-1})W_{\text{read}}^\ell$. If every Π_ℓ, $\ell = 0, \ldots, L$, lies in $\mathcal{G}$, Lemma A.2 gives $N(h'\Pi_{\ell-1}) = N(h')P_{\ell-1}$, so the reads $P_{\ell-1}W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell \Pi_\ell^{-1}$ and the readout $P_L D$, since the readout reads $N(h_L)$, reproduce the induction of that proof, and the result is a pre-RMSNorm residual network. Conversely, $G_{\ell-1}^{-1}A_\ell G_\ell = I$ for every ℓ gives $G_\ell = \Pi_\ell^{-1}G_0$, so in the variables $h'_\ell = h_\ell G_\ell$ block ℓ reads $N(h'_{\ell-1}G_0^{-1}\Pi_{\ell-1})W_{\text{read}}^\ell$, and the readout reads $N(h'_L G_0^{-1}\Pi_L)$. In a pre-RMSNorm residual network every read has the form $N(h')W$; as the reads and the readout have rank d_{model}, this gives matrices M_ℓ with $N(h'G_0^{-1}\Pi_\ell) = N(h')M_\ell$ for all h', and Lemma A.2 puts every $G_0^{-1}\Pi_\ell$ in $\mathcal{G}$. For $\ell = 0$ this puts G_0 in $\mathcal{G}$, and so every Π_ℓ lies in $\mathcal{G}$.

Part (iii). Apply Proposition A.1 to the pre-RMSNorm network with learned skips: the skip A_ℓ absorbs any change of basis, and the normalized reads restrict it to $\mathcal{G}$, which is $\mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$ and $\mathrm{O}(d_{\text{model}})$ when $\varepsilon > 0$ (Lemma A.2), so each of the $L+1$ hidden states carries its own copy of $\mathcal{G}$. The network with learned skips has Ld_{model}^2 more parameters than the residual network and L more copies of $\mathcal{G}$ in its symmetry group. If both satisfy the equality condition, the difference of their functional dimensions is therefore $L\,(d_{\text{model}}^2 - \dim \mathcal{G})$, which is $L\,d_{\text{model}}(d_{\text{model}}+1)/2$ when $\varepsilon > 0$ and $L\,(d_{\text{model}}(d_{\text{model}}+1)/2 - 1)$ when $\varepsilon = 0$. The difference between the residual and the skipless network is $L \dim \mathcal{G}$ by Theorem A.13(iii), which is $L\,d_{\text{model}}(d_{\text{model}}-1)/2$ when $\varepsilon > 0$ and $L\,(d_{\text{model}}(d_{\text{model}}-1)/2 + 1)$ when $\varepsilon = 0$. □

Remark A.11 (Other starting points and tied readouts). The recursion of part (i) can also start at any h_k with any orthogonal Q_k, running forward as above for $\ell > k$ and backward for $\ell \leq k$ with the QR factorization $A_\ell Q_\ell = Q_{\ell-1}R_\ell$ or the polar decomposition $A_\ell Q_\ell = Q_{\ell-1}S_\ell$; the embedding absorbs Q_0. With Q_k the rotation of Proposition A.9(i) for one submatrix of a read or a write of h_k, that submatrix becomes triangular too, and $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters are removed in all. If Q_0 and Q_L are fixed, or tied to each other as by a tied readout $D = E^\top$, which forces $Q_L = Q_0$, the recursions from h_0 forward and from h_L backward meet at any one skip, which keeps its full matrix; without normalization the same holds in $\mathrm{GL}(d_{\text{model}})$, with identities in place of the R_ℓ and with $G_L = G_0^{-\top}$ for a tied readout.

We recall Theorem 3.4 from Section 3.4.

Theorem 3.4 (Orthogonal changes of basis under RMSNorm)**.** *Consider pre-RMSNorm networks.*
(i) *In a skipless network, let d_{model} columns of the read of h_ℓ form a matrix $W_1 = QR$, with Q orthogonal and R upper triangular. The change of basis Q at h_ℓ preserves the function and turns W_1 into R; at all $L+1$ hidden states, they remove $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters.*
(ii) *The changes of basis that preserve the function at all parameters are orthogonal, and in a residual network they agree at all hidden states. They form a group of dimension $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ in a skipless network and $d_{\text{model}}(d_{\text{model}}-1)/2$ in a residual one.*
(iii) *In a residual network, let d_{model} columns of any one read form a matrix $W_1 = QR$. The same change of basis Q at all hidden states preserves the function and turns W_1 into R.*
(iv) Triangular skips. *If every learned skip A_ℓ is invertible, there are orthogonal $Q_1, \dots, Q_L$, with $Q_0 = I$, such that every $Q_{\ell-1}^\top A_\ell Q_\ell$ is upper triangular, and the network with these skips, the reads $Q_{\ell-1}^\top W_{\text{read}}^\ell$, the writes $W_{\text{write}}^\ell Q_\ell$ and the readout $Q_L^\top D$ computes the same function.*

Proof of Theorem 3.4. For $\varepsilon > 0$, $\mathcal{G} = \mathrm{O}(d_{\text{model}})$ (Lemma A.2). Part (i) is Proposition A.9(i). In part (ii), the changes of basis are orthogonal by Lemma A.4 for the skipless network and by Theorem A.5(iii) for the residual one, where they also agree at all hidden states, and the dimensions are Theorem A.13(i) with $\dim \mathcal{G} = d_{\text{model}}(d_{\text{model}}-1)/2$. Part (iii) is Proposition A.9(ii), and part (iv) is Theorem A.10(i). □

A.5 LayerNorm reduces to RMSNorm (Proposition A.12)

The reduction of LayerNorm to RMSNorm is standard: the centering folds into the adjacent weights (Elhage et al., 2021), and centering every write gives an RMSNorm network whose hidden states are mean-free (Ashkboos et al., 2024a; Theus et al., 2025). We state it with the constants that the counts need. It is used in Section 3.4 and for ViT-B in Section 3.4 and Appendix D.

Proposition A.12 (LayerNorm reduces to RMSNorm)**.** *Let $Z = I - \frac{1}{d_{\text{model}}}\mathbf{1}\mathbf{1}^\top$ and $\mathcal{V} = \mathbf{1}^\perp$, and consider a pre-LayerNorm network, skipless or residual, whose reads and readout see $\mathrm{LN}(h) = \gamma \odot hZ/\sqrt{\|hZ\|^2/d_{\text{model}} + \varepsilon} + \beta$, with h, γ and β row vectors.*
(i) *Replacing each read W by $\mathrm{diag}(\gamma)W$ with the added bias βW, and each write W, write bias b and embedding E by WZ, bZ and EZ, gives a network that computes the same function, whose hidden states lie in $\mathcal{V}$, and whose reads see $N(h) = h/\sqrt{\|h\|^2/d_{\text{model}} + \varepsilon}$.*
(ii) *In an orthonormal basis of $\mathcal{V}$, this network is a pre-RMSNorm network of dimension $d_{\text{model}} - 1$, with every read multiplied by $\sqrt{d_{\text{model}}/(d_{\text{model}}-1)}$ and ε replaced by $\varepsilon d_{\text{model}}/(d_{\text{model}}-1)$.*
(iii) *If the gains γ have nonzero entries, the difference in redundancy between the skipless and the residual version of the LayerNorm network equals the difference for the reduced networks of (ii).*

Overview. LayerNorm sees only the mean-free part of the stream, so the mean can be removed from every write without changing the function. On the mean-free subspace LayerNorm is RMSNorm up to constants. The map from the parameters of the LayerNorm network to those of the reduced network has a derivative that is onto, and it is the same for the skipless and the residual version, so it preserves the difference in redundancy.

Proof. (i) Since $\mathrm{LN}(h) = \gamma \odot N(hZ) + \beta$, a read satisfies $\mathrm{LN}(h)\,W = N(hZ)\,\mathrm{diag}(\gamma)W + \beta W$. Every hidden state is a sum of writes, directly in a skipless network and through the skips in a residual one, so replacing every write, write bias and the embedding by its image under Z replaces every hidden state h by hZ. Since $Z^2 = Z$, every read then sees the same $N(hZ)$ as before, and the function is unchanged; the hidden states lie in $\mathcal{V}$. (ii) Let the columns of $\bar{U} \in \mathbb{R}^{d_{\text{model}} \times (d_{\text{model}}-1)}$ be an orthonormal basis of $\mathcal{V}$ and $h = y\bar{U}^\top$ with a row vector y. Then $\|h\|^2/d_{\text{model}} + \varepsilon = \frac{d_{\text{model}}-1}{d_{\text{model}}}\big(\|y\|^2/(d_{\text{model}}-1) + \varepsilon d_{\text{model}}/(d_{\text{model}}-1)\big)$, so $N(y\bar{U}^\top) = \sqrt{d_{\text{model}}/(d_{\text{model}}-1)}\,N'(y)\bar{U}^\top$, where N' is RMSNorm of dimension $d_{\text{model}}-1$ with constant $\varepsilon d_{\text{model}}/(d_{\text{model}}-1)$. A read $N(y\bar{U}^\top)W$ is the read $\sqrt{d_{\text{model}}/(d_{\text{model}}-1)}\,\bar{U}^\top W$ of $N'(y)$, and a write WZ is the write $W\bar{U}$ in the coordinates y. (iii) The reduced parameters are $\sqrt{d_{\text{model}}/(d_{\text{model}}-1)}\,\bar{U}^\top \mathrm{diag}(\gamma)W$ and $b + \beta W$ for each read, $W\bar{U}$ and $b\bar{U}$ for each write, $E\bar{U}$ for the embedding, and the analogous expressions for the readout. For γ with nonzero entries each is linear and onto in W, b or E, so the derivative of the map from the

parameters of the LayerNorm network to those of the reduced network is onto. The LayerNorm network computes the reduced network's function composed with this map, so by the chain rule every batch Jacobian of the LayerNorm network is that of the reduced network times this derivative, and at corresponding parameters both have the same functional dimension, and the redundancy of the LayerNorm network is that of the reduced network plus the difference of the parameter counts. That difference is the same for the skipless and the residual versions, which have the same shapes. □

Section 6 of the notebook (Appendix E) checks each step on a residual pre-LayerNorm network with $d_{\text{model}} = 6$: folding the gains changes the function by a relative 2×10^{-16}, centering the writes by 2×10^{-16}, and an orthogonal change of basis on $\mathcal{V}$ by 8×10^{-16}, while a generic orthogonal change that moves $\mathbf{1}$ changes it by 0.75.

A.6 Symmetry counts and upper bounds (Theorem 4.3)

Theorem 4.3 and Theorem 3.3(ii) are parts of the following theorem. A symmetry inside a block (Section 4.2) changes the read and the write of that block and leaves its map unchanged. So it preserves the function of a skipless and of a residual network alike, with or without normalization, and the total dimension s of these symmetries is the same number for both. The changes of basis and the symmetries inside blocks act *locally freely* at θ if no continuous family of them, other than the identity, fixes θ.

Theorem A.13 (Symmetry counts and upper bounds)**. (i)** *Without normalization or with pre-RMSNorm and $\varepsilon > 0$, the changes of basis that preserve the function at all parameters form a group of dimension $(L+1)\dim \mathcal{G}$ in a skipless network and $\dim \mathcal{G}$ in a residual one, so the L skips remove $L \dim \mathcal{G}$ dimensions of symmetry. For $\varepsilon = 0$, the changes of basis in $\mathcal{G}$ at the hidden states preserve the function of a skipless network, and they preserve that of every residual network only if they agree at all hidden states.*
(ii) *If in both networks the symmetries act locally freely at generic parameters, the functional dimension at these parameters is at most $P - (L+1)\dim \mathcal{G} - s$ without skips and $P - \dim \mathcal{G} - s$ with them.*
(iii) *If in addition both networks satisfy the equality condition, the skips raise the functional dimension by exactly $L \dim \mathcal{G}$, which is $L d_{\text{model}}^2$ without normalization.*

For $\varepsilon = 0$ and positively homogeneous blocks, a skipless network has more symmetries than the changes of basis in $\mathcal{G}$: all of $\mathrm{GL}(d_{\text{model}})$ at $h_0, \ldots, h_{L-1}$ (Lemma A.3). Its functional dimension then lies below the bound of (ii), so it does not satisfy the equality condition; for one block the skip adds at least d_{model}^2 (Theorem B.1(iv)).

Overview. Part (i) counts copies of $\mathcal{G}$: one per hidden state without skips, by Proposition A.1 and Lemma A.4, and one with them, by Theorem A.5(iii). Part (ii) bounds the rank of every batch Jacobian by the number of parameters minus the number of independent directions in which the symmetries move the parameters, which is the dimension of the symmetries when they act locally freely; Lemma A.14 proves that they do when the embedding has rank d_{model} and every block has width at least d_{model}. Part (iii) turns both bounds into equalities under the equality condition and subtracts. In the last row of Table 3, the change of basis $I + \tau X$ moves the fixed submatrix $W_1 = I$ by $-\tau X$, so its directions fill the fixed entries: fixing them removes d_{model}^2 parameters and no rank.

Lemma A.14 (Locally free action)**.** *Consider an MLP network, skipless or residual, without normalization or with pre-RMSNorm, with vanilla MLP blocks with $\tanh$, GELU, ReLU or ReLU^2 activations, or with gated blocks with any gate, and the symmetries inside blocks of Section 4.2. If the embedding has rank d_{model} and every block has width at least d_{model}, then at generic parameters the changes of basis and the symmetries inside blocks act locally freely. The same holds if the readout has rank d_{model} and every block has width at least d_{model}. With an embedding of rank d_{model}, it also holds for networks with attention blocks whose heads have width $d_h \leq d_{\text{model}}$ and total width $H d_h \geq d_{\text{model}}$.*

Table 3: Parameters, dimension of the symmetries and, under the equality condition, functional dimension of the networks of Section 2, without normalization or biases; P counts the parameters of the skipless network. Each step of the gauge fix removes as many parameters as dimensions of symmetry, while deleting the identity units removes $2Ld_{\text{model}}^2$ parameters but only Ld_{model}^2 dimensions of symmetry. Given the value for the residual network, those of the mixed network, of learned skips and of the last row hold at corresponding parameters without further assumption (Theorem A.5(ii), Proposition A.9(ii)).

Network	Parameters	Symmetry	Functional dimension
skipless	P	$(L+1)d_{\text{model}}^2 + s$	$P - (L+1)d_{\text{model}}^2 - s$
mixed, with trainable identity units	$P + 2Ld_{\text{model}}^2$	$(2L+1)d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
learned skips	$P + Ld_{\text{model}}^2$	$(L+1)d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
residual	P	$d_{\text{model}}^2 + s$	$P - d_{\text{model}}^2 - s$
residual, one submatrix fixed	$P - d_{\text{model}}^2$	s	$P - d_{\text{model}}^2 - s$

Proof. Permutations and sign flips are isolated, so it suffices to consider changes of basis and rescalings close to the identity. Let changes of basis $G_0, \dots, G_L$ and positive rescalings c_ℓ of hidden units (for ReLU and ReLU^2), of gates and values (for ReGLU) or of values (for the other gated blocks) fix a generic θ; for $\tanh$ and GELU there are no c_ℓ. The embedding transforms as $E \mapsto EG_0$, and E has rank d_{model}, so $G_0 = I$. Suppose $G_{\ell-1} = I$. The read of block ℓ then transforms as $W_{\text{read}}^\ell \mapsto W_{\text{read}}^\ell \operatorname{diag}(c_\ell)$, with entries 1 on the columns that are not rescaled, and no column of a generic W_{read}^ℓ vanishes, so $c_\ell = \mathbf{1}$. The write then transforms as $W_{\text{write}}^\ell \mapsto W_{\text{write}}^\ell G_\ell$, and W_{write}^ℓ has rank d_{model} because the block has width at least d_{model}, so $G_\ell = I$. By induction only the identity fixes θ. In a residual network the G_ℓ are equal (Theorem A.5(iii)) and the same argument applies. The second statement follows by the mirror induction from the readout, $D \mapsto G_L^{-1} D$: rescalings act freely on the rows of a generic write, except that for ReGLU a row fixes only the product cc' of the gate and value scales of its unit; each read, of rank d_{model} and with generic columns, then forces $G_{\ell-1} = I$ and the remaining scales to be 1. With RMSNorm the changes of basis lie in $\mathrm{CO}(d_{\text{model}})$ or $\mathrm{O}(d_{\text{model}})$, and the argument is unchanged. For an attention block, once $G_{\ell-1} = I$ the changes of basis inside its heads multiply $W_{Q,i}$ and $W_{V,i}$ on the right. At generic parameters these have full column rank, as $d_h \le d_{\text{model}}$, so the changes of basis inside the heads are the identity. The write, which stacks the $W_{O,i}$, then transforms as $W_O \mapsto W_O G_\ell$ and has rank d_{model}, as $Hd_h \ge d_{\text{model}}$, so $G_\ell = I$. □

Remark A.15 (The width condition; attention blocks). The width condition is needed: at a hidden state of a skipless network that is written and read at rank $m < d_{\text{model}}$, the changes of basis that fix the parameters form a family of dimension $(d_{\text{model}} - m)^2$, so they move the parameters in only $2d_{\text{model}}m - m^2$ independent directions, as Zhao et al. (2023, Proposition C.3) compute for one hidden layer. The full group of the symmetries inside attention heads is characterized by Wang and Wang (2025). A numerical check on small skipless transformers confirms a locally free action, also for heads of total width $Hd_h < d_{\text{model}}$ followed by an MLP block of width at least d_{model}, which the induction does not reach (Section 6 of the notebook, Appendix E).

Proof of Theorem A.13. Parts (i) and (ii). If a family of transformations of the parameters leaves the function unchanged, then the function does not change to first order along any direction in which the family moves θ, so these directions lie in the kernel of every batch Jacobian at θ, and the functional dimension is at most P minus their number. For a skipless network, Proposition A.1 gives $L+1$ independent copies of $\mathcal{G}$, one per hidden state, and the directions in which they move θ are linearly independent of one another and of those of the symmetries inside blocks at generic parameters: the change of basis at h_ℓ multiplies W_{write}^ℓ on the right and $W_{\text{read}}^{\ell+1}$ on the left, the symmetries inside block ℓ multiply W_{read}^ℓ on the right and W_{write}^ℓ on the left, and by hypothesis they act locally freely at generic parameters (Lemma A.14). The number of independent directions is therefore $(L+1)\dim\mathcal{G} + s$. For a residual network, Theorem A.5(iii) leaves one copy, and the same argument gives $\dim\mathcal{G} + s$. For part (i), no other changes of basis preserve the function at all parameters: without normalization $\mathcal{G}$ is all of $\mathrm{GL}(d_{\text{model}})$, and for $\varepsilon > 0$ this is Lemma A.4 for the skipless network and Theorem A.5(iii) for the residual one. *Part (iii).* Under the hypothesis, the

inequalities of part (ii) are equalities, so the generic functional dimensions are $P-(L+1)\dim\mathcal{G}-s$ and $P-\dim\mathcal{G}-s$, and the difference is $L\dim\mathcal{G}$, which is d_{model}^2 per skip without normalization, and under RMSNorm $d_{\text{model}}(d_{\text{model}}-1)/2$ when $\varepsilon>0$ and $d_{\text{model}}(d_{\text{model}}-1)/2+1$ when $\varepsilon=0$ (Lemma A.2). □

We recall Theorem 4.3 from Section 4.2.

Theorem 4.3 (Upper bounds)**.** *Let $\mathcal{G}$ be the group of changes of basis of one hidden state, and s the dimension of the symmetries inside the blocks, the same with and without skips. With or without pre-RMSNorm, at generic parameters, the functional dimension is at most $P-(L+1)\dim\mathcal{G}-s$ without skips and at most $P-\dim\mathcal{G}-s$ with identity skips.*

Proof of Theorem 4.3. Within the scope of Section 2, the symmetries act locally freely at generic parameters (Lemma A.14). So the theorem is Theorem A.13(ii), and the gain under the equality condition is Theorem A.13(iii). □

The next proposition relates the equality condition to identifiability (Definition 4.5).

Proposition A.16 (The equality condition and local identifiability)**.** *Let the blocks be analytic, with or without pre-RMSNorm, within the scope of Section 2. A skipless or a residual network satisfies the equality condition if and only if, at generic parameters, it is locally identifiable up to its symmetries. Every symmetry that is needed is then close to the identity.*

Proof. Let k be the dimension of the symmetries of the network, $(L+1)\dim\mathcal{G}+s$ without skips and $\dim\mathcal{G}+s$ with them. For a finite batch X, write $F_X(\theta)$ for the outputs on X, and let r be the largest rank that the Jacobian of F_X attains on any batch at any parameters. Fix a batch X that attains it. The outputs are analytic in θ, so the rank on X is r outside the zero set of the sum of the squares of the minors of size r, a set of measure zero. Hence r is the functional dimension at generic parameters, $r\leq P-k$ by Theorem 4.3, and the equality condition is $r=P-k$.

Take θ at which the rank on X is r and the symmetries act locally freely, as they do at generic parameters (Lemma A.14). By lower semicontinuity the rank on X is r on a neighborhood U of θ. By the constant rank theorem, after shrinking U, the set $Z=\{\theta'\in U : F_X(\theta')=F_X(\theta)\}$ is a connected manifold of dimension $P-r$, whose tangent space at each point is the kernel of the Jacobian. For any further input x, the Jacobian on $X\cup\{x\}$ has rank r on U as well, so it has the same kernel, and $\theta'\mapsto F_{\theta'}(x)$ is constant on Z. So $Z=\{\theta'\in U : F_{\theta'}=F_\theta\}$. The changes of basis that preserve the function, the rescalings and the changes of basis inside the heads form a Lie group of dimension k, which acts smoothly. It acts locally freely at θ, so $g\mapsto g\cdot\theta$ is an immersion near the identity, and it maps a neighborhood of the identity onto a manifold $O\subset Z$ of dimension k.

If $r=P-k$, then O is open in Z, since the two have the same dimension. So every θ' close to θ with $F_{\theta'}=F_\theta$ is the image of θ under a symmetry close to the identity. Conversely, let every θ' in a neighborhood of θ in Z be the image of θ under a symmetry. With the permutations and the sign flips, the symmetries form a Lie group of dimension k with finitely many components, and the images of θ under it form the image of a manifold of dimension k under a smooth map. By Sard's theorem this image contains no open subset of a manifold of larger dimension, so $P-r\leq k$, which with $r\leq P-k$ is the equality condition. □

A.7 Depth (Theorems 4.7, 4.8 and 4.9)

This subsection uses Lemmas B.2 and B.3 and the proof of Theorem B.1 from Appendix B, which does not depend on it. Theorem A.19 below and Theorem 4.7 rest on two lemmas. The first turns a gap in functional dimension into disjointness. The second shows that with small writes the Jacobian of a residual network reaches the largest rank that its symmetries allow, by the argument of Theorem B.1 for one block.

Lemma A.17 (A rank criterion)**.** *Let $\theta\mapsto F_\theta$, $\theta\in\mathbb{R}^P$, and $\eta\mapsto G_\eta$, $\eta\in\mathbb{R}^{P'}$, be two families of networks with the same inputs and outputs, whose outputs are real-analytic in the parameters. Let a Lie group act smoothly on $\mathbb{R}^{P'}$ without changing G_η, with orbits of dimension at least k at generic η.*

If at some θ_0 the Jacobian of $\theta \mapsto (F_\theta(x))_{x\in X}$ has rank greater than $P' - k$ on some finite batch X, then for almost every θ no η satisfies $F_\theta = G_\eta$.

Proof. Write $F_X(\theta), G_X(\eta) \in \mathbb{R}^n$ for the outputs on X. At generic η the map G_X is constant along an orbit of dimension at least k, so its Jacobian has rank at most $r_0 = P' - k$; the rank is lower semicontinuous, so the bound holds at every η. The Jacobian of F_X reaches its largest rank r^*, which exceeds r_0 by hypothesis, outside the zero set of the sum of the squares of its minors of size r^*. This function is analytic and not identically zero, so its zero set has measure zero. Take θ outside it. By the constant rank theorem, F_X maps a neighborhood U of θ onto an embedded manifold Y of dimension r^*, and after shrinking U a linear projection $\pi : \mathbb{R}^n \to \mathbb{R}^{r^*}$ maps Y diffeomorphically onto an open set. The map $\pi \circ G_X$ has rank at most $r_0 < r^*$, so all its values are critical, and by Sard's theorem its image has measure zero. So $G_X(\mathbb{R}^{P'}) \cap Y$ has measure zero in Y, and so does its preimage under F_X in U, which contains every θ' in U with $F_{\theta'} = G_\eta$ for some η. Countably many such U cover the complement of the zero set. □

Lemma A.18 (Small writes)**.** *Let $d_{\text{model}} \geq 2$, let E and D have rank d_{model}, and let each block, in any order and of any width, be an attention block with heads of width at most d_{model}, causal or not, on sequences of at least two tokens, or an MLP block, with all MLP blocks* tanh*, all GELU, all SwiGLU or all GeGLU blocks. At generic parameters, multiply every write, and every write bias, by τ. For all small $\tau \neq 0$ the residual network without biases has functional dimension $P - d_{\text{model}}^2 - s$ without normalization and $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$ under pre-RMSNorm with $\varepsilon > 0$. Without normalization and attention blocks, biases lower the first value by $(L+1)d_{\text{model}}$. Without an embedding, a readout, biases and normalization it has functional dimension $P - s$.*

Proof. Write the write of block ℓ, for an attention block its output projection, as τV_ℓ. Let $W = [\, W_{\text{read}}^1 \mid \cdots \mid W_{\text{read}}^L \,]$ place the reads of the MLP blocks side by side and V stack their V_ℓ, so that $n \mapsto f(nW)V$ is one wide block that holds all units of the network, and let the heads of all attention blocks form one attention block. Divide the columns of the Jacobian that belong to the reads, for attention blocks the queries, keys and values, and to the read biases by τ, which keeps its rank, and let $\tau \to 0$. On a batch of inputs the limit is a linear map J_0 on $\mathbb{R}^P$; as in Appendix B, a direction η is flat if $J_0\,\eta = 0$ on every input. For analytic blocks $J_0\,\eta$ is real-analytic in the input, so a direction that is flat on an open set of inputs is flat on all inputs; the arguments below use rays and sequences of equal tokens, which need not lie in that set.

MLP blocks without normalization. The stream is $h_\ell = xE + b_E + O(\tau)$, so the output is

$$y = xED + b_E D + b_D + \tau \sum_{\ell=1}^{L} \Big(f\big((xE + b_E)W_{\text{read}}^\ell + b_{\text{read}}^\ell\big)V_\ell + b_{\text{write}}^\ell \Big) D + O(\tau^2),$$

with the write biases written as $\tau b_{\text{write}}^\ell$. The derivatives with respect to E, D and the biases of the embedding and the readout keep only the terms without τ in the limit. The derivative along a direction η_V of the writes τV_ℓ is the term of order τ with η_V in place of τV. The derivative along a direction η_W of the reads is of order τ, and divided by τ it tends to the derivative of the term of order τ. Hence $J_0\,\eta = \mathrm{D}_\phi G(\phi, x)\,\psi$ for the one-block function G of Lemma B.3 at $M = ED$, $\gamma = b_E D + b_D$, the columns c_i of EW, the entries β_i of $b_E W + b_{\text{read}}$ and the rows a_i of VD, in the direction ψ with

$$\psi_M = \eta_E D + E\eta_D, \qquad \psi_\gamma = \eta_{b_E} D + b_E \eta_D + \eta_{b_D} + \textstyle\sum_\ell \eta_{b_{\text{write}}^\ell} D,$$
$$\psi_C = E\eta_W, \qquad \psi_\beta = b_E \eta_W + \eta_{b_{\text{read}}}, \qquad \psi_A = \eta_V D,$$

where ψ_C, ψ_β and ψ_A collect the ψ_{c_i}, ψ_{β_i} and ψ_{a_i}, and the values of the gated units enter as the gates do. Unlike for one block, ψ_C has no term $\eta_E W$, so Lemma B.2 is not needed below. At generic parameters the hypotheses of Lemma B.3 hold, as for one block (Appendix B). For a flat η the lemma gives $\psi_M = 0$, $\psi_\gamma = 0$ and a rescaling at every unit. As E and D have rank d_{model}, $\psi_M = 0$ gives $\eta_E = EX$ and $\eta_D = -XD$ for some $X \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$; ψ_C and ψ_A make η_W and η_V a rescaling of the units, which fixes $\eta_{b_{\text{read}}}$ through ψ_β; and $\psi_\gamma = 0$ fixes η_{b_D} once η_{b_E} and the L write biases are chosen. So the flat directions form a space of dimension $d_{\text{model}}^2 + s$, and $d_{\text{model}}^2 + (L+1)d_{\text{model}} + s$ with biases. Without an embedding and a readout, $M = I$ is fixed, and only the rescalings are left.

Attention blocks without normalization. Without biases, $y = xED + \tau \sum_\ell g_\ell(xE)D + O(\tau^2)$, where g_ℓ is block ℓ with the write V_ℓ. So $J_0\,\eta$ is the derivative of the residual network with one block that holds all heads and units, in the direction with the entries ψ_M, ψ_C and ψ_A above and, for each head i,

$$\psi_{B_i} = E\big(\eta_{Q,i}W_{K,i}^\top + W_{Q,i}\eta_{K,i}^\top\big)E^\top/\sqrt{d_h}, \qquad \psi_{C_i} = E\big(\eta_{V,i}W_{O,i} + W_{V,i}\eta_{O,i}\big)D,$$

at $B_i = EW_{Q,i}W_{K,i}^\top E^\top/\sqrt{d_h}$ and $C_i = EW_{V,i}W_{O,i}D$ as in Appendix B, where $W_{O,i}$ holds the rows of V_ℓ that belong to head i of block ℓ. On inputs whose tokens all equal x, head i gives every token xC_i, and its derivative is $x\psi_{C_i}$, since all its scores move alike. There the equation is that of the units with the linear part $M + \sum_i C_i$, and Lemma B.3 gives a rescaling at every unit. The units then drop out on every input. On inputs whose tokens other than the second equal the first, head i gives the second token the output of (10) with $k = 1$ if its block is causal and $k = T - 1$ otherwise; the constants $-\frac{1}{2}\log k$ act as biases of the $\tanh$ units, which Lemma B.3 allows, so Step 2 for attention blocks in Appendix B gives $\psi_M = 0$, $\psi_{B_i} = 0$ and $\psi_{C_i} = 0$. Cancelling E on the left and $E^\top$ or D on the right, the last two leave, as in Step 1 there, the directions of the changes of basis between the queries and keys and between the values and outputs of each head. So the flat directions form a space of dimension $d_{\text{model}}^2 + s$, and without an embedding and a readout of dimension s.

Under pre-RMSNorm. The stream is $h_\ell = xE + O(\tau)$, every block reads $N(xE) + O(\tau)$, and the output is

$$y = N(xE)D + \tau\,\mathrm{D}N(xE)\,\Big(\textstyle\sum_\ell g_\ell\big(N(xE)\big)\Big)D + O(\tau^2),$$

with N applied to every token. So $J_0\,\eta = \mathrm{D}N(h)\,\big(x\eta_E + k(n)\big)D + N(h)\eta_D$, with $h = xE$, $n = N(h)$ and $k(n)$ the derivative of the wide block along η at the input n. This is the equation of the proof of Theorem B.1(ii) with $h_1 = xE$ and $\mathrm{D}_\theta h_1\,\eta = x\eta_E + k(n)$, and its Steps 1 to 5 go through, with $k(n)$ bounded in Step 2 since n stays in the ball, and with the units handled first, through equal tokens, in Step 4. In Step 3 the direction η^Y now has the entries EY and $-YD$ at E and D and no others, as $\mathrm{D}N(h)\,(hY) = N(h)Y$ for antisymmetric Y. So the flat directions are these, the rescalings of the units and the changes of basis inside the heads, a space of dimension $d_{\text{model}}(d_{\text{model}} - 1)/2 + s$.

In each case the kernel of J_0 shrinks as the batch grows and becomes stationary at the flat directions, so on some finite batch J_0 has rank P minus this dimension. By lower semicontinuity the Jacobian of the residual network on this batch has at least this rank for all small $\tau \neq 0$. Its symmetries bound the rank by the same number, since they move the parameters in independent directions when E has rank d_{model} and the queries and values of every head have full column rank: the global change of basis (Theorem A.5(iii)), the translations (Lemma A.8), the rescalings of the units and the changes of basis inside the heads. □

Theorem 4.9 is a case of the following theorem.

Theorem A.19 (GELU and analytic networks at depth)**. (i)** *For GELU blocks and almost every choice of weights, a composition of $L \geq 2$ residual blocks equals no composition of L blocks without skips of the same widths.*
(ii) *With an embedding and a readout, the same holds for networks of analytic blocks of any depth $L \geq 1$, also under pre-RMSNorm with $\varepsilon > 0$.*
(iii) Too few extra units. *For GELU blocks with an embedding and a readout, without normalization, and almost every choice of weights, a residual network of L blocks equals no skipless network of L blocks with, in total, fewer than $Ld_{\text{model}}/2$ more hidden units.*

Proof of Theorem A.19(i) and (ii). By Lemma A.18, a residual network without biases has functional dimension $P - d_{\text{model}}^2 - s$ at some parameters, where s counts the changes of basis inside the heads and the rescalings of the values of gated units, and a network of GELU blocks without an embedding and a readout has functional dimension P. In a skipless network of the same widths, with embedding E' and readout D', the changes of basis at h_0 and h_L, those inside the heads and the rescalings have orbits of dimension $2d_{\text{model}}^2 + s$ at generic parameters, where E' and D' have rank d_{model} and the

queries and values of every head have full column rank. As $P - d_{\text{model}}^2 - s > P - 2d_{\text{model}}^2 - s$, Lemma A.17 gives the claim. Without an embedding, a readout and attention blocks, for $L \geq 2$ the changes of basis at h_1 have orbits of positive dimension, since every block has at least one unit, so Lemma A.17 applies again. Under pre-RMSNorm with $\varepsilon > 0$ the networks stay analytic, Lemma A.18 gives the residual network functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$, and the orthogonal changes of basis at h_0 and h_L, which commute with the normalization, have orbits of dimension $d_{\text{model}}(d_{\text{model}} - 1) + s$ in the skipless network; as $d_{\text{model}} \geq 2$, Lemma A.17 applies once more. □

Proof of Theorem A.19(iii). Let the residual network have blocks with $m_1, \ldots, m_L$ hidden units and P parameters, and let a skipless network have blocks with $m'_1, \ldots, m'_L$ hidden units. If some $m'_\ell < d_{\text{model}}$, the hidden state h_ℓ of the skipless network lies in a subspace of dimension m'_ℓ, so the Jacobian of its output with respect to the input has rank below d_{model} at every input. That of the residual network is $ED + O(\tau)$ when its writes are multiplied by τ, of rank d_{model} for small τ, so by analyticity it has rank d_{model} at some input for almost every choice of weights, and the two networks differ. So let every $m'_\ell \geq d_{\text{model}}$. The skipless network then has $P' = P + 2d_{\text{model}} \sum_\ell (m'_\ell - m_\ell)$ parameters, and its changes of basis at $h_0, \ldots, h_L$ have orbits of dimension $(L+1)d_{\text{model}}^2$ at generic parameters (Lemma A.14). By Lemma A.18 the residual network has functional dimension $P - d_{\text{model}}^2$ at some parameters, which exceeds $P' - (L+1)d_{\text{model}}^2$ exactly when $\sum_\ell (m'_\ell - m_\ell) < Ld_{\text{model}}/2$. For each of the finitely many such choices of widths, Lemma A.17 gives a set of weights of measure zero, and so does their union. □

We recall Theorem 4.9 from Section 4.4.

Theorem 4.9 (Different function classes)**.** **(i)** *For analytic blocks, with or without pre-RMSNorm, a residual network at generic parameters equals no skipless network of the same widths.*
(ii) Too few extra units. *For networks of GELU blocks without normalization, a residual network of L blocks at generic parameters equals no skipless network of L blocks with, in total, fewer than $Ld_{\text{model}}/2$ more hidden units.*

Proof of Theorem 4.9. Part (i) is Theorem A.19(ii), and part (ii) is Theorem A.19(iii). □

We recall Theorems 4.7 and 4.8 from Section 4.3.

Theorem 4.7 (Equality for residual networks at any depth)**.** *Let the blocks be analytic, with or without pre-RMSNorm. The residual network satisfies the equality condition: at generic parameters its functional dimension is $P - \dim \mathcal{G} - s$. So the skips raise the functional dimension by at least $L \dim \mathcal{G}$, and by exactly $L \dim \mathcal{G}$ when the skipless network satisfies the equality condition too.*

Proof of Theorem 4.7. The equality condition. Let r be P minus the dimension of the symmetries of the residual network, the value in Lemma A.18. At generic parameters the symmetries move the parameters in independent directions, so the rank of the Jacobian is at most r on every batch. Lemma A.18 gives parameters and a finite batch on which it equals r. On this batch the rank is therefore at least r outside the zero set of the sum of the squares of the minors of size r, an analytic function that is not identically zero, so this zero set has measure zero. At generic parameters the functional dimension is thus r, which is the equality condition.

The gain. When every block has width at least d_{model}, the functional dimension of the skipless network is at most $P - (L+1)\dim \mathcal{G} - s$ (Theorem A.13(ii), Lemma A.14). This gives the gain of at least $L \dim \mathcal{G}$, with equality when the skipless network satisfies the equality condition. □

Theorem 4.8 (Exact gain for $\tanh$ and GELU)**.** *Without normalization, in the two cases below the skipless network satisfies the equality condition too, so the skips raise the functional dimension by exactly Ld_{model}^2.*
(i) *The blocks are $\tanh$ blocks.*
(ii) *The blocks are GELU blocks with at least $2d_{\text{model}}$ hidden units.*

Proof of Theorem 4.8. By Theorem 4.7 the residual network satisfies the equality condition, so it remains to show it for the skipless network; the gain is then Ld_{model}^2.

Part (i), $\tanh$. With read biases b_ℓ and write biases c_ℓ, which may vanish, the skipless network computes

$$x^0 = x, \qquad x^\ell = \tanh\left(x^{\ell-1}A_\ell + \beta_\ell\right) \ (1 \le \ell \le L), \qquad y = x^L A_{L+1} + \beta_{L+1},$$

with $A_1 = EW_1$, $A_\ell = V_{\ell-1}W_\ell$ and $A_{L+1} = V_L D$, where W_ℓ and V_ℓ are the read and the write of block ℓ, and $\beta_1 = b_E W_1 + b_1$, $\beta_\ell = c_{\ell-1}W_\ell + b_\ell$ and $\beta_{L+1} = c_L D + b_D$. Two such networks that agree on an open set of inputs agree on all inputs, by analyticity. As $\sigma(z) = \tanh(z/2)$ is injective, two such networks have the same output exactly when the nets of Fefferman (1994) with the hidden layers $x^\ell = \sigma(2x^{\ell-1}A_\ell + 2\beta_\ell)$ and the output layer $\sigma(x^L A_{L+1} + \beta_{L+1})$ do, and then so do, for every input coordinate and every output, the nets with one input and one output that keep only these. At generic parameters every entry of every A_ℓ is nonzero, and no ratio of the weights that one unit sends to two units of a layer of width m is a fraction p/q with $1 \le q \le 100m^2$. These conditions are open, and they are those of his Theorem I.B.9, on nets with one input and one output, which puts no condition on the thresholds. So near generic parameters, two networks with the same output give, for every input coordinate and every output, nets with one input and one output that agree up to permutations of the units of each hidden layer and sign flips. These are the identity: at the first layer where a unit is moved or flipped, the weights it receives would move by a fixed amount, since none vanishes and the weights that one unit sends to two units differ in absolute value. So the two networks have the same A_ℓ and β_ℓ. When E, D and every read and write have rank d_{model}, the parameters with the same A_ℓ and β_ℓ are those reached by the changes of basis at the $L+1$ hidden states, and with biases by the translations there.

By analyticity, the functional dimension at generic parameters is the largest rank r that the Jacobian attains on any batch at any parameters. Take generic parameters and a batch on which the rank is r. Near these parameters the rank on this batch is at least r, by lower semicontinuity, and so equal to r. Its kernel is then the space of directions that are flat for every input, and the parameters with the same outputs on the batch form a manifold of dimension $P - r$ along which the function does not change. So $P - r$ is at most the dimension of the symmetries, and the symmetries give the reverse inequality (Theorem A.13(ii), Lemma A.8).

Part (ii), GELU. Let block ℓ have $m_\ell + d_{\text{model}}$ units, with $m_\ell \ge d_{\text{model}}$, and take a residual network with blocks of m_ℓ units at generic parameters, with every write multiplied by τ as in Lemma A.18. Since $\sigma(z) - \sigma(-z) = z$, each of its blocks equals a block without skip with d_{model} more units: for d_{model} units i whose reads w_i form an invertible matrix, add the row γ_i of its inverse to the write of unit i, and add a unit with the read $-w_i$ and the write $-\gamma_i$. This construction is smooth in the residual parameters near these, where the chosen reads stay invertible, and gives a skipless network that computes the residual one. By the chain rule, the columns of the skipless Jacobian therefore span those of the residual Jacobian, which, with the read columns divided by τ, tend to the limit J_0 of Lemma A.18, of rank $P_R - d_{\text{model}}^2$ on some finite batch, where $P_R = P - 2Ld_{\text{model}}^2$ counts the parameters of the residual network. As $\sigma(z) - \sigma(-z) = z$ gives $\sigma'(-z) = 1 - \sigma'(z)$, and every block is the identity at $\tau = 0$, the reads of the added units give the further columns

$$\big(\sigma'(xEw_i) - 1\big)(xE\delta_i)\,\gamma_i D, \qquad \delta_i \in \mathbb{R}^{d_{\text{model}}},$$

in the limit. Suppose that $J_0\,\eta$ plus a sum of these vanishes for all x. Each added term has the Gaussian part of the term of ψ_{c_i} in Lemma B.3, with $E\delta_i$ in place of ψ_{c_i} and the row $\gamma_i D$ in place of a_i. In the proof of that lemma for GELU, on the rays $x = tn$, the coefficient of t^2 for the slowest unit j becomes $u_j\big((n\psi_{c_j})\,a_j + (nE\delta_j)\,\gamma_j D\big)$. At generic parameters $\gamma_j D$ is not parallel to a_j, so $nE\delta_j = 0$ for all n in an open set, and $\delta_j = 0$ as E has rank d_{model}; the proof then goes on as before. So, on some finite batch, the added reads raise the rank in the limit by Ld_{model}^2, to $P_R - d_{\text{model}}^2 + Ld_{\text{model}}^2 = P - (L+1)d_{\text{model}}^2$. By lower semicontinuity the skipless network has this rank for small τ, which is the most its symmetries allow, and an analytic network keeps its largest rank at generic parameters. □

Lemma A.20 (ReLU-type blocks with small writes)**.** *Let the blocks be ReLU, ReLU^2 or ReGLU blocks, without normalization or biases, let E and D have rank d_{model}, and let inputs range over all of $\mathbb{R}^{1\times d_{\text{in}}}$, or over an open set that contains 0. At generic parameters, multiply every write by τ. For all small $\tau \neq 0$, the Jacobian of the residual network has rank $P - d_{\text{model}}^2 - s$ on some finite batch on which no pre-activation vanishes. So the residual network has functional dimension $P - d_{\text{model}}^2 - s$ on an open set of weights that contains these, and at almost every point of this set the skips add at least Ld_{model}^2 when every block has width at least d_{model}.*

Proof. We follow the proof of Lemma A.18 for MLP blocks without normalization. The limit J_0 is again the derivative of the one-block function G of Lemma B.3, whose units are those of all blocks, with the pre-activations xEW_{read}^{ℓ}. On each region of inputs on which these pre-activations keep their signs, $J_0\,\eta$ is a polynomial in x, and the regions are cones with apex 0. So a direction that is flat on a neighborhood of 0 is flat at every input at which no pre-activation vanishes, and by Lemma B.3, which covers these units, the flat directions form a space of dimension $d_{\text{model}}^2 + s$, as in Lemma A.18. Fix a finite batch of such inputs, inside the set of inputs, on which the kernel of J_0 is this space; it exists, since the kernels decrease as the batch grows.

With the writes τV_ℓ, the pre-activations of block ℓ are $h_{\ell-1}W_{\text{read}}^{\ell}$ with $h_{\ell-1} = xE + O(\tau)$. So for $|\tau|$ below some $\tau_0 > 0$, which depends on the parameters and on the batch, every pre-activation on the batch has the sign of its limit. The outputs on the batch are then polynomial in the parameters and in τ, the Jacobian on the batch, with its read columns divided by τ, tends to J_0, and by lower semicontinuity it has rank at least $P - d_{\text{model}}^2 - s$ for all small $\tau \neq 0$.

Near such parameters the signs on the batch do not change, E has rank d_{model} and no read of a unit vanishes. There the change of basis of the hidden states and the rescalings of the units move the parameters in $d_{\text{model}}^2 + s$ independent directions, which lie in the kernel of the Jacobian on every batch on which no pre-activation vanishes. So the rank on the batch, which is at least $P - d_{\text{model}}^2 - s$ by lower semicontinuity, is exactly $P - d_{\text{model}}^2 - s$, and it is the functional dimension. The gain follows from Theorem A.13(ii), which bounds the functional dimension of the skipless network by $P - (L+1)d_{\text{model}}^2 - s$ at generic parameters. □

For ReLU blocks the count fails on another open set of weights, where a block folds the hidden state and units of later blocks are never active (Appendix A.8).

A.8 ReLU blocks: folding and dead units

This section explains why the count of Lemma A.20 holds on one open set of weights and fails on another, so that it does not hold at almost every parameter. The blocks are ReLU blocks, without normalization. An MLP block acts on each token separately, so we take inputs with one token, $x \in \mathbb{R}^{1\times d_{\text{in}}}$, and hidden states $h \in \mathbb{R}^{1\times d_{\text{model}}}$. In this section inputs range over all of $\mathbb{R}^{1\times d_{\text{in}}}$, and E has rank d_{model}.

Active and dead units. Unit i of block ℓ has the read r_i, column i of W_{read}^{ℓ}, and the write $\hat{r}_i$, row i of W_{write}^{ℓ}. Its *pre-activation* at the input x is the number $z_i(x) = h_{\ell-1}(x)\,r_i$, and the unit adds $\mathrm{ReLU}(z_i(x))\,\hat{r}_i$ to the output of the block. The unit is *active* at x if $z_i(x) > 0$. Where $z_i(x) < 0$, the unit adds nothing, and the derivatives of the output with respect to r_i and $\hat{r}_i$ vanish. A unit is *dead* if it is active at no input. A dead unit can be deleted, or its write changed, without changing the function.

Definition A.21 (Folding)**.** Let $\Phi_\ell(h) = h + B_\ell(h)$ be the map of $\mathbb{R}^{1\times d_{\text{model}}}$ computed by residual block ℓ. The block *folds* if Φ_ℓ is not injective: two different hidden states give the same next hidden state.

For example, take $m = d_{\text{model}}$, $W_{\text{read}} = -I$ and $W_{\text{write}} = 2I$. Then $B(h) = 2\,\mathrm{ReLU}(-h)$, and coordinate j of $h + B(h)$ is h_j if $h_j \geq 0$ and $-h_j$ if $h_j < 0$. So the block maps h to $(|h_1|, \ldots, |h_{d_{\text{model}}}|)$. It folds every coordinate axis at 0, like a sheet folded along a line, and its image is the nonnegative orthant. Wherever no coordinate vanishes, its Jacobian is $\mathrm{diag}(\pm 1)$, so a block can fold although its Jacobian is invertible wherever it exists.

Proposition A.22 (Blocks that do not fold keep units alive)**.** *Consider a residual network of ReLU blocks without normalization.*
(i) *If block ℓ does not fold, Φ_ℓ is a homeomorphism of $\mathbb{R}^{1\times d_{\text{model}}}$.*
(ii) *If no block folds, every unit with a nonzero read is active on a nonempty open set of inputs and inactive on another. In particular, no unit with a nonzero read is dead.*
(iii) *A block with $\|W_{\text{read}}\|\,\|W_{\text{write}}\| < 1$, in spectral norm, does not fold.*

Proof. (i) The map Φ_ℓ is continuous, and $\Phi_\ell(ch) = c\,\Phi_\ell(h)$ for $c > 0$, since there are no biases. As $\Phi_\ell(0) = 0$ and Φ_ℓ is injective, $\Phi_\ell(h) \neq 0$ for $h \neq 0$. So $c_0 = \min_{\|h\|=1}\|\Phi_\ell(h)\|$ is positive, and $\|\Phi_\ell(h)\| \geq c_0\|h\|$. Hence the preimage of a bounded set is bounded, and the image of Φ_ℓ is closed. By invariance of domain, an injective continuous map of $\mathbb{R}^{d_{\text{model}}}$ to itself is open. So the image is also open, hence all of $\mathbb{R}^{1\times d_{\text{model}}}$, and the inverse is continuous. (ii) The map $x \mapsto xE$ is onto $\mathbb{R}^{1\times d_{\text{model}}}$, and by (i) so is the continuous map $x \mapsto h_{\ell-1}(x)$. The sets $\{h : hr_i > 0\}$ and $\{h : hr_i < 0\}$ are open and not empty, and so are their preimages. (iii) As ReLU is 1-Lipschitz, $\|B_\ell(h) - B_\ell(h')\| \leq \|W_{\text{read}}\|\,\|W_{\text{write}}\|\,\|h-h'\|$. If $\Phi_\ell(h) = \Phi_\ell(h')$, then $\|h-h'\| = \|B_\ell(h') - B_\ell(h)\|$, so $h = h'$. □

By (iii), small enough writes never fold, which is the regime of Lemma A.20. The condition of (iii) is the one under which Behrmann et al. (2019) invert residual networks (their Theorem 1), and it is only sufficient. With $W_{\text{read}} = I$ and $W_{\text{write}} = 2I$, the block maps each coordinate h_j to h_j if $h_j \leq 0$ and to $3h_j$ if $h_j > 0$, so it does not fold, although $\|W_{\text{read}}\|\,\|W_{\text{write}}\| = 2$.

Proposition A.23 (Folding blocks can kill units and lower the functional dimension)**.** *Consider residual networks of $L \geq 2$ ReLU blocks without normalization.*
(i) *There is a nonempty open set of weights on which block 1 folds and block 2 has a dead unit.*
(ii) *If ν units are dead, then on every batch on which no pre-activation vanishes, the Jacobian has rank at most $P - d_{\text{model}}^2 - s - (2d_{\text{model}} - 1)\nu$.*

Proof. (i) Let block 1 hold the d_{model} units of the example above, with $W_{\text{read}} = -I$ and $W_{\text{write}} = 2I$. Give its other units zero writes. Let a unit of block 2 have a read r whose entries are all negative. Then $\Phi_1(h)\,r < 0$ for every $h \neq 0$, since $\Phi_1(h)$ is a nonzero vector with no negative entry. By compactness of the unit sphere, $\max_{\|h\|=1}\Phi_1(h)\,r < 0$ persists under small changes of all the weights. Since the perturbed Φ_1 is still positively homogeneous, $\Phi_1(h)\,r < 0$ then holds for every $h \neq 0$. As $h_1(x) = \Phi_1(xE)$ and $\Phi_1(0) = 0$, the unit is active at no input, so it is dead. And Φ_1 misses the half-space $\{h : hr > 0\}$, so it is not onto, and block 1 folds by Proposition A.22(i). (ii) The columns of the Jacobian that belong to the read and the write of a dead unit vanish, $2d_{\text{model}}\nu$ columns in all. The other columns are those of the network with the dead units deleted. That network has $P - 2d_{\text{model}}\nu$ parameters and $s - \nu$ unit rescalings, since s counts one rescaling per ReLU unit. Its function does not change under the change of basis of the hidden states and these rescalings. At generic parameters they act with orbits of dimension $d_{\text{model}}^2 + s - \nu$, since E has rank d_{model}. On the open set of parameters at which no pre-activation vanishes on the batch, the outputs on the batch are polynomial in the parameters. So, as in the proof of Lemma A.17, the rank is at most $P - 2d_{\text{model}}\nu - d_{\text{model}}^2 - (s - \nu)$ at generic points of this set, and by lower semicontinuity at all of them. □

So on the open set of Proposition A.23(i) the residual network has functional dimension below $P - d_{\text{model}}^2 - s$. For ReLU networks, the bound of Theorem 4.3 is therefore attained on one open set of weights (Lemma A.20) and missed on another.

Remark A.24 (Units that are always active). A unit that is active at every input with $xE \neq 0$ lowers the bound in the same way, on the open set of weights on which it stays so; a unit of block 2 whose read has positive entries is one, after block 1 of Proposition A.23(i). It adds a linear map of rank one to the skip of its block, which for generic weights is then an invertible learned skip, and by Theorem 3.2 the network equals a residual network without this unit, which has $2d_{\text{model}}$ fewer parameters and one rescaling less. On this open set the Jacobian is that of the smaller network times the derivative of a smooth map of the weights, so its rank is at most $(P - 2d_{\text{model}}) - d_{\text{model}}^2 - (s - 1)$.

Without skips, units also die in ReLU^2 and ReGLU blocks, and at every width.

Proposition A.25 (Dead units without skips)**.** *Consider skipless networks of $L \geq 2$ ReLU, ReLU^2 or ReGLU blocks without normalization. There is a nonempty open set of weights on which a unit of block 2 is dead. On it the functional dimension is at most $P - (L+1)d_{\text{model}}^2 - s - (2d_{\text{model}} - 1)$, and at most $P - (L+1)d_{\text{model}}^2 - s - (3d_{\text{model}} - 2)$ for ReGLU, so the equality condition fails there.*

Proof. Let unit k of block 1 have the read g_k and the write $\hat{r}_k$, and let a unit of block 2 have the read r. For ReGLU, g_k and r are gate reads, and v_k is the value read of unit k.

ReLU and ReLU^2. At $h_0 = u$ the pre-activation of the unit of block 2 is $\sum_k \sigma(ug_k)\,\hat{r}_k r$, with $\sigma \geq 0$. It is positive at no input when $\hat{r}_k r < 0$ for all k, which holds on a nonempty open set of weights.

ReGLU. The pre-activation of the gate is $z(u) = \sum_k \mathrm{ReLU}(ug_k)\,(uv_k)\,\hat{r}_k r$, and the sign of a term depends on the value uv_k. We choose weights at which every gate opens where its value is positive. Let m be the width of block 1 and $0 < \delta < 1/(md_{\text{model}})$. Let $g_k = e_{i(k)}$, a standard basis column, and $v_k = e_{i(k)} - \delta \sum_{j \neq i(k)} e_j$, with every index $i \leq d_{\text{model}}$ used by some unit, and let $\hat{r}_k r = -1$ for all k. Then

$$z(u) = -\sum_k \mathrm{ReLU}(u_{i(k)})\Big(u_{i(k)} - \delta \sum_{j \neq i(k)} u_j\Big) \;\leq\; -(1 - \delta m d_{\text{model}}) \sum_{i\,:\,u_i > 0} u_i^2,$$

so $z(u) < 0$ if u has a positive entry, and $z(u) = 0$ if it has none. Suppose that, at weights that converge to these, the unit is active at unit rows u_n, and let u be a limit of the u_n. Then $z(u) \geq 0$, so u has no positive entry. For large n, a unit k with $u_{i(k)} < 0$ has a closed gate at u_n. A unit k with $u_{i(k)} = 0$ has a positive value at u_n, since its value at u is $-\delta \sum_{j \neq i(k)} u_j > 0$, so its term is not positive. Hence the pre-activation at u_n is not positive, a contradiction. The pre-activation is positively homogeneous in u, so the unit is dead at all weights close to the chosen ones.

The rank. As in the proof of Proposition A.23(ii), the columns of the Jacobian that belong to the dead unit vanish, $2d_{\text{model}}$ of them, or $3d_{\text{model}}$ for ReGLU. The other columns are those of the network without this unit, whose symmetries are the changes of basis of its $L+1$ hidden states and $s-1$ rescalings, or $s-2$ for ReGLU. □

With skips, no unit of a ReLU^2 or ReGLU block with a nonzero read is dead. Such a block is of second order at 0: its Lipschitz constant on a ball of radius ρ is $O(\rho)$. So $h \mapsto h + B(h)$ maps a small ball onto a neighborhood of 0, and every hidden state of a residual network takes all values near 0.

Whether almost every residual ReLU network whose blocks all satisfy $\|W_{\text{read}}\|\,\|W_{\text{write}}\| < 1$ has functional dimension $P - d_{\text{model}}^2 - s$ is open.

B The one-block theorem

This appendix states and proves the general form of Theorem 4.6. With one block, the skipless and the residual network are

$$x \mapsto f(xEW_{\text{read}})W_{\text{write}}D \qquad \text{and} \qquad x \mapsto \big(xE + f(xEW_{\text{read}})W_{\text{write}}\big)D$$

without normalization, and under pre-RMSNorm

$$x \mapsto N\big(f(N(xE)W_{\text{read}})W_{\text{write}}\big)D \qquad \text{and} \qquad x \mapsto N\big(xEA + f(N(xE)W_{\text{read}})W_{\text{write}}\big)D,$$

with the identity skip $A = I$. A learned skip $A \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is full or upper triangular.

Theorem B.1 (One block)**.** *Let $d_{\text{model}} \geq 2$, let the input and output dimensions be at least d_{model}, let inputs range over all of $\mathbb{R}^{T \times d_{\text{in}}}$, and consider networks with one block: a* tanh*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU or GeGLU block, or an attention block of H heads of width $d_h \leq d_{\text{model}}$, causal or not, on sequences of at least two tokens.*

(i) *Let the network have no normalization, with or without biases, and let the block have width at least d_{model}, where the width of an attention block is Hd_h. For generic parameters the skipless*

network has functional dimension $P - 2d_{\text{model}}^2 - s$ and the residual network $P - d_{\text{model}}^2 - s$; with biases on the embedding, the block and the readout, counted in P, both values are $2d_{\text{model}}$ lower, and $2d_{\text{model}} + 2Hd_h$ lower for attention. Both networks satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.
(ii) *Let the network be pre-RMSNorm with $\varepsilon > 0$, without biases. For generic parameters the residual network has functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - s$, which is the equality condition. If the block has width at least d_{model}, the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2$.*
(iii) *Let the network be pre-RMSNorm without biases, with a learned skip A, full or upper triangular. For generic parameters and $\varepsilon > 0$, its functional dimension exceeds that of the residual network of part (ii) by exactly $d_{\text{model}}(d_{\text{model}} + 1)/2$, and it satisfies the equality condition. For $\varepsilon = 0$ it exceeds that of the residual network with $\varepsilon = 0$ by at most $d_{\text{model}}(d_{\text{model}} + 1)/2 - 1$, with equality for ReLU, ReLU^2 and ReGLU blocks.*
(iv) *Let the network be pre-RMSNorm with $\varepsilon = 0$, without biases, with a ReLU, ReLU^2 or ReGLU block of width at least d_{model}. For generic parameters the residual network has functional dimension $P - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$, and the skipless network at most $P - d_{\text{model}}^2 - d_{\text{model}}(d_{\text{model}} - 1)/2 - 1 - s$, with equality for ReLU and ReLU^2 blocks. So the skip raises the functional dimension by at least d_{model}^2, and by exactly d_{model}^2 for ReLU and ReLU^2 blocks. The redundancy is then that of the symmetries: besides those inside the block, $\mathrm{GL}(d_{\text{model}})$ at h_0 and $\mathrm{CO}(d_{\text{model}})$ at h_1 without the skip, and one $\mathrm{CO}(d_{\text{model}})$ with it.*
(v) *Let the network be pre-RMSNorm with $\varepsilon = 0$, without biases, with a $\tanh$, GELU, SwiGLU, GeGLU or attention block. For generic parameters the skip raises the functional dimension by at least $d_{\text{model}}(d_{\text{model}} - 1)/2 + 1$, the dimension of $\mathrm{CO}(d_{\text{model}})$.*

In part (ii), for the skipless network we prove only the bound $P - d_{\text{model}}(d_{\text{model}} - 1) - s$ of Theorem A.13(ii), which our checks attain (Appendix E). With parts (ii) and (iii), a learned skip under RMSNorm with $\varepsilon > 0$ adds at least d_{model}^2 over no skip if the block has width at least d_{model}, as an identity skip does without normalization. Part (i) extends to a learned absolute position embedding, $h_0 = xE + E_{\text{pos}}$, for an attention block without biases, with P counting E_{pos}, and to grouped-query attention (*Extensions of part (i)* below). With rotary embeddings, the change of basis between the queries and the keys of a head shrinks to an abelian group of dimension d_h (Tran et al., 2026, Theorem 4.2), which changes s; we do not treat this case.

Notation and flat directions. We write $F(\theta, x)$ for the output of the network at the parameters $\theta \in \mathbb{R}^P$ and the input x, and $\mathrm{D}_\theta F(\theta, x)$ for the Fréchet derivative of $F(\cdot, x)$ at θ, a linear map on $\mathbb{R}^P$. The derivative is written with an upright D so that it is not confused with the readout matrix D of the network. For a direction $\eta \in \mathbb{R}^P$, $\mathrm{D}_\theta F(\theta, x)\,\eta$ is the derivative of $\tau \mapsto F(\theta + \tau\eta, x)$ at $\tau = 0$. We write η_E, η_D, η_{read}, $\eta_{\text{write}}, \ldots$ for the entries of η at the positions of E, D, W_{read}, $W_{\text{write}}, \ldots$, and $\mathrm{D}_\theta Q\,\eta$ for the derivative along η of a matrix Q that depends on the parameters; for instance $\mathrm{D}_\theta M\,\eta = \eta_E D + E\eta_D$ for $M = ED$, by the product rule. Call η *flat* if $\mathrm{D}_\theta F(\theta, x)\,\eta = 0$ for every input x, and let $\mathcal{K}(\theta)$ be the space of flat directions. For the ReLU-type activations, $F(\theta, x)$ is differentiable in θ at the inputs x at which no hidden unit sits at its kink, which form an open set of full measure, and "every input" means every such input. With $\varepsilon = 0$, it also excludes the inputs at which a state that is normalized vanishes. For $\tanh$, GELU, SwiGLU, GeGLU and attention blocks, $\mathrm{D}_\theta F(\theta, x)\,\eta$ is real-analytic in x, so a direction that is flat on an open set of inputs is flat on all inputs, and the theorem holds for inputs in any open set. For the ReLU-type blocks it does not: on a ball of inputs that meets no kink the network is a polynomial in x. For a batch of inputs $x_1, \ldots, x_n$, the Jacobian of $\theta \mapsto (F(\theta, x_1), \ldots, F(\theta, x_n))$ is the matrix of the linear map $\eta \mapsto (\mathrm{D}_\theta F(\theta, x_1)\,\eta, \ldots, \mathrm{D}_\theta F(\theta, x_n)\,\eta)$; it has P columns, and its kernel consists of the directions with $\mathrm{D}_\theta F(\theta, x_k)\,\eta = 0$ for the inputs of the batch. Adding inputs can only shrink this kernel, and a decreasing chain of subspaces of $\mathbb{R}^P$ stops, so some finite batch has kernel exactly $\mathcal{K}(\theta)$. By rank–nullity, the functional dimension at θ is therefore

$$P - \dim \mathcal{K}(\theta).$$

The theorem thus amounts to finding the flat directions. A statement holds for *generic* parameters if it

fails only where some nonzero polynomial in the parameters, or some nonzero real-analytic function of them, vanishes, a set of measure zero.

Overview. For an MLP block, the parameters enter the function only through $M = ED$, $A = W_{\mathrm{write}}D$ and $C = EW_{\mathrm{read}}$; from part (iii) on, A is the learned skip again. Step 1 finds the directions that keep these matrices fixed: they are the directions in which the changes of basis move the parameters, two of them without the skip and one with it. Step 2 shows that a flat direction changes M, A and C only along the symmetries of the hidden units: each unit is isolated, by its kink for the ReLU-type activations, and for $\tanh$, GELU, SwiGLU and GeGLU by how fast it approaches its limit on rays $x = tn$ with $t \to \infty$. Step 3 counts. For an attention block the same steps run with one factorization per head, and inputs of two tokens turn Step 2 into the case of $\tanh$. For part (ii), large inputs first remove the final normalization and then separate the skip from the block. Part (iii) runs the same steps with a learned skip, which leaves a symmetric change of basis at h_0 in the input of the block; homogeneity or rays remove it.

Part (i) for MLP blocks

Reduction. With biases b_E, b_{read}, b_{write} and b_D, the skipless network computes $F(\theta, x) = f(xC + \beta)A + \gamma_0$ and the residual network $F(\theta, x) = xM + f(xC + \beta)A + \gamma$, where

$$M = ED, \qquad C = EW_{\mathrm{read}}, \qquad A = W_{\mathrm{write}}D,$$
$$\beta = b_E W_{\mathrm{read}} + b_{\mathrm{read}}, \qquad \gamma_0 = b_{\mathrm{write}}D + b_D, \qquad \gamma = b_E D + \gamma_0.$$

Without biases $\beta = 0$ and $\gamma = \gamma_0 = 0$. For a vanilla block let c_i be the columns of C, one per hidden unit, a_i the rows of A, β_i the entries of β, and $z_i = xc_i + \beta_i$, so that $F(\theta, x) = xM + \gamma + \sum_i \sigma(z_i)\, a_i$. For generic parameters E, D, W_{read} and W_{write} have rank d_{model}, every a_i is nonzero, and the c_i are nonzero and pairwise non-collinear: they lie in the column space of E, of dimension $d_{\mathrm{model}} \geq 2$, where two generic vectors are not collinear. For the gated blocks ReGLU, SwiGLU and GeGLU, in the same way, the value weights (e_i, β_i') of each unit, introduced below, have $e_i \neq 0$ and are not proportional to its gate weights (c_i, β_i).

Step 1: changes of basis. We need the following lemma.

Lemma B.2 (The directions that keep M, A and C)**.** *Let E, D, W_{read} and W_{write} have rank d_{model}. A direction η keeps C and A fixed to first order, $\mathrm{D}_\theta C\,\eta = 0$ and $\mathrm{D}_\theta A\,\eta = 0$, if and only if*

$$\eta_E = EX, \qquad \eta_{\mathrm{read}} = -XW_{\mathrm{read}}, \qquad \eta_{\mathrm{write}} = W_{\mathrm{write}}Y, \qquad \eta_D = -YD$$

for some $X, Y \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{model}}}$, and it keeps M fixed as well, $\mathrm{D}_\theta M\,\eta = 0$, if and only if moreover $X = Y$. With biases, it keeps β and γ_0, respectively β and γ, fixed as well if and only if moreover $\eta_{b_E} = b_E X + g_0$, $\eta_{b_{\mathrm{read}}} = -g_0 W_{\mathrm{read}}$, $\eta_{b_{\mathrm{write}}} = b_{\mathrm{write}}Y + g_1$ and $\eta_{b_D} = -g_1 D$, respectively $\eta_{b_D} = -(g_0 + g_1)D$, for some row vectors $g_0, g_1 \in \mathbb{R}^{1 \times d_{\mathrm{model}}}$. The matrices X, Y and vectors g_0, g_1 are determined by the direction, so these directions form a space of dimension $2d_{\mathrm{model}}^2$ without the skip and d_{model}^2 with it, and $2d_{\mathrm{model}}$ more with biases.

Proof. Since W_{read} has rank d_{model}, it has a right inverse R, with $W_{\mathrm{read}}R = I$, and E, of rank d_{model} with d_{model} columns, has a left inverse. From $\mathrm{D}_\theta C\,\eta = \eta_E W_{\mathrm{read}} + E\eta_{\mathrm{read}} = 0$, multiplying on the right by R gives $\eta_E = EX$ with $X = -\eta_{\mathrm{read}}R$; then $E\eta_{\mathrm{read}} = -EXW_{\mathrm{read}}$, and cancelling E on the left gives $\eta_{\mathrm{read}} = -XW_{\mathrm{read}}$. In the same way, D has a right inverse and W_{write} a left inverse, and $\mathrm{D}_\theta A\,\eta = \eta_{\mathrm{write}}D + W_{\mathrm{write}}\eta_D = 0$ gives $\eta_{\mathrm{write}} = W_{\mathrm{write}}Y$ and $\eta_D = -YD$. Conversely these directions give $\mathrm{D}_\theta C\,\eta = EXW_{\mathrm{read}} - EXW_{\mathrm{read}} = 0$ and $\mathrm{D}_\theta A\,\eta = 0$. In the residual network $\mathrm{D}_\theta M\,\eta = \eta_E D + E\eta_D = E(X - Y)D$, which vanishes if and only if $X = Y$, by cancelling E on the left and D on the right. With biases, set $g_0 = \eta_{b_E} - b_E X$ and $g_1 = \eta_{b_{\mathrm{write}}} - b_{\mathrm{write}}Y$. Then $\mathrm{D}_\theta \beta\,\eta = \eta_{b_E}W_{\mathrm{read}} + b_E\eta_{\mathrm{read}} + \eta_{b_{\mathrm{read}}} = g_0 W_{\mathrm{read}} + \eta_{b_{\mathrm{read}}}$, which vanishes exactly when $\eta_{b_{\mathrm{read}}} = -g_0 W_{\mathrm{read}}$, and in the same way $\mathrm{D}_\theta \gamma_0\,\eta = g_1 D + \eta_{b_D}$ and, with $Y = X$, $\mathrm{D}_\theta \gamma\,\eta = (g_0 + g_1)D + \eta_{b_D}$. Finally $\eta_E = EX$ determines X, $\eta_D = -YD$ determines Y, and then η_{b_E} and $\eta_{b_{\mathrm{write}}}$ determine g_0 and g_1. □

These are the directions in which the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1 move the parameters, and, with biases, the translations at the two hidden states (Lemma A.8).

Step 2: one hidden layer. We need a lemma on the function

$$G(\phi, x) = xM + \gamma + \textstyle\sum_i \sigma(z_i)\, a_i, \qquad z_i = xc_i + \beta_i,$$

of the matrices $\phi = \big(M, \gamma, (c_i, \beta_i, a_i)_i\big)$ themselves. For a direction ψ of ϕ, with entries ψ_M, ψ_γ, ψ_{c_i}, ψ_{β_i} and ψ_{a_i}, the Fréchet derivative of $G(\cdot, x)$ at ϕ gives

$$\mathrm{D}_\phi G(\phi, x)\,\psi = x\psi_M + \psi_\gamma + \sum_i \big[\sigma(z_i)\,\psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\,a_i\big]. \tag{9}$$

A gated unit computes $\sigma(z_i)\, w_i\, a_i$ with a gate $z_i = xc_i + \beta_i$ and a value $w_i = xe_i + \beta_i'$, where $\sigma(z) = \max(z, 0)$ for ReGLU, $z/(1 + e^{-z})$ for SwiGLU and $z\Phi(z)$ for GeGLU, so that ϕ also contains the e_i and β_i', and the terms of the unit in $\mathrm{D}_\phi G(\phi, x)\,\psi$ are $\sigma(z_i)\, w_i\, \psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\, w_i\, a_i + \sigma(z_i)\,(x\psi_{e_i} + \psi_{\beta_i'})\, a_i$.

Lemma B.3 (A flat direction only rescales units)**.** *Let every $a_i \neq 0$ and let the c_i be nonzero and pairwise non-collinear; for the gated blocks let also every $e_i \neq 0$, with (e_i, β_i') not proportional to (c_i, β_i). If $\mathrm{D}_\phi G(\phi, x)\,\psi = 0$ for all x, then $\psi_M = 0$, $\psi_\gamma = 0$, and for every unit i:*
for $\tanh$ *and GELU, $\psi_{a_i} = 0$, $\psi_{c_i} = 0$ and $\psi_{\beta_i} = 0$;*
for ReLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$ and $\psi_{a_i} = -\lambda_i a_i$ for some number λ_i;
for ReLU^2, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$ and $\psi_{a_i} = -2\lambda_i a_i$;
for ReGLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$, $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i (e_i, \beta_i')$ and $\psi_{a_i} = (\alpha_i - \lambda_i) a_i$ for some numbers λ_i and α_i;
for SwiGLU and GeGLU, $\psi_{c_i} = 0$, $\psi_{\beta_i} = 0$, $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i (e_i, \beta_i')$ and $\psi_{a_i} = \alpha_i a_i$ for some number α_i.

Proof. Let $H_i = \{x : z_i = 0\}$, a hyperplane since $c_i \neq 0$. Since the c_i are pairwise non-collinear, the H_i are pairwise non-parallel, so every H_i contains points that lie on no other H_k. We use one fact from linear algebra: an affine function that vanishes on an open piece of H_i is a multiple of z_i, since it vanishes on all of H_i and H_i has codimension one. We write $J(x) = \mathrm{D}_\phi G(\phi, x)\,\psi$, a function of x alone, which vanishes by assumption.

ReLU. Here $\sigma' = \mathbf{1}[z > 0]$ jumps at 0. Near a point x_0 of H_i that lies on no other H_k, every term of (9) is continuous except $\mathbf{1}[z_i > 0](x\psi_{c_i} + \psi_{\beta_i})\, a_i$, which jumps by $(x_0\psi_{c_i} + \psi_{\beta_i})\, a_i$ as x crosses H_i at x_0. As $J = 0$ is continuous, this jump vanishes, and since $a_i \neq 0$, the affine function $x\psi_{c_i} + \psi_{\beta_i}$ vanishes on an open piece of H_i, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$. The terms of unit i then add up to $\max(z_i, 0)\,(\psi_{a_i} + \lambda_i a_i)$, since $z\,\mathbf{1}[z > 0] = \max(z, 0)$. Across H_i, the gradient of J with respect to x jumps by the matrix $c_i(\psi_{a_i} + \lambda_i a_i)$, which must vanish, so $\psi_{a_i} = -\lambda_i a_i$. With every unit removed in this way, $J(x) = x\psi_M + \psi_\gamma = 0$ for all x gives $\psi_M = 0$ and $\psi_\gamma = 0$.

ReLU^2. The same argument one derivative higher: $\sigma'(z) = 2\max(z, 0)$ is continuous, and across H_i the gradient of $2\max(z_i, 0)(x\psi_{c_i} + \psi_{\beta_i})\, a_i$ jumps by $2c_i(x_0\psi_{c_i} + \psi_{\beta_i})\, a_i$, which forces $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$. The terms of unit i then add up to $\max(z_i, 0)^2(\psi_{a_i} + 2\lambda_i a_i)$, whose second derivative jumps across H_i, so $\psi_{a_i} = -2\lambda_i a_i$.

ReGLU. As (e_i, β_i') is nonzero and not proportional to (c_i, β_i), the affine function w_i does not vanish on all of H_i, so it is nonzero on an open piece of H_i that meets no other H_k; we take x_0 there. Across H_i the terms of unit i jump by $(x_0\psi_{c_i} + \psi_{\beta_i})\, w_i(x_0)\, a_i$, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i (c_i, \beta_i)$ as before. The terms of unit i then add up to $\max(z_i, 0)\, V_i(x)$ with the affine function $V_i(x) = w_i(x)\,(\psi_{a_i} + \lambda_i a_i) + (x\psi_{e_i} + \psi_{\beta_i'})\, a_i$. The jump of their gradient across H_i is $c_i V_i(x_0)$, so V_i vanishes on an open piece of H_i, and $V_i(x) = z_i\, \nu_i$ for a row vector ν_i. The terms of unit i then add up to $\max(z_i, 0)^2\, \nu_i$, whose second derivative jumps across H_i, so $\nu_i = 0$ and $V_i = 0$ for all x: $w_i(x)\,(\psi_{a_i} + \lambda_i a_i) = -(x\psi_{e_i} + \psi_{\beta_i'})\, a_i$. Both sides are products of an affine function of x and a fixed row vector, and $a_i \neq 0$, so $\psi_{a_i} + \lambda_i a_i = \alpha_i a_i$ and $(\psi_{e_i}, \psi_{\beta_i'}) = -\alpha_i (e_i, \beta_i')$ for some number α_i. Then $\psi_M = 0$ and $\psi_\gamma = 0$ as for ReLU.

$\tanh$. We restrict (9) to a ray $x = tn$, $t > 0$, with a direction n such that the numbers $u_k = nc_k$ are all nonzero and $|u_j| < |u_k|$ for one unit j and all $k \neq j$. For each j and each sign $s = \pm 1$, the directions n with $su_j > 0$ and this property form a nonempty open set: start from a point of the hyperplane $nc_j = 0$ where no other nc_k vanishes, and move a little to the side s. For $z > 0$,

$$\tanh z = 1 - 2e^{-2z} + r(z), \quad 0 \le r(z) \le 2e^{-4z},$$
$$\tanh' z = 4e^{-2z} - q(z), \quad 0 \le q(z) \le 8e^{-4z},$$

from $\tanh z = (1 - e^{-2z})/(1 + e^{-2z})$ and $\tanh' = 1 - \tanh^2$, and $\tanh$ is odd. On the ray $z_k = u_k t + \beta_k$, so for large t the terms of unit k in (9) are

$$s_k \psi_{a_k} + e^{-2|u_k|t}\, e^{-2s_k\beta_k} \big[-2s_k \psi_{a_k} + 4\,(t\, n\psi_{c_k} + \psi_{\beta_k})\, a_k \big] + O\big((1+t)\, e^{-4|u_k|t}\big),$$

with s_k the sign of u_k. All terms of $J(tn)$ are therefore $t\, n\psi_M$, a constant, the bracket of unit j times $e^{-2|u_j|t}$, and terms that are $O\big((1+t)e^{-\rho t}\big)$ with $\rho = \min\big(4|u_j|, \min_{k\neq j} 2|u_k|\big) > 2|u_j|$. As $J(tn) = 0$ for all t, dividing by t and letting $t \to \infty$ gives $n\psi_M = 0$; then the constant vanishes; then multiplying by $e^{2|u_j|t}$ and letting $t \to \infty$ shows that the bracket of unit j, a polynomial of degree one in t, tends to zero, so both of its coefficients vanish: $(n\psi_{c_j})\, a_j = 0$ and $2\psi_{\beta_j} a_j = s\, \psi_{a_j}$. As $a_j \neq 0$, $n\psi_{c_j} = 0$ for all n in an open set, so $\psi_{c_j} = 0$; and the two signs $s = \pm 1$ give $\psi_{a_j} = 2\psi_{\beta_j} a_j = -2\psi_{\beta_j} a_j$, so $\psi_{\beta_j} = 0$ and $\psi_{a_j} = 0$. This holds for every unit j, and then $J(x) = x\psi_M + \psi_\gamma = 0$ gives $\psi_M = 0$ and $\psi_\gamma = 0$.

GELU. Here $\sigma(z) = z\Phi(z)$, with Φ the standard normal distribution function and $\varphi = \Phi'$ its density. The tail $\bar\Phi = 1 - \Phi$ satisfies $\varphi(z)(1/z - 1/z^3) \le \bar\Phi(z) \le \varphi(z)/z$ for $z > 0$, as one checks by differentiating, and $\Phi(-z) = 1 - \Phi(z)$ carries these bounds to $z < 0$. From these, as $|z| \to \infty$ on either side,

$$\sigma(z) = \max(z, 0) - \varphi(z) + O\big(\varphi(z)/z^2\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + \varphi(z)\,(z - 1/z) + O\big(\varphi(z)/|z|^3\big).$$

Take a ray $x = tn$ as for $\tanh$. The terms of unit k in (9) split into a polynomial of degree at most one in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and the Gaussian part

$$\varphi(z_k) \Big[u_k\, (n\psi_{c_k})\, t^2 + \big(u_k \psi_{\beta_k} + \beta_k\, n\psi_{c_k}\big)\, t + \beta_k \psi_{\beta_k} - (n\psi_{c_k})/u_k \Big] a_k \\ - \ \varphi(z_k)\, \psi_{a_k} \ + \ O\big(\varphi(z_k)/t\big).$$

Gaussian parts tend to zero faster than any power of t, so as $J(tn) = 0$ for all t, the polynomial part vanishes, and so does the sum of the Gaussian parts. For $k \neq j$, $\varphi(z_k)/\varphi(z_j) = \exp\big(-\frac{1}{2}(u_k^2 - u_j^2)\, t^2 + O(t)\big)$ tends to zero faster than any power of t, since $u_k^2 > u_j^2$. Dividing the sum of the Gaussian parts by $\varphi(z_j)$ and letting $t \to \infty$ therefore makes the coefficients of t^2, t and 1 in the bracket of unit j vanish, in this order: $u_j (n\psi_{c_j})\, a_j = 0$, so $n\psi_{c_j} = 0$; then $u_j \psi_{\beta_j}\, a_j = 0$, so $\psi_{\beta_j} = 0$; then $\psi_{a_j} = 0$. As for $\tanh$, $n\psi_{c_j} = 0$ on an open set gives $\psi_{c_j} = 0$, and finally $\psi_M = 0$ and $\psi_\gamma = 0$.

SwiGLU and GeGLU. Take a ray $x = tn$ as for $\tanh$, and write $v_k = ne_k$, so that $w_k = v_k t + \beta'_k$; as $e_j \neq 0$, also $v_j \neq 0$ on an open set of such n. For SwiGLU, $\sigma(z) = z - z/(1 + e^z)$ gives, as $|z| \to \infty$ with s the sign of z,

$$\sigma(z) = \max(z, 0) - |z| e^{-|z|} + O\big(|z| e^{-2|z|}\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + s\,(|z| - 1)\, e^{-|z|} + O\big(|z| e^{-2|z|}\big).$$

The terms of unit k in (9) then split into a polynomial in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and a rest that is $e^{-|z_k|}$ times a polynomial in t, up to $O(t^3 e^{-2|z_k|})$; for GeGLU, with the expansions of GELU, the rest is $\varphi(z_k)$ times a polynomial in t and $1/z_k$, up to $O(\varphi(z_k)/t)$. As for $\tanh$ and GELU, the polynomial parts vanish, and dividing the sum of the rests by $e^{-|z_j|}$, respectively

$\varphi(z_j)$, and letting $t \to \infty$ shows that the rest of the slowest unit j vanishes. For SwiGLU this rest is $e^{-|z_j|}$ times

$$s\,(|z_j|-1)\,(t\,n\psi_{c_j}+\psi_{\beta_j})\,w_j\,a_j - |z_j|\big((t\,n\psi_{e_j}+\psi_{\beta'_j})\,a_j + w_j\,\psi_{a_j}\big),$$

with s the sign of u_j and $|z_j| = |u_j|t + s\beta_j$, a polynomial of degree three in t, whose coefficients therefore vanish. The coefficient of t^3 is $s|u_j|\,(n\psi_{c_j})\,v_j\,a_j$, so $\psi_{c_j} = 0$ as for $\tanh$. The coefficient of t^2 is then $-|u_j|\big((n\psi_{e_j})\,a_j + v_j(\psi_{a_j} - s\psi_{\beta_j}a_j)\big)$. As it vanishes for all n in an open set, $\psi_{e_j} = -\alpha e_j$ and $\psi_{a_j} - s\psi_{\beta_j}a_j = \alpha a_j$ for a number α, the same for both signs since ψ_{e_j} is; the two signs then give $\psi_{\beta_j} = 0$. The coefficient of t is then $-|u_j|(\psi_{\beta'_j} + \alpha\beta'_j)\,a_j$, so $\psi_{\beta'_j} = -\alpha\beta'_j$. For GeGLU the rest of unit j is $\varphi(z_j)$ times the same bracket with $z_j - 1/z_j$ in place of $s(|z_j|-1)$ and 1 in place of $|z_j|$, and its coefficients of t^3, t^2, t and 1 give, in this order, $\psi_{c_j} = 0$, $\psi_{\beta_j} = 0$, $(\psi_{e_j}, \psi_{a_j}) = (-\alpha e_j, \alpha a_j)$ and $\psi_{\beta'_j} = -\alpha\beta'_j$. In both cases the terms of unit j then vanish; this holds for every unit, and $\psi_M = 0$ and $\psi_\gamma = 0$ as for $\tanh$. □

The directions left by the lemma are exactly those of the rescalings of the units: for ReLU, scaling the input weights and the bias of unit i by $1+\tau\lambda_i$ and its output weights by $1-\tau\lambda_i$ changes the function only at second order in τ.

Step 3: counting. Let η be flat. The network depends on θ only through ϕ, its M, γ and the c_i, β_i and a_i formed from C, β and A, so by the chain rule $\mathrm{D}_\theta F(\theta,x)\,\eta = \mathrm{D}_\phi G(\phi,x)\,(\mathrm{D}_\theta\phi\,\eta)$, and Lemma B.3 applies to $\psi = \mathrm{D}_\theta\phi\,\eta$. The derivatives of M, A, C, β and γ along η are thus those of a rescaling of the hidden units. A rescaling of the units by numbers λ_i is also a direction in parameter space: it scales column i of W_{read} and entry i of b_{read} by λ_i and row i of W_{write} by $-\lambda_i$ for ReLU, by $-2\lambda_i$ for ReLU^2, and similarly for ReGLU and for the values of SwiGLU and GeGLU, and along it C, β and A have the same derivatives. Subtracting it from η leaves a direction along which M, A, C, β and γ have derivative zero, which Lemma B.2 describes. So every flat direction is the sum of a direction of the changes of basis and translations of Lemma B.2 and a rescaling of the units, and conversely all of these are flat, since they come from symmetries. The two kinds are independent: a rescaling that is also a direction of Lemma B.2 has $\eta_E = EX = 0$, so $X = 0$, and then $W_{\mathrm{read}}\,\mathrm{diag}(\lambda) = 0$, so every $\lambda_i = 0$, because no column of W_{read} vanishes. Hence $\dim\mathcal{K}(\theta) = 2d_{\mathrm{model}}^2 + s$ without the skip and $d_{\mathrm{model}}^2 + s$ with it, with $2d_{\mathrm{model}}$ more for each with biases, where s is the number of rescalings: m for ReLU, ReLU^2, SwiGLU and GeGLU, $2m$ for ReGLU, none for $\tanh$ and GELU. The functional dimension $P - \dim\mathcal{K}(\theta)$ has the values of the theorem. The symmetries account for all of $\mathcal{K}(\theta)$, which is the equality condition, and the difference between the two networks is d_{model}^2.

Part (i) for attention blocks

Reduction. Let $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\mathrm{model}}\times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h\times d_{\mathrm{model}}}$ be the projections of head i, and let

$$M = ED, \qquad B_i = EW_{Q,i}W_{K,i}^\top E^\top/\sqrt{d_h}, \qquad C_i = EW_{V,i}W_{O,i}D, \qquad S = \textstyle\sum_i C_i,$$

with M only in the residual network. Head i gives token t the scores $x_tB_ix_s^\top$, and the output at token t is $x_tM + \sum_i\sum_s a_{ts}^{(i)}\,x_sC_i$, so the parameters enter only through these matrices. We use inputs of two kinds: with a causal mask, any sequence, whose second token sees only the first two; without a mask, sequences whose tokens other than the second all equal the first. Let k be the number of tokens other than the second that the second token sees, $k = 1$ with a mask and $k = T-1$ without. With $u = x_2$ and $v = x_2 - x_1$, head i puts the weight $1/(1+k\,e^{-uB_iv^\top})$ on the second token and the rest on the copies of the first, and $1/(1+e^{-z}) = \frac12 + \frac12\tanh(z/2)$ gives the output at the second token

$$y_2 = uK - \tfrac12\,vS + \tfrac12\sum_i \tanh\big(\tfrac12(uB_iv^\top - \log k)\big)\,vC_i, \qquad K = M+S. \tag{10}$$

With biases b_E on the embedding, $b_{Q,i}$, $b_{K,i}$, $b_{V,i}$ on the queries, keys and values, b_O on the output projection and b_D on the readout, $uB_iv^\top$ becomes $(uB_i+\beta_i)v^\top$ with $\beta_i = (b_EW_{Q,i} + b_{Q,i})W_{K,i}^\top E^\top/\sqrt{d_h}$, and the constant $\gamma = \sum_i (b_EW_{V,i}+b_{V,i})W_{O,i}D + b_OD + b_D$ is added, plus b_ED in the residual network. The key biases drop out, since they shift all scores of a token equally, and the value biases enter only through γ, since the weights of a token sum to one.

Step 1: the directions that keep M, B_i and C_i. For generic parameters E and D have rank d_{model}; every $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}^\top$ has rank d_h, since $d_h \le d_{\text{model}}$; and the queries $[W_{Q,1},\dots,W_{Q,H}] \in \mathbb{R}^{d_{\text{model}}\times Hd_h}$, the keys $[W_{K,1},\dots,W_{K,H}]$ and the output projection $W_O \in \mathbb{R}^{Hd_h\times d_{\text{model}}}$ of all heads have rank d_{model}, since $Hd_h \ge d_{\text{model}}$. Write $\eta_{Q,i}$, $\eta_{K,i}$, $\eta_{V,i}$ and $\eta_{O,i}$ for the entries of a direction η at the positions of $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}$, and consider a direction with $\mathrm{D}_\theta M\,\eta = 0$ and, for every head, $\mathrm{D}_\theta B_i\,\eta = 0$ and $\mathrm{D}_\theta C_i\,\eta = 0$. Multiplying $\mathrm{D}_\theta B_i\,\eta = 0$ on the left by the projection $I - EE^+$ onto the orthogonal complement of the column space of E, where E^+ is the pseudo-inverse, leaves $(I-EE^+)\,\eta_E\,W_{Q,i}W_{K,i}^\top E^\top = 0$. As $W_{K,i}^\top E^\top$ has full row rank, $(I-EE^+)\,\eta_E\,W_{Q,i} = 0$ for every head, and as the queries of all heads have rank d_{model}, $\eta_E = EX$ with $X = E^+\eta_E$. In the same way, multiplying $\mathrm{D}_\theta C_i\,\eta = 0$ on the right by the projection $I - D^+D$ gives $\eta_D = -YD$ with $Y = -\eta_D\,D^+$, because $EW_{V,i}$ has full column rank and the output projection has rank d_{model}. Substituting, and cancelling E on the left and $E^\top$ or D on the right,

$$(\eta_{Q,i} + XW_{Q,i})W_{K,i}^\top + W_{Q,i}(\eta_{K,i}+XW_{K,i})^\top = 0,$$
$$(\eta_{V,i}+XW_{V,i})W_{O,i} + W_{V,i}(\eta_{O,i} - W_{O,i}Y) = 0.$$

If P_1 and P_2 have full column rank and $Z_1P_2^\top + P_1Z_2^\top = 0$, then multiplying on the right by a right inverse of $P_2^\top$ gives $Z_1 = P_1\Gamma$ for a square matrix Γ, and then $Z_2 = -P_2\Gamma^\top$. Hence $\eta_{Q,i} = -XW_{Q,i} + W_{Q,i}\Gamma_i$, $\eta_{K,i} = -XW_{K,i} - W_{K,i}\Gamma_i^\top$, $\eta_{V,i} = -XW_{V,i} + W_{V,i}\Gamma_i'$ and $\eta_{O,i} = W_{O,i}Y - \Gamma_i'W_{O,i}$: the directions of the changes of basis $I+\tau X$ at h_0 and $I+\tau Y$ at h_1, and of $I+\tau\Gamma_i$ between the queries and keys and $I+\tau\Gamma_i'$ between the values and outputs of head i. In the residual network $\mathrm{D}_\theta M\,\eta = E(X-Y)D = 0$ gives $X = Y$. These directions are independent: $EX = 0$ forces $X = 0$, $YD = 0$ forces $Y = 0$, and then $W_{Q,i}\Gamma_i = 0$ and $W_{V,i}\Gamma_i' = 0$ force $\Gamma_i = \Gamma_i' = 0$. With biases, $\mathrm{D}_\theta\beta_i\,\eta = 0$ gives $\eta_{b_{Q,i}} = b_{Q,i}\Gamma_i - gW_{Q,i}$ with $g = \eta_{b_E} - b_EX$, since $W_{K,i}^\top E^\top$ has full row rank, and $\mathrm{D}_\theta\gamma\,\eta = 0$ determines η_{b_D}. This leaves g, η_{b_O} and the entries of η at the key and value biases free, $2d_{\text{model}} + 2Hd_h$ dimensions, which the translations at h_0 and h_1, the shifts of the key biases and the value biases moved into the output bias span.

Step 2: two tokens. For generic parameters and generic v the vectors $B_iv^\top$ are nonzero and pairwise non-collinear, since $d_{\text{model}} \ge 2$, and every vC_i is nonzero. For such a v, (10) is, as a function of u, a residual network with one hidden layer of $\tanh$ units, one per head, with input weights $\frac{1}{2}B_iv^\top$, biases $\frac{1}{2}(\beta_iv^\top - \log k)$, output weights $\frac{1}{2}vC_i$, linear part K and constant term $\gamma - \frac{1}{2}vS$. Along a flat direction η these have the derivatives $\frac{1}{2}(\mathrm{D}_\theta B_i\,\eta)v^\top$, $\frac{1}{2}(\mathrm{D}_\theta\beta_i\,\eta)v^\top$, $\frac{1}{2}v(\mathrm{D}_\theta C_i\,\eta)$, $\mathrm{D}_\theta K\,\eta$ and $\mathrm{D}_\theta\gamma\,\eta - \frac{1}{2}v(\mathrm{D}_\theta S\,\eta)$, and Lemma B.3 for $\tanh$ shows that all of these vanish. This holds for all v in an open set, so $\mathrm{D}_\theta B_i\,\eta = 0$, $\mathrm{D}_\theta\beta_i\,\eta = 0$ and $\mathrm{D}_\theta C_i\,\eta = 0$; hence $\mathrm{D}_\theta S\,\eta = 0$, $\mathrm{D}_\theta M\,\eta = \mathrm{D}_\theta K\,\eta - \mathrm{D}_\theta S\,\eta = 0$ and $\mathrm{D}_\theta\gamma\,\eta = 0$.

Step 3: counting. Every direction of Step 1 is flat, since it comes from a symmetry, and by Step 2 every flat direction, already on the inputs of (10), is one of them. So $\dim\mathcal{K}(\theta)$ is $2d_{\text{model}}^2 + 2Hd_h^2$ without the skip and $d_{\text{model}}^2 + 2Hd_h^2$ with it, and $2d_{\text{model}} + 2Hd_h$ more with biases.

Part (ii)

Write $h_1 = xE + g(x)$, with $g(x) = f(N(xE)W_{\text{read}})W_{\text{write}}$ the output of the block, and $y = N(h_1)D$, with N applied to every token; for attention, x has T tokens, and for an MLP block one suffices. Let η be flat, and write $\mathrm{D}_\theta h_1\,\eta$ and $\mathrm{D}_\theta y\,\eta$ for the derivatives of h_1 and y along η at a given input, and $\mathrm{D}N(h)$ for the Fréchet derivative of N at h.

Step 1: the final normalization. Split $\eta_D = -YD + Z$, where $Y \in \mathbb{R}^{d_{\text{model}}\times d_{\text{model}}}$ and the rows of Z are orthogonal to the row space of D, which has dimension d_{model}. At every token, by the chain

rule,

$$0 \;=\; \mathrm{D}_\theta y\,\eta \;=\; \big(\mathrm{D}N(h_1)\,(\mathrm{D}_\theta h_1\,\eta) - N(h_1)Y\big)D + N(h_1)Z,$$

and the two terms lie in complementary subspaces, so both vanish. As x varies, the rows $N(h_1)$ span $\mathbb{R}^{d_{\mathrm{model}}}$, since at large inputs they approach $\sqrt{d_{\mathrm{model}}}\,xE/\|xE\|$ and xE takes every direction, so $Z = 0$; and since D has full row rank, $\mathrm{D}N(h_1)\,(\mathrm{D}_\theta h_1\,\eta) = N(h_1)Y$. With $\rho^2 = \|h\|^2/d_{\mathrm{model}} + \varepsilon$, the derivative of N at h is $\mathrm{D}N(h)\,k = k/\rho - h\,\langle h, k\rangle/(d_{\mathrm{model}}\rho^3)$, which vanishes only for $k = 0$ when $\varepsilon > 0$, and the unique $\mathrm{D}_\theta h_1\,\eta$ with $\mathrm{D}N(h_1)\,(\mathrm{D}_\theta h_1\,\eta) = N(h_1)Y$ is

$$\mathrm{D}_\theta h_1\,\eta = h_1 Y + \frac{h_1 Y_s h_1^\top}{d_{\mathrm{model}}\varepsilon}\,h_1, \qquad Y_s = \tfrac{1}{2}(Y + Y^\top),$$

as one checks by substituting $k = h_1 Y + \mu h_1$ and solving for the number μ.

Step 2: large inputs. Fix w with $wE \neq 0$, for attention one such row for every token, and let $x = tw$ with $t \to \infty$. The block reads $N(twE)$, which stays in the ball of radius $\sqrt{d_{\mathrm{model}}}$, so $g(tw)$ stays bounded, and for attention the scores are bounded too. The derivative of $g(tw)$ along η is bounded as well: by the formula of Step 1, $\|\mathrm{D}N(h)\,k\| \le 2\sqrt{d_{\mathrm{model}}}\,\|k\|/\|h\|$, which is $O(1)$ for $h = twE$ and $k = tw\eta_E$. Hence $\mathrm{D}_\theta h_1\,\eta = t\,w\eta_E + O(1)$ at the input tw, while $h_1(tw) = t\,wE + O(1)$ and the right-hand side of Step 1 is $t^3\,(wEY_sE^\top w^\top)\,wE/(d_{\mathrm{model}}\varepsilon) + O(t^2)$. So $wEY_sE^\top w^\top = 0$ for all w in an open set, hence $EY_sE^\top = 0$, and $Y_s = 0$ because E has rank d_{model}. The matrix Y is antisymmetric, and $\mathrm{D}_\theta h_1\,\eta = h_1 Y$.

Step 3: the skip. An antisymmetric Y is the derivative at $\tau = 0$ of the orthogonal changes of basis $I + \tau Y + O(\tau^2)$ that the residual network shares between h_0 and h_1 (Lemma A.2). They define a direction η^Y with the entries EY, $-YW_{\mathrm{read}}$, $W_{\mathrm{write}}Y$ and $-YD$ at E, W_{read}, W_{write} and D, along which the stream h_1 has the derivative $h_1 Y$. The direction $\eta' = \eta - \eta^Y$ is flat, with $\eta'_D = 0$ and $\mathrm{D}_\theta h_1\,\eta' = 0$ at every input. At the inputs tw, $\mathrm{D}_\theta h_1\,\eta' = t\,w\eta'_E + O(1)$, so $w\eta'_E = 0$ for all w in an open set, and $\eta'_E = 0$.

Step 4: the block. With $\eta'_E = 0$, the condition $\mathrm{D}_\theta h_1\,\eta' = 0$ says that the derivative of $n \mapsto f(nW_{\mathrm{read}})W_{\mathrm{write}}$ along the entries of η' at the weights of the block vanishes for every n in the image of $x \mapsto N(xE)$, which is the open ball of radius $\sqrt{d_{\mathrm{model}}}$, since E has rank d_{model} and N maps $\mathbb{R}^{d_{\mathrm{model}}}$ onto that ball for $\varepsilon > 0$. For $\tanh$, GELU, SwiGLU, GeGLU and attention this derivative is, along every line through the ball, a function of one variable given by a convergent power series around every point; it vanishes on an interval, hence on the whole line, so it vanishes for all inputs. For the ReLU-type activations without biases the kinks lie on hyperplanes through the center of the ball, and the arguments of Lemma B.3 are local. So Lemma B.3, with E and D replaced by the identity and $M = 0$, makes the entries of η' at each hidden unit a rescaling. For attention, each token $N(x_tE)$ ranges over the ball, and Step 2 for attention blocks, with E and D replaced by the identity and $M = 0$, keeps every $W_{Q,i}W_{K,i}^\top$ and $W_{V,i}W_{O,i}$ fixed to first order; by Step 1 there, η' is a direction of the changes of basis inside the heads.

Step 5: counting. The flat directions are thus those of the orthogonal change of basis and of the symmetries inside the block. An antisymmetric Y with $EY = 0$ vanishes, since E has rank d_{model}, and the symmetries inside the block move generic parameters in independent directions, so $\dim \mathcal{K}(\theta) = d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + s$. If the block has width at least d_{model}, the skipless network with the same shapes has functional dimension at most $P - d_{\mathrm{model}}(d_{\mathrm{model}} - 1) - s$ by Theorem A.13(ii), Lemma A.14, so the skip adds at least $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$.

Part (iii)

Write $B(n) = f(nW_{\mathrm{read}})W_{\mathrm{write}}$ for the block as a function of its input n and of its weights, applied to every token for attention, so that $h_1 = xEA + g(x)$ with $g(x) = B(N(xE))$, and $y = N(h_1)D$. For a direction ψ of the weights of the block, $\partial_\psi B(n)$ is the derivative of $B(n)$ along ψ at fixed n, and $\mathrm{D}B(n)\,k$ is the derivative of B at n along k at fixed weights. Let η be flat. For an upper triangular A, view the network as the network with a full A restricted to triangular skips: its flat directions are those of the full network whose entry η_A is upper triangular.

Steps 1 and 2 of part (ii) run with EA in place of E, which has rank d_{model} since A is invertible for generic parameters: $\eta_D = -YD$ with Y antisymmetric, and $\mathrm{D}_\theta h_1\,\eta = h_1 Y$ at every input.

Step 3: the skip. At the inputs tw of Step 2, $\mathrm{D}_\theta h_1\,\eta = t\,w(\eta_E A + E\eta_A) + O(1)$ and $h_1 Y = t\,wEAY + O(1)$, so $\eta_E A + E\eta_A = EAY$. Multiplying on the left by $I - EE^+$ gives $(I - EE^+)\,\eta_E = 0$, as A is invertible, so $\eta_E = EX$ with $X = E^+\eta_E$, and then $\eta_A = AY - XA$. At every input $\mathrm{D}_\theta h_1\,\eta = xEAY + \mathrm{D}_\theta g\,\eta$ and $h_1 Y = xEAY + g(x)Y$, so $\mathrm{D}_\theta g\,\eta = g(x)Y$. By the formula for $\mathrm{D}N$ in Step 1 of part (ii), which gives the same for $\varepsilon = 0$, the input $n = N(xE)$ of the block has the derivative

$$\mathrm{D}N(xE)\,(xEX) = nX - q(n)\,n, \qquad q(n) = nX_s n^\top / d_{\text{model}}, \quad X_s = \tfrac{1}{2}(X + X^\top).$$

Step 4: the symmetric part of X. Write $X = X_a + X_s$ with X_a antisymmetric. The rotations of h_0 and h_1 with the derivatives X_a and Y are symmetries. They define a flat direction η^{rot} with the entries EX_a at E, $AY - X_a A$ at A, $-X_a W_{\text{read}}$ at W_{read}, $W_{\text{write}} Y$ at W_{write} and $-YD$ at D. The flat direction $\eta' = \eta - \eta^{\text{rot}}$ has the entries EX_s, $-X_s A$ and 0 at E, A and D, so Step 3 applies to it with X_s in place of X and 0 in place of Y, and $\mathrm{D}_\theta g\,\eta' = 0$. The change nX_s of the input moves the block as the entries $X_s W_{\text{read}}$ at W_{read} do; for attention, X_s moves the queries, keys and values in the same way. So, with ψ the entries of η' at the weights of the block plus these,

$$\partial_\psi B(n) = q(n)\,\mathrm{D}B(n)\,n \tag{11}$$

for every n in the image of N, the open ball of radius $\sqrt{d_{\text{model}}}$. For attention, $q(n)\,n$ stands for the matrix with the rows $q(n_t)\,n_t$, one per token.

We show that $q = 0$, hence $X_s = 0$. For ReLU, ReLU^2 and ReGLU, $B(\lambda n) = \lambda^k B(n)$ for $\lambda > 0$, with $k = 1$ or 2, so the right side of (11) is $k\,q(n)B(n)$. Replacing n by λn with $0 < \lambda \le 1$ multiplies the left side by λ^k and the right side by λ^{k+2}, so both vanish, and $q = 0$ wherever $B \neq 0$. For the other blocks both sides are analytic in n, so (11) holds for all n, and we take $n = t\hat{n}$ with a fixed row $\hat{n}$ and $t \to \infty$, as in the proof of Lemma B.3. For GELU the right side is $t^3 q(\hat{n}) \max(\hat{n}W_{\text{read}}, 0) W_{\text{write}}$ up to terms that decay faster than any power of t, and the left side is a polynomial of degree at most one in t up to such terms, so $q(\hat{n}) \max(\hat{n}W_{\text{read}}, 0) W_{\text{write}} = 0$. SwiGLU and GeGLU units $\sigma(nc_i)\,(ne_i)\,a_i$ give $2t^4 q(\hat{n}) \sum_{i:\,\hat{n}c_i > 0} (\hat{n}c_i)(\hat{n}e_i)\,a_i$ against a polynomial of degree two. For $\tanh$ the right side is $4t^3 q(\hat{n}) \sum_i (\hat{n}c_i)\,e^{-2|\hat{n}c_i|t} a_i$ up to terms $O(t^3 e^{-4|\hat{n}c_i|t})$; for the unit j with the smallest $|\hat{n}c_j|$, the left side contributes a polynomial of degree one times $e^{-2|\hat{n}c_j|t}$ (proof of Lemma B.3), so $q(\hat{n})(\hat{n}c_j)\,a_j = 0$. For attention, on the two-token inputs of Step 2 for attention blocks with $u = t\hat{n}$ and v fixed, the right side is $t^3 q(\hat{n})\,\hat{n}S + O(t^2)$ and the left side is $O(t)$. In every case q vanishes on an open set for generic parameters, so $X_s = 0$.

Step 5: counting. With $X_s = 0$, the direction η' moves only the weights of the block, and by (11) $B(n)$ has derivative zero along it on the whole ball. So η' lies in the space $\mathcal{K}_B$ of such directions, which is the space found in Step 4 of part (ii), and every direction of $\mathcal{K}_B$ is flat. Hence $\mathcal{K}(\theta)$ is spanned by the directions η^{rot} and $\mathcal{K}_B$, as for the residual network with its single rotation, and for generic parameters of the block $\mathcal{K}_B$ has the same dimension in both networks. The directions η^{rot} are independent of $\mathcal{K}_B$ and of each other, since $\eta_E = EX_a$ and $\eta_D = -YD$ determine X_a and Y. For a full A they have dimension $d_{\text{model}}(d_{\text{model}} - 1)$. For an upper triangular A, $AY - X_a A$ must be upper triangular, which determines Y from X_a: for generic A the map from antisymmetric Y to the entries of AY below the diagonal is invertible, as it is for $A = I$. They then have dimension $d_{\text{model}}(d_{\text{model}} - 1)/2$. So $\mathcal{K}(\theta)$ is larger than for the residual network by $d_{\text{model}}(d_{\text{model}} - 1)/2$, respectively 0, while the network has d_{model}^2, respectively $d_{\text{model}}(d_{\text{model}} + 1)/2$, more parameters. In both cases the functional dimension is larger by $d_{\text{model}}(d_{\text{model}} + 1)/2$. For the blocks of the theorem, $\mathcal{K}_B$ consists of the symmetries inside the block (Step 4 of part (ii)), so the equality condition holds.

The case $\varepsilon = 0$. Now N maps onto the sphere of radius $\sqrt{d_{\text{model}}}$, and the scalings of h_0 and h_1 are symmetries too, since $N(ch) = N(h)$ for $c > 0$. In Step 1, $\mathrm{D}N(h)$ vanishes along h, so $\mathrm{D}_\theta h_1\,\eta = h_1 Y + \mu\,h_1$ with a number μ at each input. Y is again antisymmetric, since $N(h_1)Y$ must be orthogonal to h_1, which is $t\,wEA + O(1)$ at the inputs tw. At these inputs g and its derivative along η do not depend on t, since $N(twE) = N(wE)$. The terms of order t give

$w\Lambda = \mu_w\, wEA$ with $\Lambda = \eta_E A + E\eta_A - EAY$ and a number μ_w, for all w in an open set. Adding to w a vector w_0 with $w_0 E = 0$ leaves wEA unchanged, so $w_0\Lambda$ is a multiple of wEA for every such w, hence $w_0\Lambda = 0$, and $\Lambda = E\Lambda'$ with $\Lambda' = E^+\Lambda$. Every vector $v = wE$ of an open set is then a left eigenvector of $\Lambda' A^{-1}$, which is therefore cI for a number c, as in the proof of Lemma A.2: $\eta_E A + E\eta_A = EA(Y + cI)$. The constant terms then give $\mu = c$ wherever wEA and $g(w)$ are linearly independent. Steps 3 and 4 run with $Y + cI$ in place of Y and with $X_s = c_0 I + X_0$, where X_0 has trace zero; the scalings, with the derivatives c_0 at h_0 and c at h_1, join the rotations in η^{rot}; and (11) holds with X_0 in place of X_s on the sphere. For the residual network the same steps give $\eta_E = E(Y + cI)$, so $X_s = cI$ and $X_0 = 0$. Its flat directions are the rotation and the scaling of all hidden states, and the space $\mathcal{K}_B$ of directions of the block along which B has derivative zero on the sphere. For the learned skip, η^{rot} and $\mathcal{K}_B$ are flat, which gives at least $d_{\mathrm{model}}(d_{\mathrm{model}} - 1) + 2$, respectively $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 2$, dimensions besides $\mathcal{K}_B$, against $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$ for the residual network. So the difference is at most $d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2 - 1$. For ReLU, ReLU^2 and ReGLU, both sides of (11) are positively homogeneous, of degrees k and $k + 2$, so $\|n\|^2\, \partial_\psi B(n) = k d_{\mathrm{model}}\, q(n) B(n)$ for all $n \neq 0$. On each region where no nc_k changes sign, both sides are polynomials in n. Across the hyperplane $nc_i = 0$, at a point of no other such hyperplane, only the terms of unit i change, so they satisfy the identity on their own: $\|n\|^2$ times a polynomial equals $k d_{\mathrm{model}}\, q(n)$ times $(nc_i)\, a_i$, $(nc_i)^2 a_i$ or $(nc_i)(ne_i)\, a_i$. Since $\|n\|^2$ has no real linear factor, it divides $q(n)$, so X_0 is a multiple of I, and $X_0 = 0$ because its trace vanishes. Then Step 5 gives equality.

Part (iv)

With $\varepsilon = 0$, $N(h) = \sqrt{d_{\mathrm{model}}}\, h/\|h\|$ depends only on the direction of h, and $B(\lambda n) = \lambda^k B(n)$ for $\lambda > 0$, so $N(B(N(h))) = N(B(h))$: the skipless network computes $x \mapsto N(B(xE))D$. Its normalization before the block drops out, which leaves the changes of basis $\mathrm{GL}(d_{\mathrm{model}})$ at h_0 and $\mathrm{CO}(d_{\mathrm{model}})$ at h_1, of dimension $d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$, and the residual network keeps one $\mathrm{CO}(d_{\mathrm{model}})$ for both hidden states (Theorem A.5(iii)).

The residual network. The case $\varepsilon = 0$ of part (iii), with $A = I$, shows that a flat direction is the sum of a direction of the rotation and scaling of all hidden states and a direction ψ of the weights of the block along which B has derivative zero on the sphere of radius $\sqrt{d_{\mathrm{model}}}$. This derivative is positively homogeneous of degree k in n, so it vanishes for all $n \neq 0$, and Lemma B.3, with E and D replaced by the identity and $M = 0$, makes ψ a rescaling of the units. So $\dim \mathcal{K}(\theta) = d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1 + s$.

The skipless network: the bound. The changes of basis in $\mathrm{GL}(d_{\mathrm{model}})$ at h_0 and in $\mathrm{CO}(d_{\mathrm{model}})$ at h_1 and the symmetries inside the block preserve the function. They move the parameters along the directions with the entries EX at E and $-XW_{\mathrm{read}}$ at W_{read}, with the entries $W_{\mathrm{write}}(Y + cI)$ at W_{write} and $-YD$ at D for an antisymmetric Y and a number c, and along the rescalings of the units. For generic parameters these directions are independent, as in Step 3 of part (i). So $\dim \mathcal{K}(\theta) \geq d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1 + s$, which is the bound.

The skipless network: equality for ReLU and ReLU^2. Let $\sigma(z) = \max(z, 0)^k$ with $k = 1$ or 2, and write $h_1 = B(xE) = \sum_i \sigma(xc_i)\, a_i$, with c_i the columns of $C = EW_{\mathrm{read}}$ and a_i the rows of W_{write}. The network is defined on the set Ω of inputs at which some unit is active: for generic parameters $h_1 \neq 0$ there, and $h_1 = 0$ elsewhere. Indeed, let S be the set of units active at x, and let w_i be the columns of W_{read}. If $|S| < d_{\mathrm{model}}$, the rows a_i, $i \in S$, are linearly independent. If $|S| \geq d_{\mathrm{model}}$ and $k = 1$, $h_1 = xE \sum_{i \in S} w_i a_i$, and this $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrix is invertible. If $|S| \geq d_{\mathrm{model}}$ and $k = 2$, the entries of h_1 are d_{model} quadratic forms in xE. A common zero other than 0, real or complex, makes their resultant vanish. The resultant is a polynomial in the weights, and it is nonzero at $w_i = e_i$, $a_i = e_i^\top$ for d_{model} units of S and $a_i = 0$ for the others, where the forms are the squares of the coordinates. Let η be flat, and let ψ_{c_i} and ψ_{a_i} be the derivatives of c_i and a_i along η. As in the case $\varepsilon = 0$ of part (iii), $\eta_D = -YD$ with Y antisymmetric, since h_1 takes all values in an open set, and $\mathrm{D}_\theta h_1\, \eta = h_1 Y + \mu\, h_1$ with a number $\mu(x)$ at each input of Ω. So

$$J(x) := \sum_i \Big[\sigma(xc_i)\, (\psi_{a_i} - a_i Y) + \sigma'(xc_i)\, (x\psi_{c_i})\, a_i \Big] = \mu(x)\, h_1(x) \qquad \text{on } \Omega. \tag{12}$$

The kinks stay. Fix a unit i and a point x_0 with $x_0 c_i = 0$, on no other hyperplane $x c_j = 0$, at which another unit is active. For generic parameters and x_0 in an open dense set of such points, $h_1(x_0)$ is not parallel to a_i. Near x_0, on each side of the hyperplane, $\mu = \langle J, h_1\rangle / \|h_1\|^2$ is a smooth function with limits at x_0. For $k = 1$, h_1 is continuous at x_0, so across the hyperplane the right side of (12) jumps by a multiple of $h_1(x_0)$, and the left side by $(x_0\psi_{c_i})\, a_i$; both jumps vanish. For $k = 2$, both sides, μ and the derivative of h_1 in x are continuous at x_0. The derivative in x of the left side jumps by the matrix $2\,(x_0\psi_{c_i})\, c_i a_i$, and that of the right side by a column times the row $h_1(x_0)$; two such matrices are equal only if both vanish. In both cases $x_0\psi_{c_i} = 0$ on an open subset of the hyperplane, so $\psi_{c_i} = \lambda_i c_i$ for a number λ_i.

One scale. As $\sigma'(z)\, z = k\, \sigma(z)$, (12) now reads

$$\sum_i \sigma(x c_i)\, \big(b_i - \mu(x)\, a_i\big) = 0 \quad \text{on } \Omega, \qquad b_i = \psi_{a_i} - a_i Y + k\lambda_i a_i,$$

and $J = \sum_i \sigma(x c_i)\, b_i$ and h_1 are continuous, so μ is continuous on Ω. On a region where only unit j is active, $b_j = \mu(x)\, a_j$, so μ is constant. On a region where at least two units are active, the entries of J and of h_1 are homogeneous polynomials of degree k in x, and for generic parameters two entries h_a and h_b of h_1 have no common factor. From $J_a h_b = J_b h_a$, h_a divides J_a, so $J = c\, h_1$ for a number c, and $\mu = c$ on the region. So μ is continuous on Ω and constant on every region. The set Ω is connected: it is the union of the open half-spaces $\{x : x c_i > 0\}$, and any two of them meet, as the c_i are pairwise non-collinear. Hence $\mu = c$ on Ω for one number c, and $\mathrm{D}_\theta h_1\, \eta = h_1(Y + cI)$ at every input, since both sides vanish where no unit is active.

Counting. Subtracting the change of basis $I + \tau(Y + cI)$ at h_1, with the entries $W_{\mathrm{write}}(Y + cI)$ at W_{write} and $-Y D$ at D, leaves a direction along which $f(xC)W_{\mathrm{write}}$ has derivative zero, which moves C and W_{write} only along a rescaling of the units, by Lemma B.3 with $M = 0$. After subtracting that rescaling as well, Lemma B.2 gives $\eta_E = EX$ and $\eta_{\mathrm{read}} = -XW_{\mathrm{read}}$ for a matrix $X \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{model}}}$, the change of basis at h_0. So $\dim \mathcal{K}(\theta) = d_{\mathrm{model}}^2 + d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1 + s$ for ReLU and ReLU^2 blocks, and the difference to the residual network is d_{model}^2.

Part (v)

With $\varepsilon = 0$, $N(h) = \sqrt{d_{\mathrm{model}}}\, h/\|h\|$ ignores the length of each token, so the block reads only rows on the sphere of radius $\sqrt{d_{\mathrm{model}}}$. Let V_θ be the space of directions that move only the weights of the block and leave its output unchanged to first order at every input whose rows lie on this sphere; it contains the symmetries inside the block. We show that the flat directions of the residual network are those of $\mathrm{CO}(d_{\mathrm{model}})$ and those of V_θ, while the skipless network has at least two copies of $\mathrm{CO}(d_{\mathrm{model}})$ and V_θ. The bound then holds whatever V_θ is.

The residual network. Write $h_1 = xE + g(x)$, with $g(x) = f(N(xE)W_{\mathrm{read}})W_{\mathrm{write}}$, and $y = N(h_1)D$, and let η be flat. Step 1 of part (ii) runs as before, except that with $\varepsilon = 0$ the derivative $\mathrm{D}N(h)$ projects onto the vectors orthogonal to h and scales by $\sqrt{d_{\mathrm{model}}}/\|h\|$, so its kernel is spanned by h. Hence $N(h_1)Y$ is orthogonal to h_1; as h_1 takes every value in an open set, $Y_s = 0$, and $\mathrm{D}_\theta h_1\, \eta = h_1 Y + \mu(x)\, h_1$ with a number $\mu(x)$, for attention one per token. Subtracting the rotation η^Y of Step 3 there leaves a flat η' with $\eta'_D = 0$ and $\mathrm{D}_\theta h_1\, \eta' = \mu'(x)\, h_1$.

Take the inputs $x = tw$ with $t > 0$. Since N is homogeneous of degree 0, the block output $g(tw) = g(w)$ and its derivative $\gamma(w)$ along η' do not depend on t, so

$$t\, w\eta'_E + \gamma(w) \;=\; \mu'(tw)\, \big(t\, wE + g(w)\big) \qquad \text{for all } t > 0.$$

For generic parameters wE and $g(w)$ are linearly independent for all w in an open set, since otherwise the block would map every point of the sphere to a multiple of itself. Comparing the parts along wE, along $g(w)$ and outside their span gives $w\eta'_E = c_w\, wE$ and $\gamma(w) = c_w\, g(w)$ with a number c_w. Then $w\eta'_E$ is parallel to wE for all w, so η'_E vanishes on the kernel of E, $\eta'_E = ET$ with vT parallel to v for every v, and $\eta'_E = cE$ with one number c. The scaling $E \mapsto (1+\tau)E$, $W_{\mathrm{write}} \mapsto (1+\tau)W_{\mathrm{write}}$ multiplies h_1 by $1 + \tau$ and is the scale in $\mathrm{CO}(d_{\mathrm{model}})$. Subtracting c times its direction leaves a flat η'' that moves only the weights of the block, with $\mathrm{D}_\theta g\, \eta'' = 0$ at the inputs tw. Their normalized

rows cover an open set of the sphere, and the derivative of the block along η'' is analytic on the sphere, so $\eta'' \in V_\theta$. The directions of $\mathrm{CO}(d_{\mathrm{model}})$ move E by $E(Y + cI)$, which vanishes only for $Y = 0$ and $c = 0$. So $\mathcal{K}(\theta)$ is the direct sum of these directions and V_θ, and the residual network has functional dimension $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - \dim V_\theta$.

The skipless network. Its block also reads only rows on the sphere, so the directions of V_θ are flat, and so are the changes of basis in $\mathrm{CO}(d_{\mathrm{model}})$ at h_0 and at h_1 (Proposition A.1). These directions are independent. A combination moves E by $E(Y_0 + c_0 I)$ and D by $-Y_1 D$, so $Y_0 = 0$, $c_0 = 0$ and $Y_1 = 0$. What is left is a multiple of the scaling $W_{\mathrm{write}} \mapsto (1 + \tau) W_{\mathrm{write}}$ plus a direction of V_θ, and the scaling lies in V_θ only if the block vanishes on the sphere. So the skipless network has functional dimension at most $P - 2 \dim \mathrm{CO}(d_{\mathrm{model}}) - \dim V_\theta$.

Generic parameters. The dimension of V_θ depends only on the weights of the block and takes its smallest value v for generic parameters. The generic functional dimensions are therefore $P - \dim \mathrm{CO}(d_{\mathrm{model}}) - v$ with the skip and at most $P - 2 \dim \mathrm{CO}(d_{\mathrm{model}}) - v$ without it, and the skip adds at least $\dim \mathrm{CO}(d_{\mathrm{model}}) = d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2 + 1$.

We recall Theorem 4.6 from Section 4.3.

Theorem 4.6 (Equality for one block)**.** *Let $L = 1$, with a* tanh*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU, GeGLU or attention block.*
(i) *Without normalization, the skipless and the residual network satisfy the equality condition, and the skip raises the functional dimension by exactly d_{model}^2.*
(ii) *Under RMSNorm, the residual network satisfies the equality condition, and the skip raises the functional dimension by at least $d_{\mathrm{model}}(d_{\mathrm{model}} - 1)/2$. A learned skip raises it by exactly $d_{\mathrm{model}}(d_{\mathrm{model}} + 1)/2$ more than an identity skip.*

Proof of Theorem 4.6. Part (i) is Theorem B.1(i), and part (ii) is Theorem B.1(ii) and (iii). □

Extensions of part (i)

Grouped-query attention. If the H query heads share G heads of keys and values, the same proof applies with B_i and C_i formed from the keys and values of the group of head i. Step 1 uses only the queries and the output projection of all heads, and in its last part a direction that keeps the products $W_{Q,i} W_{K,g}^\top$ over the heads i of group g has one Γ for the whole group, since $W_{K,g}$ has full column rank; likewise for the values. So $s = 2Gd_h^2$, and the biases add $2d_{\mathrm{model}} + 2Gd_h$.

Position embeddings. Let a learned position embedding add the row e_t of E_{pos} to token t of h_0, in a network with an attention block and no biases, and write η_{e_t} for the entry of a direction η at the position of e_t. As E has rank d_{model}, choose ϵ_t with $\epsilon_t E = e_t$: the network computes the network without position embedding on the tokens $\tilde{x}_t = x_t + \epsilon_t$. Along η, $h_{0,t} = \tilde{x}_t E$ has the derivative $\tilde{x}_t \eta_E + \nu_t$ with $\nu_t = \eta_{e_t} - \epsilon_t \eta_E$. This is the derivative of the network without position embedding along the other entries of η together with a shift κ_t of each token $\tilde{x}_t$, where $\kappa_t E = \nu_t$. Fix a token t that sees at least one other token, and inputs for which the tokens $\tilde{x}_s$, $s \neq t$, that token t sees all equal one vector w. The output at token t is symmetric in these tokens, so its derivative sees their shifts only through their mean $\bar{\kappa}$, and it is (10) with $u = \tilde{x}_t$, $v = u - w$ and k the number of these tokens. The shifts move u by κ_t and v by $\kappa_t - \bar{\kappa}$, which adds $\frac{1}{2} B_i (\kappa_t - \bar{\kappa})^\top$ to the derivatives of the input weights, $\frac{1}{2} \kappa_t B_i v^\top$ to those of the biases and $\frac{1}{2} (\kappa_t - \bar{\kappa}) C_i$ to those of the output weights of the tanh units, and Step 2 again makes each of these derivatives vanish. For all v this gives $\mathrm{D}_\theta B_i \, \eta = 0$, $\mathrm{D}_\theta C_i \, \eta = 0$, $\kappa_t B_i = 0$ and $B_i (\kappa_t - \bar{\kappa})^\top = 0$ for every head, and since the queries and the keys of all heads have rank d_{model}, $\nu_t = \kappa_t E = 0$ and $\bar{\kappa} E = 0$. This holds for every token $t \geq 2$, and for every token without a mask; with a mask, $\bar{\kappa} = \kappa_1$ at $t = 2$, so $\nu_1 = 0$ as well. So $\eta_{e_t} = \epsilon_t \eta_E = e_t X$ by Step 1, the change of basis at h_0 applied to E_{pos}, and the values of part (i) hold with P counting E_{pos}. A fixed encoding, which is not trained, must instead be preserved by the change of basis at h_0; if its rows span a space of dimension r, this removes $r d_{\mathrm{model}}$ dimensions of symmetry from both networks, and the skip still adds d_{model}^2.

C The symmetry group of a transformer with its embedding and readout

Wang and Wang (2025) characterize the symmetry group of a transformer block. For a full transformer they state that the group is the product of the groups of its blocks, since residual connections and LayerNorm prevent any coupling across layers (Wang and Wang, 2025, Theorem 7). They do not define the group of the full model. We take the model to be the map from tokens to outputs, so that the embedding and the readout are among its parameters, as in the models they name (GPT-2, BERT, LLaMA and Qwen) and in their end-to-end checks, which compare the logits and generated tokens of GPT-2 models. In this case two transformations of the whole stream, applied at every hidden state from the embedding to the readout, are counterexamples.

Table 4: Relative change of the output under the two transformations of the whole stream, worst case over three random transformers ($d_{\text{model}} = 8$, two heads, three layers, GELU MLPs of width 16, generic gains and biases). The notebook also checks the assumptions of Wang and Wang (2025) before and after each transformation, and that both transformations change the function of the bare stack of blocks. Section 7 of the notebook (Appendix E).

Model	Global permutation	Global rotation
pre-LayerNorm, untied readout	6×10^{-16}	2×10^{-15}
pre-RMSNorm, untied readout	5×10^{-16}	9×10^{-16}
pre-LayerNorm, tied readout	5×10^{-16}	fails, at least 0.16
post-LayerNorm, tied readout	9×10^{-16}	not applicable

1. A *global permutation* of the stream coordinates, with every read, write, normalization gain and bias permuted along. It preserves the function at every parameter value, for every placement of the normalization and with a tied or an untied readout. Verma and Elbayad (2024, Section 3.3) apply it to merge BERT models, embeddings included.
2. A *global rotation* Q, orthogonal and, for LayerNorm, fixing the all-ones vector. In a pre-norm model with an untied readout it preserves the function when every read W after a normalization with gain g becomes $\operatorname{diag}(g)^{-1}Q^{\top}\operatorname{diag}(g)\,W$ and every LayerNorm bias b becomes $b\operatorname{diag}(g)^{-1}Q\operatorname{diag}(g)$; the gains are unchanged. This is the construction behind Theorem 1 of Ashkboos et al. (2024a).

Both transformations change the token embedding: its columns, the coordinates of the stream, are permuted or rotated. No element of the product of per-block groups changes the embedding, so both lie outside that product. The LayerNorm transformers of Table 4 satisfy the assumptions of Wang and Wang (2025), including pairwise distinct gains and generic biases, and so do the transformed parameters.

Their proof transforms the output of one block and argues that the residual term prevents the transformation (Wang and Wang, 2025, Theorem 30); applied from the embedding onward, the transformation acts on the residual term as well. Their lemma on LayerNorm holds the gain and the bias fixed (Wang and Wang, 2025, Lemma 26); the permutation moves both, and in a pre-norm block the normalized vector is only read, so the rotation does not need LayerNorm to commute with it. For a stack of blocks that acts on a fixed input and is read in fixed coordinates, both transformations change the function, and the counterexample does not apply.

The two transformations are instances of the one change of basis that the skips leave (Theorem A.5(iii)), restricted to what the normalization lets through: the rotation needs a normalization that reads the whole vector, and the permutation also commutes with per-channel gains. With a tied readout the rotation fails: the readout after the final gain would have to become $\operatorname{diag}(g)^{-1}Q^{\top}\operatorname{diag}(g)\,E^{\top}$, while tying fixes it to $Q^{\top}E^{\top}$.

D Counts at model scale

Table 5 applies the exact reduction of Proposition A.9 to published models. Without skips, every hidden state keeps its own change of basis, and each turns one submatrix into a triangular matrix; with skips and an untied readout, the whole network keeps one (Section 3.3). Under RMSNorm each change of basis is orthogonal and frees $d_{\text{model}}(d_{\text{model}}-1)/2$ parameters, and under LayerNorm the same holds with $d_{\text{model}}-1$ in place of d_{model} (Proposition A.12). Llama-3.1 uses RMSNorm (Grattafiori et al., 2024), and its parameter counts are from the released weights; for the two ViT-B, with $d_{\text{model}}=768$ and $L=24$ sub-layers, we assume standard pre-LayerNorm blocks, which reduce to RMSNorm of dimension 767. The skipless ViT-B has 25 hidden states: one before each of its 24 blocks and one before the readout. One triangular 767×767 submatrix of a read per hidden state removes $25\cdot767\cdot766/2=7.3$M parameters. The residual ViT-B keeps one orthogonal change of basis, and so does a pre-RMSNorm language model whose readout is not tied to its embedding. Each removes the entries below the diagonal of one $d_{\text{model}}\times d_{\text{model}}$ submatrix.

Table 5: Parameters that can be removed from trained networks without changing their function. Section 8 of the notebook (Appendix E).

Model	Skips	Parameters	Removable exactly
ViT-B of Ji et al. (2026)	no	86.6M	7.3M (8.5 %)
ViT-B	yes	86.6M	0.29M (0.3 %)
Llama-3.1-8B	yes	8.0B	8.4M (0.10 %)
Llama-3.1-70B	yes	70.6B	34M (0.05 %)
Llama-3.1-405B	yes	405.9B	134M (0.03 %)

The gap between the two ViT-B, 24 copies of $\mathrm{O}(767)$ or 7.1M dimensions of symmetry, 8.1 percent of the parameters, is what the skips remove (Theorem A.13), and under the equality condition what the residual network gains in functional dimension. In general, a transformer layer with an MLP of width $4d_{\text{model}}$ has about $12d_{\text{model}}^2$ parameters, and under normalization the skips of its two sub-layers remove about d_{model}^2 dimensions of symmetry, a twelfth of the parameters of the blocks.

Inside the attention blocks, with or without skips, each head also has a change of basis between its queries and keys and one between its values and outputs: $12\cdot2Hd_h^2=1.2$M dimensions for ViT-B. With the one copy of $\mathrm{O}(767)$ that the residual ViT-B keeps, this gives 1.5M, the count of van Nierop (2024, Eq. 17).

E Numerical checks and reproducibility

All computations are in double precision, in a notebook that will be released in a public Git repository, coming soon; its sections are listed at the end of this appendix. Unless stated otherwise, weights have independent $\mathcal{N}(0,1/\text{fan-in})$ entries, inputs have independent $\mathcal{N}(0,1)$ entries, and networks compared with and without skips share their weights; the notebook gives the few exceptions.

Exactness checks. For Theorems A.5 and A.10 and Proposition A.9 the transformed weights are constructed as in the proofs, and both networks are evaluated on random inputs; the reported number is $\|f'-f\|/\|f\|$ over the batch, and values near 10^{-15} are floating-point rounding.

The sweep. At depth six, with biases, the sweep of Theorem A.5(i) turns identity units with trainable U and V into identity skips, with U and V equal to I up to 2×10^{-15} and the function unchanged up to 2×10^{-15}, and its forward version (Remark A.6) absorbs learned affine skips with skip biases up to 10^{-15}. In Theorem A.10(ii) a scaled orthogonal skip is absorbed to 10^{-9}, which is the RMSNorm constant $\varepsilon=10^{-8}$ acting on the overall scale (Lemma A.2), and a generic skip fails at 0.43.

Homogeneous blocks under RMSNorm. For Lemma A.3, on skipless ReLU, ReLU^2 and ReGLU networks of the size of Figure 3 with $\varepsilon=10^{-1}$, 10^{-3} and 10^{-6}, a change of basis $e^{0.3S}$ at h_0, h_1 or h_2, or the scaling $e^{0.3}$ at h_3, multiplies each output by the factor Λ of the proof up to 5×10^{-12}, and

with $\varepsilon = 0$ leaves the function unchanged up to 10^{-13}; the derivatives match part (ii) up to 4×10^{-12} of the output (Section 17 of the notebook).

Triangular skips and tied readouts. The checks of Section 3 use a pre-RMSNorm network with $d_{\text{model}} = 8$, $L = 5$ $\tanh$ blocks of width 16 and learned skips near the identity, with $\varepsilon = 10^{-2}$. Making every skip triangular changes the function by 5×10^{-16}, and making it symmetric positive definite by 4×10^{-15}. With a readout tied to the embedding through the gain of the final normalization, making the first four skips triangular and leaving the last one full changes it by 7×10^{-16} and removes 112 parameters; making all five triangular and inserting one 8×8 matrix before the gain also changes it by 7×10^{-16} and removes $140 - 64 = 76$, and using also the rotation that this matrix frees at h_0, on a triangular block of the embedding, and the gain, which it absorbs, changes it by 10^{-15} and removes 112, no more than without the matrix. Without normalization and with a tied readout, setting the first four skips and 8 rows of the embedding to the identity changes the function by 10^{-13} and removes 320 parameters. For identity skips, a global rotation changes the tied network by 0.22, and one inserted matrix restores it to 8×10^{-16}. Setting $r = 2$ rows of each low-rank factor U to the identity changes the function by 3×10^{-15}, and without normalization the sweep of Theorem A.5 turns the low-rank skips into identity skips with a change of 6×10^{-16}.

Changes of basis that differ across a skip. In a separate residual network of two blocks without normalization ($d_{\text{model}} = 8$, width 16), a change of basis G with $\|I - G\| = 0.41$ at the two hidden states after the first block, so that only the skip of that block joins two bases, changes the output by 0.28, 0.25, 0.050 and 0.0046 when the branch of the first block is scaled by 0.1, 1, 10 and 100, which makes it about 0.05, 0.5, 5 and 50 times the size of the skip, of the order of $\|h\|\|I - G\|/\|B(h)\|$ when the branch dominates (Section 3.3).

Functional dimension. The parameter-to-output map is evaluated on a fixed batch of inputs, and its Jacobian with respect to all parameters is formed exactly by automatic differentiation, or by central finite differences with step 10^{-6} for the small $\tanh$ networks of Table 6. For exact Jacobians the rank is read at the largest gap of the spectrum, the largest ratio s_i/s_{i+1} of consecutive singular values, 10^5 to 10^{12} for the rows of Table 6; for finite differences at the relative threshold 10^{-7}, which lies inside the gap between the genuine singular values, above 10^{-6} of the largest, and the noise of the differences, below 10^{-8}. The rank on a batch is a lower bound on the functional dimension, and where P minus it equals the symmetry count, it is the functional dimension (Theorem A.13(ii)). The values of Theorem B.1(i) hold exactly for all five activations at $d_{\text{model}} = 4$, $m = 8$, in three seeds, with and without biases, with gaps of 10^{11} to 10^{14}; with biases the inputs must cross the kink of every unit, so these checks use inputs of scale 5. For one attention block ($d_{\text{model}} = 4$, input and output dimension 6, sequences of three tokens, $(H, d_h) = (2, 2)$, $(4, 1)$ and $(1, 4)$, causal and bidirectional, with and without biases, three seeds each), all 72 networks have the values of Theorem B.1(i), with gaps of at least 4×10^{11}. Under pre-RMSNorm with $\varepsilon = 10^{-2}$ ($d_{\text{model}} = 4$, three seeds each), the residual networks with one block have the value of Theorem B.1(ii) for all five activations and for attention, and the skipless ones have $P - d_{\text{model}}(d_{\text{model}} - 1) - s$, so the skip adds exactly $d_{\text{model}}(d_{\text{model}} - 1)/2$ in these checks. A direction that the network uses only on rare inputs, such as a rarely active unit, has small singular values, and resolving it takes far more than the minimum of about P/d_{out} inputs.

The networks of Figure 3. Every network of Figure 3 is drawn 20 times, and so are the same networks with GeGLU blocks and with GLU blocks, whose gate is the logistic sigmoid; both have the counts of SwiGLU. Except for the skipless ReGLU networks under RMSNorm, the rank equals the count in every draw, at gaps of at least 6×10^4 without normalization and 10^7 under RMSNorm with $\varepsilon = 10^{-5}$. In the skipless ReGLU networks under RMSNorm, at gaps between 40 and 2×10^4, the rank read at the largest gap is 263 in 14 draws, the count 264 in 5 and 253 in one, and with hidden units added 283 in 19 draws and the count 284 in one. With $\varepsilon = 10^{-12}$ the skipless ReGLU network has rank 233 in every draw, at gaps of at least 10^4: this is $P - Ld_{\text{model}}^2 - (d_{\text{model}}(d_{\text{model}} - 1)/2 + 1) - s$, the count of the network in which every normalization but the last is removed. The residual ReGLU network then has rank $281 = P - (d_{\text{model}}(d_{\text{model}} - 1)/2 + 1) - s$ in every draw (Section 4.5). With $\varepsilon = 10^{-2}$ the rank equals the count in every draw for all six activations, ReGLU included, at gaps of at least 2.8×10^9; the notebook also plots these spectra as in Figure 3.

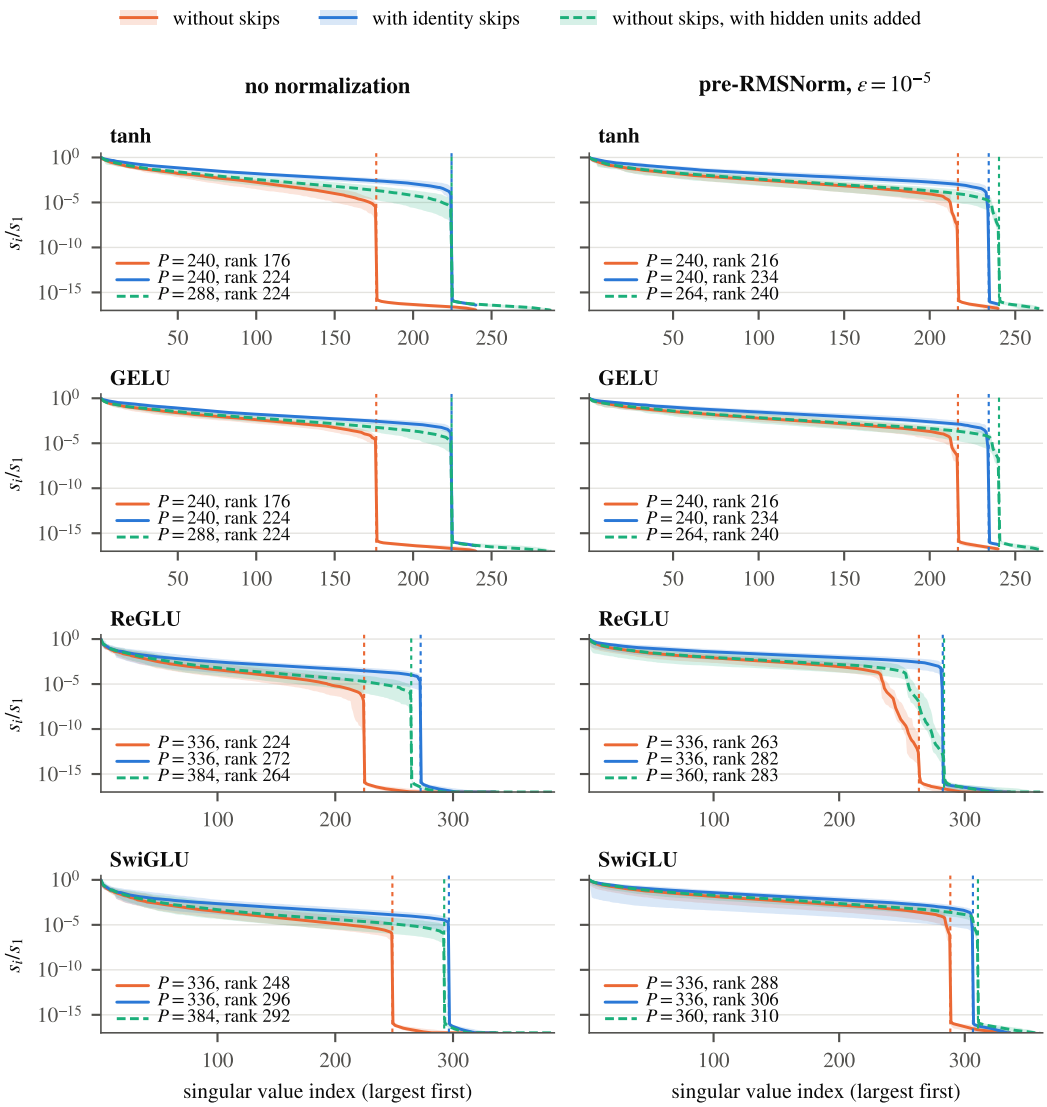


Figure 3: Singular values of the Jacobian on 4000 inputs, for the networks of Figure 1 with four kinds of blocks, and without skips with hidden units added; left without normalization, right with pre-RMSNorm. Dashed lines mark the rank. It equals the symmetry count in every draw, except for skipless ReGLU networks under RMSNorm.

The size of ε. At $\varepsilon = 0$ the skipless ReGLU network has 31 more dimensions of symmetry (Lemma A.3), and a small ε breaks them only weakly, which makes its rank hard to read at $\varepsilon = 10^{-5}$. RMSNorm divides a row $h^{(t)}$ by $\sqrt{r^2 + \varepsilon}$, where $r^2 = \|h^{(t)}\|^2/d_{\text{model}}$ is its mean square, so ε acts relative to r^2. With inputs of unit variance, the median of r^2 over the inputs, and then over the draws, is 0.74 at h_0 and between 0.15 and 0.33 after the blocks of the skipless networks, and it grows along the stream of the residual networks to between 1.6 and 1.9 at h_3; at these medians ε/r^2 lies between 5×10^{-6} and 7×10^{-5} for $\varepsilon = 10^{-5}$, and is about a thousand times larger for $\varepsilon = 10^{-2}$. In a skipless network every block reads a normalized state, so the scale of the inputs enters only at h_0; the stream of a residual network carries it to every hidden state. At $\varepsilon = 10^{-5}$ the singular values with indices 234 to 264 of the skipless ReGLU network, those of these 31 directions, lie between $10^{-4.5}$ and $10^{-12.6}$ of the largest in the median over the draws, except the last, which lies at the floor. The factors $\varepsilon/(r^2 + \varepsilon)$ that break these directions depend on the input, and they are large for the inputs whose hidden states have a small mean square: in every draw, between 11 and 592 of the 4000 inputs have a hidden state after a block with $r^2 < 10^{-3}$, where $\varepsilon/r^2 > 10^{-2}$, and in 13 of the 20 draws up

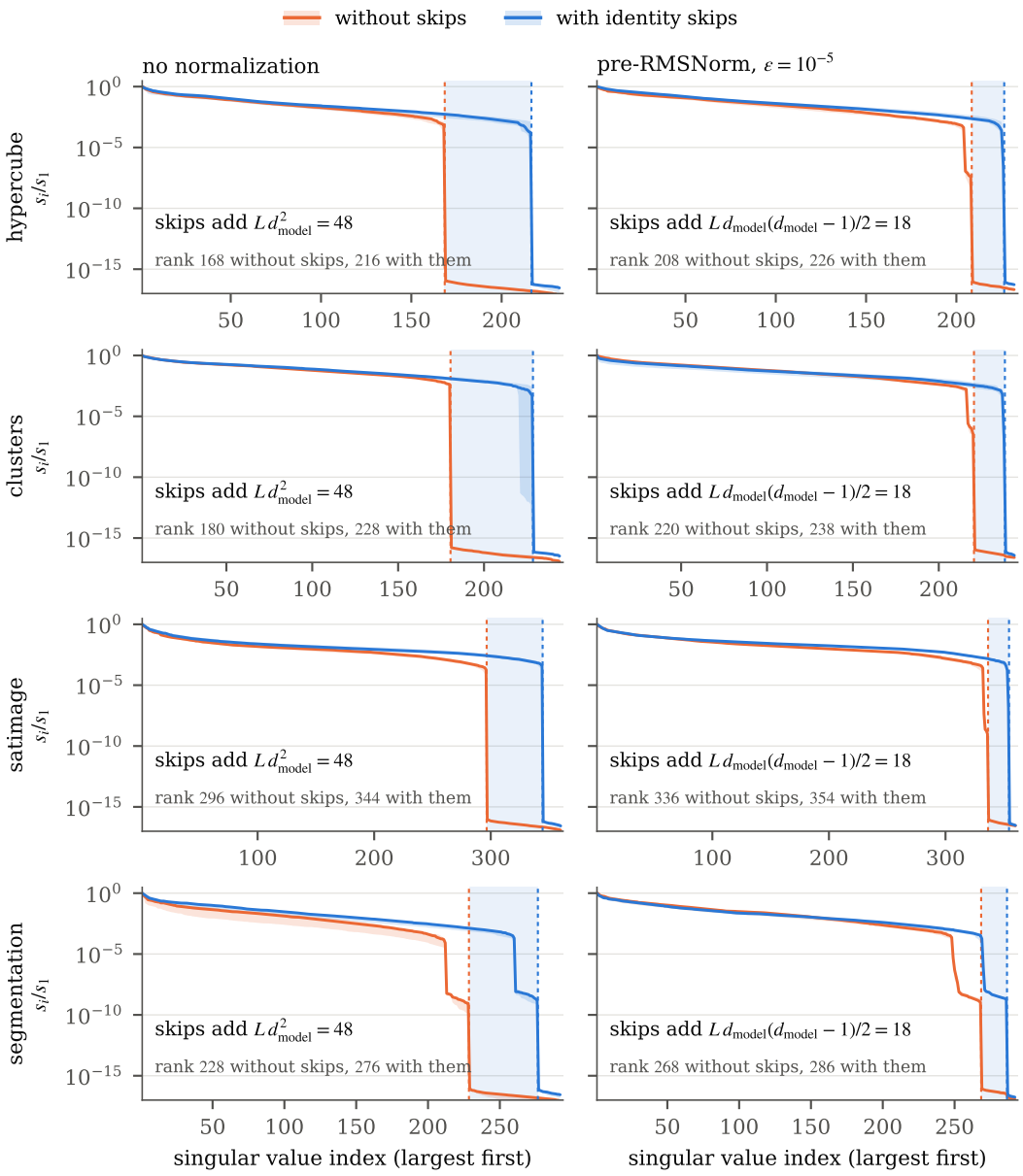


Figure 4: Singular values of the Jacobian at trained weights, on 1000 inputs of each dataset: the GELU networks of Figure 3, trained on four datasets (rows); median and range over five seeds. The skips add Ld^2_{model}, and $Ld_{\text{model}}(d_{\text{model}} - 1)/2$ under RMSNorm, as at random weights.

to 378 inputs have one that is exactly zero, with every gate closed.

Trained networks. We also train the networks of Figure 3, with tanh, GELU, ReGLU and SwiGLU blocks, on four classification datasets with at least four classes: two generated with scikit-learn, and satimage and segmentation from the Penn Machine Learning Benchmarks, with the features as inputs and one output per class. Training uses AdamW in single precision, a learning rate chosen on a validation set, early stopping and five seeds. A network counts as trained when it beats logistic regression with a tuned penalty on the test set, as 436 of 480 do. On inputs of the dataset and on Gaussian inputs alike, the rank equals the count in 398 of the 436, and the skips add 48 and 18 (Figure 4). Of the 38 exceptions, 36 are skipless ReGLU networks under RMSNorm; one residual GELU network is 8 below the count and one residual tanh network 1 below. Two further notebooks train these networks, store them and compute their Jacobians.

Table 6: Functional-dimension checks. *Rank* is the rank of the Jacobian on the batch; where the redundancy, P minus it, equals the prediction, the rank meets the upper bound of Theorem A.13(ii) and is the functional dimension. Every network has an embedding and a separate readout, and no biases unless marked; normalization is pre-norm. A *mixed* skip is a pair of trainable identity units U, V per block. The MLP networks have $d_{\text{model}} = 4$, width 8, $L = 3$ unless marked, and input and output dimension 6, except the rows marked *large*, which have $d_{\text{model}} = 16$, $L = 6$, width 16 and input and output dimension 20. The transformers have $d_{\text{model}} = 8$, two heads of width four, MLP width 16, a vocabulary of 16 tokens and two layers; each attention block has $2Hd_h^2$ symmetries inside. †The exact rank modulo a prime certifies the value at two parameter points; without skips at $L = 3$, the rank in floating point sits at the double-precision floor. ‡The symmetry count is 268; an embedding table read through RMSNorm adds $d_{\text{model}}(d_{\text{model}} + 1)/2 = 36$, because N maps $\mathbb{R}^{d_{\text{model}}}$ onto an open ball, so the rows of the table can absorb any invertible map near the identity, and with continuous inputs the measured value is 268.

Network	Skip	Normalization	P	Rank	Redundancy	Prediction
tanh	none	none	240	176	64	$(L{+}1)d_{\text{model}}^2 = 64$
tanh	identity	none	240	224	16	$d_{\text{model}}^2 = 16$
tanh	learned	none	288	224	64	$(L{+}1)d_{\text{model}}^2 = 64$
tanh	mixed	none	336	224	112	$(2L{+}1)d_{\text{model}}^2 = 112$
tanh, biases	none	none	286	206	80	$(L{+}1)(d_{\text{model}}^2{+}d_{\text{model}}) = 80$
tanh, biases	identity	none	286	254	32	$d_{\text{model}}^2 + (L{+}1)d_{\text{model}} = 32$
tanh	none	RMSNorm, $\varepsilon = 10^{-9}$	240	212	28	$(L{+}1)\dim \mathrm{CO}(4) = 28$
tanh	identity	RMSNorm, $\varepsilon = 10^{-9}$	240	233	7	$\dim \mathrm{CO}(4) = 7$
tanh	learned	RMSNorm, $\varepsilon = 10^{-9}$	288	260	28	$(L{+}1)\dim \mathrm{CO}(4) = 28$
tanh	none	RMSNorm, $\varepsilon = 10^{-2}$	240	216	24	$(L{+}1)\dim \mathrm{O}(4) = 24$
tanh	identity	RMSNorm, $\varepsilon = 10^{-2}$	240	234	6	$\dim \mathrm{O}(4) = 6$
tanh	learned	RMSNorm, $\varepsilon = 10^{-2}$	288	264	24	$(L{+}1)\dim \mathrm{O}(4) = 24$
ReLU^2	none	none	240	152	88†	$(L{+}1)d_{\text{model}}^2 + Lm = 88$
ReLU^2	identity	none	240	200	40†	$d_{\text{model}}^2 + Lm = 40$
ReGLU	none	none	336	224	112†	$(L{+}1)d_{\text{model}}^2 + 2Lm = 112$
ReGLU	identity	none	336	272	64†	$d_{\text{model}}^2 + 2Lm = 64$
ReLU^2, $L = 8$	none	none	560	352	208†	$(L{+}1)d_{\text{model}}^2 + Lm = 208$
ReLU^2, $L = 8$	identity	none	560	480	80†	$d_{\text{model}}^2 + Lm = 80$
ReGLU, $L = 8$	none	none	816	544	272†	$(L{+}1)d_{\text{model}}^2 + 2Lm = 272$
ReGLU, $L = 8$	identity	none	816	672	144†	$d_{\text{model}}^2 + 2Lm = 144$
tanh, large	none	none	3712	1920	1792	$(L{+}1)d_{\text{model}}^2 = 1792$
tanh, large	identity	none	3712	3456	256	$d_{\text{model}}^2 = 256$
tanh, large	none	RMSNorm, $\varepsilon = 10^{-2}$	3712	2872	840	$(L{+}1)\dim \mathrm{O}(16) = 840$
tanh, large	identity	RMSNorm, $\varepsilon = 10^{-2}$	3712	3592	120	$\dim \mathrm{O}(16) = 120$
transformer	identity	RMSNorm, $\varepsilon = 10^{-2}$	1280	1124	156	$\dim \mathrm{O}(8) + 4Hd_h^2 = 156$
transformer	none	RMSNorm, $\varepsilon = 10^{-2}$	1280	976	304‡	$5\dim \mathrm{O}(8) + 4Hd_h^2 = 268$

Learned skips. The tanh networks of Table 6 with learned skips or with trainable identity units have the functional dimension of the residual network, 224, as Theorem A.5(ii) requires. Under RMSNorm with $\varepsilon = 10^{-2}$, learned skips have $264 - 234 = 30 = 3\,d_{\text{model}}(d_{\text{model}} + 1)/2$ more functional dimension than identity skips, as Theorem A.10(iii) predicts. On residual networks without biases or normalization ($d_{\text{model}} = 4$, $L = 3$, width 8), using the last change of basis on d_{model} columns of the first read leaves exactly the redundancy s of the symmetries inside the blocks, the last row of Table 3.

Low-rank skips. Networks with skips $I + UV^\top$ ($L = 3$ tanh blocks of width $2d_{\text{model}}$, $d_{\text{model}} = 4$ with $r = 1, 2$ and $d_{\text{model}} = 6$ with $r = 2, 3$, exact Jacobians) have redundancy $d_{\text{model}}^2 + 2Ld_{\text{model}}r$ without normalization, all of U and V besides the global change of basis, and $d_{\text{model}}(d_{\text{model}} - 1)/2 + Lr(2r - 1)$ under RMSNorm with $\varepsilon = 10^{-2}$. Besides the global rotation and the r^2 dimensions of the $\mathrm{GL}(r)$ inside each product, each skip adds $r(r - 1)$. To first order, rotations $I + \Omega_{\ell-1}$ and $I + \Omega_\ell$ of the two hidden states that it joins change it by $(\Omega_\ell - \Omega_{\ell-1}) - \Omega_{\ell-1}UV^\top + UV^\top\Omega_\ell$, which keeps it within rank r of the identity when $\Omega_\ell - \Omega_{\ell-1}$ lies in the tangent space $\{XV^\top + UY^\top\}$ of the matrices of rank r at $UV^\top$; the antisymmetric matrices in this space have dimension $r(r - 1)$ when $2r \le d_{\text{model}}$ and U, V are generic, which the notebook confirms in every case.

Where the redundancy exceeds the count. For ReLU networks the exact Jacobian is taken with

the activation pattern held fixed. On skipless ReLU networks without biases ($d_{\text{model}} = 4$, $L = 3$, width 8), 7 of 8 random draws attain the bound of Theorem A.13(ii), redundancy 88 including the rescalings of hidden units, and all 8 residual draws attain their bound, 40. In the eighth skipless draw, five units of the last block are active on exactly the same inputs, those on which one unit of the block before is active, so there they act as one linear map of $\mathbb{R}^4$: their 40 parameters carry $40 - 16 = 24$ dimensions of redundancy instead of the 5 rescalings of the count, and the redundancy is $88 + 19 = 107$. Such coincidences need a block that reads the output of another block, so they do not occur for one block, where Theorem B.1 applies; this is consistent with Grigsby et al. (2023). On the 3000 inputs of Section 1 of the notebook, the redundancy of a second draw also exceeds the count, by 6, because one of its units is active on only 0.08 percent of the inputs; with 30,000 inputs or more it attains the bound, and the eighth draw stays at 107 (Section 15 of the notebook). Skipless networks whose first hidden state is an embedding table read through RMSNorm are also more redundant than their symmetries (Table 6).

Certificates at depth. The rows of Table 6 at $L = 8$ are certified as those at $L = 3$, with the activation pattern from ball arithmetic and the Jacobian from reverse mode (Section 16 of the notebook). Without skips these networks start from odd pairs of units, plus noise: ReLU^2 units (u, w) and $(-u, -w)$, and ReGLU units with gates u and $-u$ that share their value and write. With identity skips the weights are Gaussian, with the writes scaled by $1/L$. From Gaussian weights, the skipless ReGLU networks at $L = 8$ attain the count on 12,000 inputs, and the ReLU^2 ones do not. In each ReLU^2 draw, 1 to 5 units of the last three blocks stay off on 2×10^6 fresh inputs, and a unit that is never on leaves its 8 parameters redundant, 7 more than its rescaling. With inputs added on which the rarely active units fire, the rank stays below the count by 35, 7 and 29: 7 for each unit that stays off, and one more in the third draw.

DyT and nanoGPT. DyT replaces a normalization by $\tanh(\alpha h)$ with a learnable scale α (Zhu et al., 2025). DyT networks ($d_{\text{model}} = 4$, $L = 3$ blocks of width 8, exact Jacobians, three draws) lose $L = 3$ dimensions to the skips, the scale of each hidden state, which the learnable α of the next DyT absorbs. The nanoGPT models of Table 7 follow the `model.py` of nanoGPT (Karpathy, 2023), with token and position embeddings, pre-LayerNorm blocks with $\varepsilon = 10^{-5}$, causal attention with four heads of width two, GELU MLPs of width $4d_{\text{model}}$, a final LayerNorm and a readout, and no biases; without normalization every LayerNorm is the identity. The Jacobians are exact, over all logits of 64 random token sequences per layer. With skips the redundancy is, per layer, 144 with LayerNorm: 32 from the changes of basis inside the heads, 16 from the LayerNorm gains against the next matrix, 56 from the zero mean of each LayerNorm output, seen by every column of the next matrix, and 40 from the mean of each write, which every later LayerNorm erases. Once per model come 60 more: the global rotation, 21; the means of the 16 embedding rows, 16; the final LayerNorm against the readout, 16; and a vector moved from the position to the token embedding, 7. Without normalization it is 32 per layer and $72 = 64 + 8$ per model. For one and two layers we construct all of these symmetries, and in every draw they span the null space of the Jacobian.

Table 7: Redundancy of nanoGPT ($d_{\text{model}} = 8$, four heads of width two, a vocabulary and a context of 8, no biases, untied readout): P minus the rank of the exact Jacobian, the same in all four draws of every network and equal to the prediction. N layers make $L = 2N$ blocks. With skips, each layer is redundant by 144 with LayerNorm and by 32 without normalization, and the model once more by 60 and 72 (Appendix E).

	$N = 1$	2	3	4	Prediction
LayerNorm					
P	984	1768	2552	3336	
without skips	246	432	618	804	$144N + 60 + 21L$
with skips	204	348	492	636	$144N + 60$
No normalization					
P	960	1728	2496	3264	
without skips	232	392	552	712	$32N + 72 + 64L$
with skips	104	136	168	200	$32N + 72$

Notebook. The notebook runs on a CPU in an environment pinned with uv, with Python 3.11. Its sections follow the paper and are listed below, with their times on a CPU with two cores. The two long experiments save what Figure 3 and Table 7 need, from which the notebook draws the figure and the table in seconds unless it is asked to rerun them; the times below are those of the full computation.

Section of the notebook	What it reproduces	Time
1. The gauge fix, the counts and RMSNorm	exactness checks of Theorems A.5 and A.10 and Proposition A.9; finite-difference rows of Table 6; exact ReLU and degree-two Jacobians	1 min
2. The last row of Table 3	the last row of Table 3	23 s
3. A dominant block, and learned, tied and low-rank skips	the checks of Sections 3, 3.3 and 3.4	under 1 s
4. The redundancy of low-rank skips	the low-rank skips of Section 3.4	36 s
5. DyT reads	the DyT networks	9 s
6. Exact reductions after training	the reductions of Section 3.1 and Proposition A.12	under 1 s
7. The symmetry group of a transformer is not a product	Table 4	under 1 s
8. Counts at model scale	Table 5	under 1 s
9. One block	Theorem B.1, for MLP and attention blocks	44 s
10. Figure 3	Figures 1 and 3 and the checks of their networks	28 min
11. Exact Jacobians	the rows at $d_{\text{model}} = 16$, the transformers and the embedding table of Table 6	4 min
12. Certificates modulo a prime	the certificates at $L = 3$ in Table 6	19 s
13. nanoGPT	Table 7 and the constructed symmetries	38 min
14. Skipless networks as limits of residual networks	the limits of the companion paper	under 1 s
15. ReLU draws above the count	the ReLU draws above the count in Appendix E	2 min
16. Certificates at depth	the certificates at $L = 8$ in Table 6	5 min
17. Homogeneous blocks under RMSNorm	the checks of Lemma A.3	2 s